Policy Research 2024-19-03

2024 K-Innovation Partnership Program with Indonesia

Policy Consultation on Design and Implementation of R&D Support Policy for Improvement of Science, Technology and Innovation System in Indonesia

STEPI SCIENCE AND TECHNOLOGY POLICY INSTITUTE

BRIN BADAN RISET DAN INOVASI NASIONAL

2024 K-Innovation Partnership Program with Indonesia

Title:	Policy Consultation on Design and Implementation of R&D Support Policy for Improvement of Science, Technology and Innovation System in Indonesia
Institute:	Science and Technology Policy Institute (STEPI)
Partner Country:	Republic of Indonesia
In Cooperation with:	National Research and Innovation Agency (BRIN)
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Executive Summary

The 2024 Korea-Innovation (K-Innovation) partnership programme with Indonesia is a pivotal initiative designed to strengthen Indonesia's National Research and Innovation Agency (BRIN) by providing comprehensive policy consultations on research and innovation (R&I) infrastructure and funding systems. Implemented by the Science and Technology Policy Institute (STEPI) of Korea, this project serves as a strategic platform for cross-border knowledge transfer and collaborative efforts aimed at advancing Indonesia's national R&I ecosystem. It is part of a broader, multi-year cooperation between South Korea and Indonesia, which has demonstrated tangible outcomes since its inception in 2023.

This project represents a significant step in reinforcing Indonesia's science, technology, and innovation (STI) policies through tailored activities with STEPI of South Korea. The primary objectives of the 2024 partnership program focus on optimizing R&I infrastructure management and enhancing R&I funding systems, which are crucial for Indonesia's long-term STI growth. These objectives were addressed through an integrated approach involving policy consultation, hands-on workshops, joint research, and knowledge-sharing seminars. By leveraging South Korea's expertise in R&D support systems, the program aims to align Indonesia's R&I infrastructure and funding frameworks with international best practices.

A key component of the program is the policy advisory delivered by Korean experts, who provided insights into optimizing Indonesia's R&I infrastructure, particularly focusing on improving the R&D information systems that facilitate the management and accessibility of research facilities. Additionally, the program addresses the evolution of Indonesia's R&I funding system, focusing on fostering collaboration between public and private sectors and enhancing the competitiveness of Indonesia's R&D funding mechanisms. Through these tailored interventions, the program supports BRIN's strategic initiatives to foster a more inclusive and sustainable innovation ecosystem.

Furthermore, the program facilitated the development of specific policy recommendations and action plans to improve Indonesia's R&I funding schemes. By analyzing existing funding structures and identifying gaps in governance, the program proposes implementing more agile and transparent funding mechanisms. These recommendations align with global innovation funding trends, which increasingly emphasize flexibility, industry collaboration, and performance-based metrics. This is particularly relevant in Indonesia's efforts to diversify its funding sources and attract private sector investments into R&I.

The knowledge exchange between Korea and Indonesia has also been operationalized through capacity-building workshops in both countries. These workshops have enabled Indonesian policymakers, researchers, and analysts to gain practical exposure to South Korea's R&D management systems, particularly in areas such as industry engagement in R&D funding and the optimization of research infrastructure. The capacity-building component of the program is expected to have a lasting impact on the development of Indonesia's R&I policies, ensuring that local stakeholders are equipped with the skills and knowledge required to implement the recommended reforms effectively. In conclusion, the 2024 K-Innovation Partnership Program marks a critical milestone in the ongoing collaboration between Korea and Indonesia in STI policy. The project provides actionable policy insights and establishes a robust framework for the continuous improvement of Indonesia's R&I ecosystem. By leveraging South Korea's successful innovation practices, the program aims to catalyze the modernization of Indonesia's R&I infrastructure and funding systems, positioning the country to become a more competitive player in the global innovation landscape. This partnership highlights the importance of international cooperation in advancing STI policies and offers valuable lessons for other nations seeking to enhance their R&D capabilities. As such, the 2024 program serves as a model of international collaboration, potentially driving significant, long-term benefits for both countries' innovation agendas.

ABBREVIATION

BAPPENAS	Ministry of National Development and Plan
BATAN	National Nuclear Energy Agency
BLU-LPDP	Public Service Agency of Indonesia Endowment Fund for Education Agency
BMGF	Bill & Melinda Gates Foundation
BPPT	Agency for Assessment and Application of Technology
BRIN	National Research and Innovation Agency of Republic of Indonesia
CoCI	Chamber of Commerce and Industry
CSF	CPO Supporting Fund
DFRI	Deputy of Facilitation for Research and Innovation
DKP-BRIN	Deputy for Development Policy-BRIN
DKRI – BRIN	Deputy for Research and Innovation Policy
DLMRFSTP	Directorate of Laboratory Management, Research Facilities, and Science and -Technology Park
DRIF	Directorate for Research and Innovation Funding
DRTIPE	Directorate for Research, Technology and Innovation Policy Evaluation
DRTIPF	Directorate for Research, Technology and Innovation Policy Formulation
ELSA	E-Service for Science or E-Layanan Sains
FGD	Focused Group Discussions
GBARD	Government Budget Allocations for Research and Development
GDP	Gross Domestic Product
GII	Global Innovation Index
HPC	High Performance Computing
IAoS/AIPI	Indonesian Academy of Sciences
IDR	Indonesian Rupiah
IoT	Internet of Things

IP	Intellectual Property
IRCC/PKR	Industrial Research Collaboration Center
ISF	Indonesia Science Fund
ISMP	Information System Master Plan
JIAT	Jeonbuk Institute of Automotive Convergence Technology
KAERI	Korea Atomic Energy Research Institute
KBE	Knowledge-Based Economy
KBSI	Korea Basic Science Institute
Kemenag	Ministry of Religious Affairs
Kemendikbud-Ristek	Ministry of Education, Culture, Research and Technology
K-Innovation	Korea-Innovation
KIST	Korea Institute of Science & Technology
KISTEP	Korea Institute of Science & Technology Evaluation and Planning
KPIs	Key Performance Indicators
LAPAN	National Institute of Aeronautics and Space
LIPI	Indonesian Institute of Sciences
LPDP	Indonesia Endowment Fund for Education Agency
MoF	Ministry of Finance
Monev	Monitoring and Evaluation
MSIP	Ministry of Science, ICT and Future Planning
MSIT	Ministry of Science and ICT
NEDO	New Energy and Industrial Technology Development Organization
NFEC	National Research Facilities Equipment Center
NGOs	Non-Government Organizations
NICs	Newly Industrializing Countries
NIS	National Innovation System
NREB/APBN	National Revenue and Expenditure Budget
NRF	National Research Fund
ODA	Official Development Assistance

PCP	Project Concept Paper
Pemda	Local Government Agency
POPFMA/ BPDPKS	Palm Oil Plantation Fund Management Agency
PPPs	Public-Private Partnerships
Q&A	Question and Answer
R&D	Research and Development
R&I	Research and Innovation
RCA	Collaborative Research Action
RCPP	Research Centre for Public Policy
Renstra	Annual Strategic Development Plans
RFP	Request for Proposal
RIfAI/RIIM	Research and Innovation for Advanced Indonesia
RIGHT	Research Investment for Global Health Technology Foundation
RIRN	National Research Master Plan
RoK	Republic of Korea
RPJM	Medium-term Development Plans
RPJMN	National Medium-Term Development Plan
RPJP	Long-term Development Plans
S&T	Science and Technology
SATREP	Science and Technology Research Partnership for Sustainable Development
SME	Small Medium Enterprise
SMEs	Small Medium Enterprises
SoS/BMN	State-owned Assets
STEPI	Science Technology Policy Institute
STI	Science-Technology-Innovation
TMS	Total Monitoring System
TRL	Technology Readiness Level.
XRD	X-Ray Diffraction
ZEUS	Zone of Research Equipment Utilisation Service

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Project Overview

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Chapter 1. Project Overview

1. Project Background

Science-technology-innovation (STI) policy or innovation policy, refers to a set of governmental strategies aimed at fostering a knowledge-based economy (KBE) or science-driven economy. The primary objective of STI policy is to create the enabling conditions for an effective innovation ecosystem. Mani (2002), Patanakul & Pinto (2017), and Asmara & Kusumastuti (2021) highlight that STI policy is instrumental in driving research and development (R&D) investments, addressing market failures such as private underinvestment in R&D, and fostering collaboration between business, government, and academia. These policies enable the development of effective partnerships, enhance knowledge exchange, and encourage the alignment of interests across these sectors, leading to a more integrated innovation ecosystem.

STI policies also focus on strengthening human capital in science and technology (S&T), accelerating the commercialization and diffusion of R&D outcomes, and protecting intellectual property rights. In addition, they aim to integrate industrial, trade, and innovation policies, offer targeted incentives, reduce tax burdens, and ensure access to state-of-the-art R&D infrastructure. These measures are designed to create a robust environment for innovation, support sustainable economic growth, and drive technological advancements.

Research and development (R&D) infrastructure and funding are basic facilities required in promoting the science and innovation development in many countries. Studies of Huda et. al (2020); Jin & Lee (2020); Kim (2023) disclose that R&D infrastructure and funding are complementary elements to corroborate a science advancement and to flourish innovation practices, both are not separated from the support of government. Widely acknowledged, advanced countries like Newly Industrializing Countries (NICs), have globally-reputed R&D infrastructure and funding system to support industrialization system in their countries. Indeed, the presence of R&D infrastructures and funding system are the notable indicators to describe how a nation is classified as advanced economies or less economies.

Republic of Korea (RoK), one of NICs, has become best practices for countries around the world regarding how to create the global-scaled R&D infrastructure and funding system, including how each country can learn more from those practices from South Korea directly. As a consequence, not few developing or less economy countries visit to Korea to learn R&D infrastructure and funding system, or they invite the Korean experts of R&D infrastructure and funding system to learn the best practices of the Korea's innovation. The Global Innovation Index (GII) 2023 shows that the Republic of Korea (RoK) has the 10^{th} largest economies among 132 economies around the world (WIPO, 2023), it is the remarkable achievement that is important to be diffused to other countries that want to be advanced economies equal with NICs.

Though South Korea and Indonesia have the similar economic condition in 1960s by which both countries focus on agriculture system, but it appears distinguished since in the late 1990s when South Korea has become Asian high economies far surpassing the Indonesia's economic growth. It is reasonable that South Korea is one of key partners for Indonesia in developing science-based economy rapidly. Nowadays, multi-cooperation between Indonesia and South Korea are well established and developed year by year.

The 2024 K-Innovation initiative with Indonesia represents the multi-year collaborative efforts between the Republic of Korea and the Republic of Indonesia. Since 2014 when collaboration started with the Indonesian Institute of Sciences (LIPI) and Science Technology Policy Institute (STEPI), this collaboration has produced fruitful results in advancing of science-technology-innovation (STI) policy in Indonesia. Starting at the early 2023, the active collaboration between the National Research and Innovation Agency of Republic of Indonesia (BRIN) and STEPI is coined on the Project Concept Paper (PCP 2023-2025). It is a policy consultation provided by the Korean experts to the experts and stakeholders in the field of STI policy in Indonesia. Surely, this collaboration provides valuable insights and policy suggestions with regard to the STI issues which is facing by the Indonesian Government.

The primary focus of the 2024 K-Innovation collaboration with Indonesia revolves around policy consultation and workshop in terms of the Research and Innovation (R&I) infrastructure and the R&I funding system. The term of Research and Innovation (R&I) is preferably used to this report as BRIN prefers to use the term of R&I to R&D. Yet, both terms have similar meaning and goals namely to achieve the Indonesian national innovation system (NIS). In particular, the K-Innovation 2024 project will place special emphasises on those issues as follows:

I. Analysis of the information system supporting Research & Innovation (R&I) infrastructure policy in Indonesia, with a specific focus on providing data and information system for facilitating R&I infrastructure system.

II. Analysis of the Research & Innovation (R&I) funding system in Indonesia, with a specific focus on R&I funding mechanism that could address financial market failures to invest in R&I as well as collaboration and capacity failure among STI stakeholders.

Both issues are challenges faced by the Government of Indonesia especially in the STI sector. Government of Indonesia should design an information system to facilitate R&I infrastructure services delivered by the Deputy of R&I infrastructure. Again, the R&I funding system is one of prominent issues that requires a special attention of the Government of Indonesia. To achieve those goals, the STI policy consultation project between BRIN-Indonesia and STEPI-South Korea is elaborated into six main activities namely, defining the concept, focused group discussions (FGD), survey, in-depth interviews; and capacity building or workshops in Korea and Indonesia. How to elaborate two main things above is referred to the primary data collection and analysis, secondary data reviews, previous K-Innovation reports of BRIN-STEPI and Korean expertise' analysis. Both will be explained more in the subsequent chapters of this K-Innovation report BRIN-STEPI.

2. Project Objectives

The 2024 K-Innovation Official Development Assistance (ODA) Programme with Indonesia has three aims. First, providing policy suggestions for the Indonesian STI stakeholders in designing an information system which supports the R&I infrastructure services managed BRIN to BRINS's users inside and outside. Second, providing policy suggestions for the Indonesian STI stakeholders in designing a R&I funding system through a competition and collaboration mechanism by making reference to global best practices. Third, enhancing capacity and capability to conduct analyse and research STI policy studies through various schemes like STI policy workshop, STI policy consultation, and STI policy publication(s).

The objectives of the K-Innovation Project are a series of the Project Concept Paper (PCP) 2023-2025 focusing on R&D/R&I infrastructure and funding system in Indonesia. The long-term goal, from 2024 onwards, is to lay out a robust foundation for STI cooperation and collaboration between STEPI-South Korea and BRIN-Indonesia. Even, K-Innovation project is open to external actors of BRIN that have concern and attention to address national STI issues. Widely acknowledged, Republic of Korea has become the leading country of innovation around the world, the Korea's rapid economic growth relied on science and technology since few decades ago is the firm evidence that South Korea is able to transform from lower middle-class country to upper class country.

R&I is currently used in the BRIN's environment as a commitment to bring research results to innovation. Hence, one of BRIN's goals is to create an excellent R&I ecosystem based on globally-reputed researches.

In the context of R&I infrastructure development, creating R&I infrastructure system is a specific part of the STI policy that needs to be supported by governmental regulations and programs. Defining R&I infrastructure policy is concerned to activities which support the performance of R&D laboratories, technology standard and licensing equipped with technical research, technical assistance, quality control, and certification (World Bank, 2010); Yoon, 2018). According to Kim (2023), R&D infrastructure refers to "the facilities, resources, and related services used to conduct top-level research in a scientific field, including major scientific equipment or facilities, or databases for disseminating research data".

Referring to definition of R&I infrastructure above, the reinforcement of R&D infrastructure is concerned with facilities, databases, and research data. It means that the reinforcement of R&I infrastructure is possibly viewed from the information data system perspective inasmuch as the current R&I infrastructure services are inevitable measures from the use of information technology. Starting from the reinforcement and integrating R&I infrastructure data and then connecting them to the users of R&I infrastructures, are the initial phase to create a robust R&I infrastructure system relied on the information system.

In the context of R&I funding development, creating R&I funding system is a specific part of the STI policy that needs to be supported by governmental regulations and programs. Defining R&I funding policy is concerned to activities which support to flourish competitive and collaborative R&I culture. Besides, R&I funding policy is aimed to solve not only the bottleneck of R&D funding problems such as R&I limited funding, high taxes for doing R&D activities, inconsistencies of public R&D equipment procurement, and also market failures in R&I such as financial market failures as well as collaboration and capacity failures. In a nutshell, R&I funding policy is aimed to combat those funding issues. World Bank (2010) & Yoon (2018) provides clues of R&I funding policy in relation to provide R&D subsidies and R&D tax incentives, to simplify regulations in R&D tools procurement process, access R&D grants, and access venture capital. In addition, facilitation and acceleration for commercialization process of R&D results are pivotal governmental strategies in growing competitive and collaborative R&I culture.

For advanced countries, government plays an important role in growing R&I infrastructure and funding policies. Government should have special strategies to attract external investments in the fields of R&D infrastructure and funding. By appointing a special agency in dealing with STI issues, a hybrid model involving private sectors, government agencies, and universities should be well established to implementing both policies. Especially in the context of R&I infrastructure development, it is also possible to facilitate and simplify the

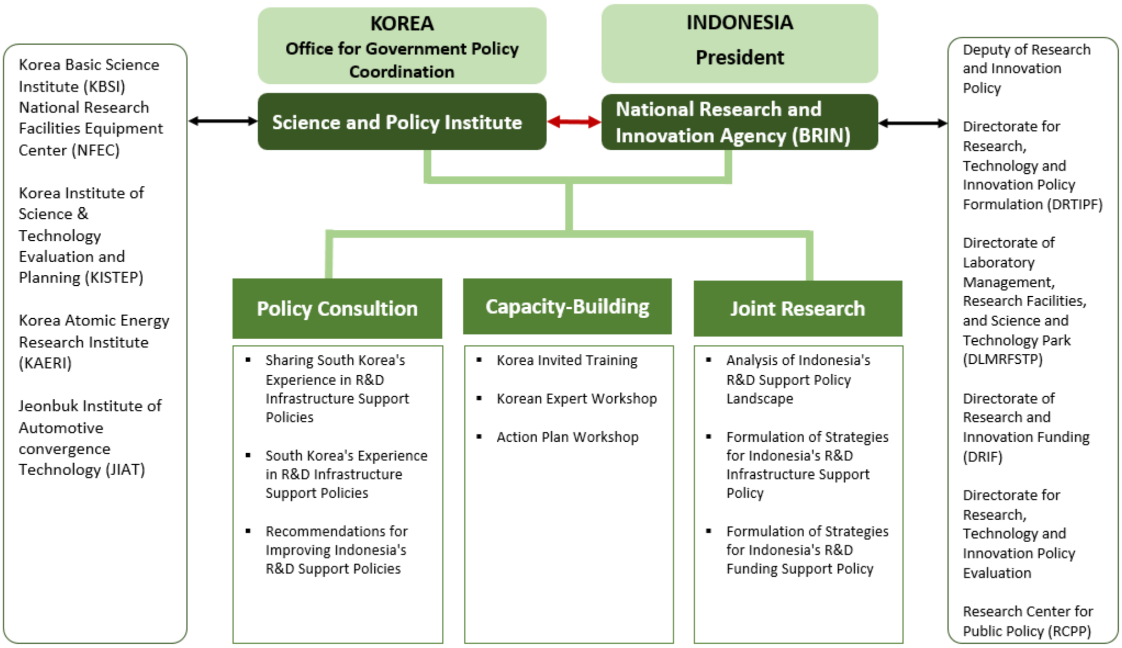
R&I infrastructure services through the support of information system. While the R&I funding development system depends on the national finance and budget system, including its regulations, it is expected to facilitate collaboration and mobilization of STI talents and also to stimulate R&I funding in the private sector by financial market.

3. Project Framework

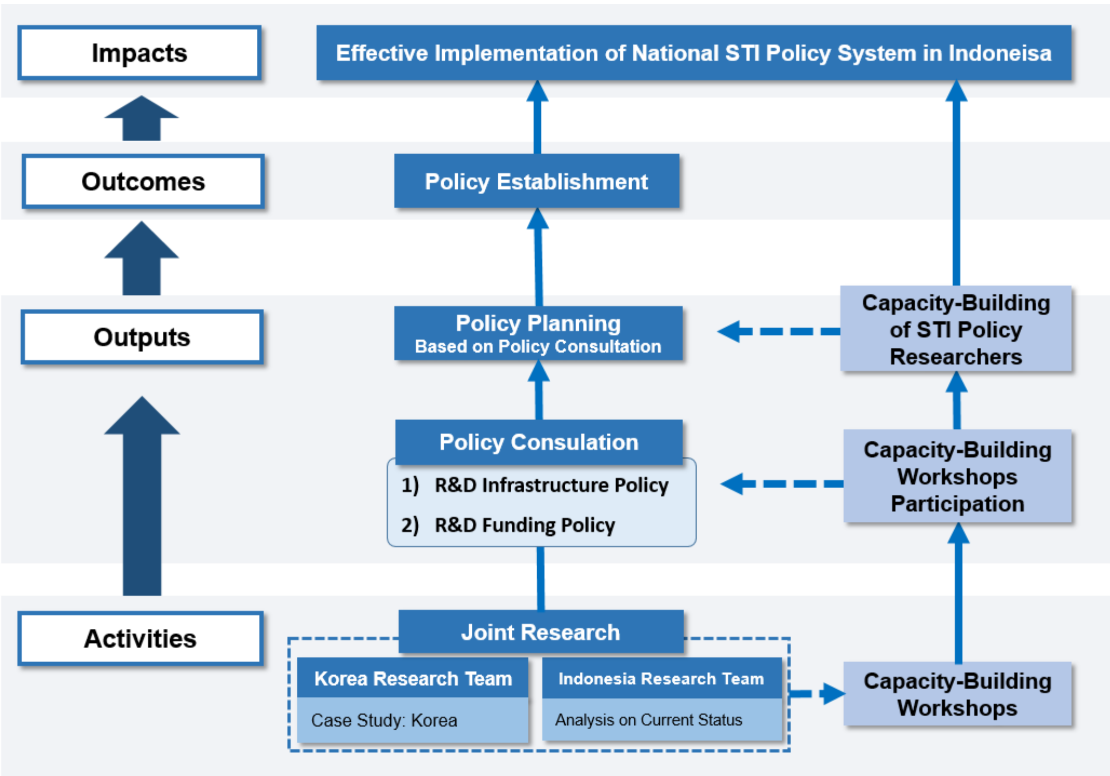
The 2024 K-Innovation Partnership Programme is delivered into five main activities: 1) a kick-off seminar, 2) a policy consultation project, 3) a joint research programme aimed at producing a policy consulting report, and 4) STI workshop for policymakers, policy analysts and researchers, and 5) a dissemination workshop. The meetings and workshops were held offline in two countries, Indonesia (Jakarta City) and South Korea (Seoul City). The first phase consisted in holding a kick-off seminar with the STEPI team and BRIN to discuss the details of the activities and to evaluate the previous year's activities. The kick-off meeting, which was held at BRIN's building in Jakarta on 13 March 2024, consisted of presentations by Korean experts on Korean practices and cases related to the topics of information systems that support the R&I infrastructure and the topics of the R&I funding scheme and management project, with the participation of Indonesian experts, and officials from BRIN, non-governmental organizations, and universities.

The Korean experts such as Korea Basic Science Institute (KBSI) and Korea Institute of Science & Technology Evaluation and Planning (KISTEP), collaborated with BRIN-Indonesia, policymakers, policy analysts and researchers for joint policy consultation and research on two themes: 1) improving public service in the field of research and innovation (R&I) infrastructures through the reinforcement of information system; 2) improving public service in the field of research and innovation (R&I) funding scheme and project management. The BRIN teams, under the supervision of the Korean experts, assigned focal points to various groups composed of Indonesian policymakers, policy analysts, and researchers.

[Figure 1-1] Schematization of the 2024 Project Structure



[Figure 1-2] Detailed Framework of the 2024 Project



4. Project Team

Led by Dr. Eun Joo Kim of STEPI, eighteen policymakers, researchers, and policy analysts from Korea and Indonesia participated in implementing the 2024 K-Innovation Project. The Korean team invited two Korean experts for a policy consultation on designing and improving research and innovation (R&I) infrastructures and funding in Indonesia. Armed with an in-depth understanding of Indonesia's STI environment and situation, and more particularly, its STI governance and policy, the Korean experts were able to facilitate collaboration with BRIN.

In Korean team, Dr. Jiho Hwang, who possesses abundant expertise in R&D funding policy, also participated in supervising the policy consultation and research on R&I funding in Indonesia. Dr. Youngjoo Kim, who possesses a wealth of expertise in R&D infrastructure policy, also participated in supervising the policy consultation and research on R&I infrastructures in Indonesia. Ms. Yoonsil Moon acted as project coordinator of the 2024 K-Innovation Project with the assistance of STEPI researchers, and the Korean-Indonesian joint research group supervised by Dr. Deok Soon Yim. Korean advisers headed the Indonesian policy consultation and research teams, each of which was assigned a focal point to ensure better communication and coordination among the Indonesian researchers.

In Indonesian team, Dr. Boediastoeti Ontowirjo took the role of advisor on 2024 K-Innovation for the Indonesian teams. Dr. Dudi Hidayat took the role of coordinator for 2024 K-Innovation, and also headed the Indonesian team on policy consultation for the research and innovation (R&I) infrastructure and funding facilitation. In terms of the R&I infrastructure policy consultation, the BRIN members of the task force are included Mr. Hendro Subagyo, Mr. Imam Muhajirin, Ms. Yustina Nita Sulistami, Ms. Leni Purwaningsih, and Mr. Anugerah Yuka Asmara. Likewise, the BRIN members of the research team for R&I funding are included Dr. Ajeng Arum Sari, Dr. Siti Annisa Silvia Rosa, Ms. Supadmi, Ms. Rahmayanti Pertiwi, Dr. Juhartono, and Ms. Nungki Indrianti.

[Table 1-1] Project Team

Name	Position/Organization	Role
Dr. Eun Joo Kim	Principal Researcher, STEPI	Project Manager
Ms. Yoonsil Moon	Principal Researcher, STEPI	Project Coordinator
Dr. Deok Soon Yim	Honorable Research Fellow, STEPI	Advisor
Dr. Boediastoeti Ontowirjo	Deputy Chairman, Deputy of Research and Innovation Policy, BRIN	Counterpart and Advisor
Dr. Dudi Hidayat	Director, Directorate for Research, Technology and Innovation Policy Formulation (DRTIPF), BRIN	Counterpart and Team Leader
Team 1. Research and Innovation (R&I) Infrastructure Policy		
Dr. Yong Joo Kim	Lead Manager/Senior Researcher, National Research Facilities and Equipment Centre (NFEC), Korea Basic Science Institute (KBSI)	Team leader
Mr. Hendro Subagyo	Head of Data and Information Center	Indonesian Team
Mr. Imam Muhajirin	Information Technology Analyst, Data and Information Center	Indonesian Team
Ms. Yustina Nita Sulistami	Coordinator, Directorate for Research, Technology and Innovation Policy Evaluation (DRTIPE), BRIN	Indonesian Team
Ms. Leni Purwaningsih	Policy Analyst, Directorate for Research, Technology and Innovation Policy Evaluation (DRTIPE), BRIN	Indonesian Team
Mr. Anugerah Yuka Asmara	Coordinator, Research Centre for Public Policy (RCPP), BRIN	Indonesian Team
Team 2. Research and Innovation (R&I) Funding Policy		
Dr. Jiho Hwang	General Director/Senior Research Fellow, Korea Institute of S&T Evaluation and Planning (KISTEP)	Team Leader
Dr. Ajeng Arum Sari	Director, Directorate for Research and Innovation Funding (DRIF)	Indonesian Team
Dr. Juhartono	Coordinator, Directorate of Research and Innovation Funding (DRIF)	Indonesian team
Ms. Nungki Indrianti	Policy Analyst, Directorate of Research and Innovation Funding (DRIF)	Indonesian Team
Dr. Siti Annisa Silvia Rosa	Policy Analyst, Directorate for Research, Technology and Innovation Policy Formulation (DRTIPF), BRIN	Indonesian Team
Ms. Supadmi	Policy Analyst, Directorate for Research, Technology and Innovation Policy Formulation (DRTIPF), BRIN	Indonesian Team
Ms. Rahmayanti Pertiwi	Policy Analyst, Directorate for Research, Technology and Innovation Policy Formulation (DRTIPF), BRIN	Indonesian Team

5. Project Activities

5.1 Kick-off Seminar

- Date/Venue: 13 - 14 March 2024 /BRIN, Jakarta, Indonesia
- Purpose: To hold an inaugural seminar to promote mutual understanding of the project's theme and annual planning agreement between the two organizations
- Participants: Over 20 people, including STEPI' project team, commissioned researchers, BRIN researchers, and other Indonesian stakeholders
- Main Contents:
  - Presentation of the purpose and detailed plan of the 2024 policy advisory
  - Status of and support system for R&D funding and infrastructure support policies in Indonesia and Korea
- Key Points:
  - Necessity of Plan to discuss ways to promote participation in Indonesia's R&D industry like the government's direction of support for large corporations
  - Correlation between government promises, political conditions, and the allocation and distribution of R&D budgets
  - Methodology of Korea's future forecasting strategy and system for setting science and technology policies and R&D missions based on that strategy
  - Utilization System of R&D Infrastructure, methods of Information gathering, operational personnel, operational systems and maintenance

- BRIN-STEPI Research Team Minutes of Meeting

1) Change of PCP contents

- Revised the PCP will be submitted to STEPI with an official letter from BRIN

2) Annual seminar and workshop in R&I funding management of BRIN and Korea

- Regular online seminars on ‘R&D funding management will be held 3 times a year in 2024 and 2025

3) Visiting Researchers to Indonesia

- Not in the project scope in 2024. But, in 2025 still in STEPI consideration

4) (Output 3) Development of Guidelines of National R&I infrastructure platform and funding schemes that serve R&I research institutions, universities, and industries

- Development of Guideline for Infrastructure Platform that can manage, control, optimization of infrastructure: Draft Guideline in 2024, Final version of Guideline in 2025
- Development of a Guideline for a Research Funding Scheme that includes funding from Industry and matches the needs and characteristics of R&I (based on Korean policy cases): Draft Guideline in 2024, Final version of Guideline in 2025

5) Request for Knowledge sharing on methodologies and indicators to measure innovation and national R&D achievement

- Could be implemented in seminars

6) Adoption of ZEUS system for R&D infrastructure management from Korea to Indonesia

- If there is a need to adopt and purchase Zone for Equipment Utilization Service (ZEUS) system, adoption and purchasing process should be discussed between government bodies of BRIN in Indonesia and Ministry of Science and ICT (MSIT) in Korea.

[Table 1-2] Kick-off Seminar Programme

Time	Programmes	Presenters
10:00-11:00	Current Status & Achievements of STI Policy in Indonesia	**Dr. *Boediastoeti Ontowirjo*** Deputy Chairman of Research and Innovation Policy - National Research and Innovation Agency
	Current Status and Achievements of STI Policy in Korea	**Dr. *Deok Soon Yim*** Consultant for International Cooperation
11:00-11:30	Question and Answer (Q&A) and Group Discussion	
11:30-13:00	Break & Lunch	
13:00-14:00	Current Status of R&D Funding Policy in Indonesia: Exploring Industry Engagement in Research and Innovation Funding	**Dr. *Ajeng Arum Sari, S.T.*** Director /Directorate of R&I Funding - National Research and Innovation Agency
	Current Status of R&D Funding policy in Korea	**Dr. *Jiho Hwang*** Director General / Senior Research Fellow Office of Strategic Technology Planning Korea Institute of S&T Evaluation and Planning
	Q&A and Group Discussion	
14:30-15:30	Current Status of R&D Infrastructure Policy in Indonesia: Infrastructure Services Through IT Systems (ELSA)	**Mr. *Hendro Subagyo, M.Eng*.** Head / Center for Data and Information
	Current Status of R&D Infrastructure policy in Korea	**Dr. *Yongjoo Kim*** Director / Office of Policy Strategy and Planning National Research Facilities and Equipment Center
15:30-16:00	Q&A and Group Discussion	
16:00-16:30	Closing Remark and Photo Session	

[Figure 1-3] Photos of Kick-off Seminar

5.2 Capacity-Building Workshop: Invitational Programme

- Date/Venue: 7 May to 11 May 2024 / Korea
- Purpose: The Capacity Building Workshop on Improvement of Research Funding and Infrastructure in Indonesia has two objectives: The first is to support policymakers, researchers, and policy analysts in identifying the research and innovation funding issues faced by BRIN, while the second is to support Indonesian policymakers, researchers, and policy analysts in designing and improving the research and innovation infrastructures delivered by BRIN by learning from South Korea’s best practices. These objectives are to be carried out in order to promote mutual economic cooperation between Indonesia and South Korea by contributing to the development of Indonesia’s science, technology and innovation (STI) capacity and improvement of the related systems and conditions.
- Participants:

[Table 1-3] List of participants in the invitational programme in Korea

	Name	Position
1	Dr. Boediastoeti Ontowirjo	Deputy Chairman for Research and Innovation Policy
2	Dr. Dudi Hidayat	Director for Policy Formulation for Research, Technology, and Innovation
3	Dr. Ajeng Arum Sari	Director for Research and Innovation Funding
4	Ms. Nungki Indrianti	Policy Analyst of Research and Innovation Funding
5	Mr. Hendro Subagyo	Head of Center for Data and Information Center
6	Mr. Imam Muhajirin	Coordinator of Data and Information Center
7	Dr. Siti Annisa Silvia Rosa	Policy Analyst of Policy Formulation for Research, Technology, and Innovation
8	Mr. Anugerah Yuka Asmara	Coordinator/Researcher of Research Centre for Public Policy

- Main Contents:
  - Formulation of action strategies for improving Indonesia's R&D Funding and infrastructure support policies
  - Learning about how to activate R&D in the Korean industry and the mechanism of R&D funding. and the case of Indonesia
  - Sharing of South Korea's policy experiences through visits to major R&D infrastructure support institutions in Korea
  - Establishment of a network of South Korean and Indonesian experts in science and technology innovation

[Table 1-4] Programme module

Module	Type	Details
R&D Support Policy and Systems	Lecture	Industrial R&D Investment Promotion Policy of Korea
	Lecture	Mechanism of R&D Funding of Korea
	Lecture	Equipment Utilization Platform about ZEUS of Korea
Korea's Development Experience	Study Visit	Hansung University
	Study Visit	Korea Institute of Science & Technology (KIST) History Museum
	Study Visit	Research Investment for Global Health Technology Foundation (RIGHT) Foundation
	Study Visit	Hyundai Motor Studios
Capacity Building	Workshop 1.	Policy Plan to Promote Indonesian Industrial R&D
	Workshop 2.	Planning Guidelines for Improving R&D Funding Policy of Indonesia
	Workshop 3.	Planning Guidelines for Improving R&D Infra System of Indonesia

[Figure 1-4] Photos of the Invitational Programme in Korea

- Performance outcomes: Survey of Satisfaction and Self-efficacy

  ① Survey Content: Relevance and satisfaction with the training (site visits, workshops, seminars) in relation to the training objectives, and personal assessment of capacity enhancement.

  ② Survey Results: All the respondents, based on the total response criteria, provided positive responses regarding their satisfaction with the training and the personal efficacy surveys.

  – Overall Results of Open-ended Surveys:

  ① Survey Content: Overall assessment of the invitational training and suggestions for improvement.

  ② Survey Results: The participants increased their understanding of Indonesia's situation through South Korea's experience. They would like more opportunities for discussions and problem-solving, and more insights into Korea’s STI policy, including the activation of R&D investment in the industry.

[Table 1-5] Results of the personal efficacy survey

Category	No.	Question	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Satisfaction Survey	1	Overall satisfaction	100%	-	-	-	-
	2	[Site Visit] Were the contents of the workshop program appropriate to the project objective?	83%	17%	-	-	-
	3-1	[Workshop 1] Were the contents of the workshops appropriate to the project objective?	67%	-	33%-	-	-
	3-2	[Workshop 2] Were the contents of the workshops appropriate to the project objective?	50%	33%	17%	-	-
	3-3	[Workshop 3] Were the contents of the workshops appropriate to the project objective?	66%	17%	17%		
	4	[Seminar] Were the contents of the workshops appropriate to the project objective?	66%	17%	17%	-	-
Personal efficacy survey	5-1	I am more fully aware of the characteristics (*Policies, Governance, Stakeholders) of the National R&D support (Funding & Infrastructure) system than I was before the workshop.	83%	17%	-	-	-
	5-2	I am more fully aware of the current status of the National R&D support (Funding & Infrastructure) system in Indonesia than I was before the workshop.	50%	33%	17%	-	-
	5-3	I have a better understanding of the policy and procedures necessary for developing the National R&D support (Funding & Infrastructure) system compared to before the workshop.	66%	17%	17%	-	-

5.3 Joint Research Workshop

- Date: 23-25 September 2024 / Jakarta, Indonesia (BRIN)
- Purpose: To share the results of the analysis of the 2024 policy research topics and discuss next year's policy advisory direction.
- Participants: All members of the Korea-Indonesia International Collaborative Research Team.
- Key Highlights:
  - The STEPI Research Team has identified and selected experts with appropriate educational background, experience, and expertise to serve as mentors for the workshop activities. These mentors will provide guidance and input for program development and evaluation, as well as mentorship and consultation for collaborative research activities and joint studies with Indonesia
  - The objectives of the 2024 Joint Research Workshop are to strengthen the overall research ecosystem in Indonesia through

    a) Increasing understanding of the importance of effective research information and funding systems in managing and utilizing research infrastructure

    b) Facilitating the exchange of knowledge and experience between South Korea and Indonesia in the development and implementation of research infrastructure and funding information systems

    c) Identifying challenges and opportunities in building a strong research infrastructure and funding information system in Indonesia

    d) Developing strategies and solutions to improve the accessibility, efficiency, and effectiveness of research infrastructure and funding through information systems

    e) Increasing the capacity of stakeholders to manage and evaluate research infrastructure and funding information system schemes

- The targets of the 2024 Joint Research Workshop are;
  a) The introduction of various innovative and effective management and administration schemes for research infrastructure information systems and research funding from South Korea
  b) Obtaining the results of identifying challenges and opportunities in developing sustainable and inclusive research infrastructure information systems management and research funding schemes in Indonesia
  c) Creating a platform for collaboration and knowledge exchange that will encourage strengthening the research ecosystem in Indonesia
  d) Creating dialogue and collaboration between relevant stakeholders to design a management scheme for research infrastructure information systems and funding that is in accordance with Indonesia's research needs and priorities
  e) Increasing the capacity of stakeholders to manage and evaluate research infrastructure information systems and research funding schemes in Indonesia

  * The discussion was divided into two teams: funding team, infrastructure team

- The funding team's main discussion include;
  a) R&D Funding-related laws and policies and BRIN's Key missions based on them
  b) Current Status and Operating System of R&D Funding
  c) R&D Funding issues/tasks and improvement directions: BRIN's R&D Funding system, BRIN's R&D project management system and industry' plan to enhance national R&D participation
- The Infrastructure team's main discussion is
  a) Functional Analysis of the Indonesia Research Infra System & Comparative Analysis of Functional Faith in Indonesia's Research Infra Policy Objectives
  b) Analysis of system functions comparison target system of ZEUS and BRIN ELSA and derivation of improvement goals

c) Explanation of Information System Master Plan (IMSP) establishment Procedure and detailed establishment plan for system improvement

• BRIN-STEPI Research Team Minutes of Meeting

1) Implementation of PCP

– This forum decided one of the outputs in this project is writing an academic paper (advised by Korean experts)

2) (Outputs 1): Policy demand survey and current status analysis of STI infrastructures policy in the National Research and Innovation Agency (BRIN) (report and policy recommendation)

– In August 2024, Report on the current status has been done and sent to STEPI. For the policy demand survey will be conducted in 2024.Visiting Researchers to Indonesia.

3) Policy Demand Survey on Report 2024

– BRIN should decide what kind of policy demand should be surveyed. Policy demand survey will be conducted on infrastructure, focus on industry and universities.

4) Dissemination forum on 2024

– The dissemination forum has been done by conducting local workshop and policy school in September 2024.

5) 2025 Project Overview

– Purpose: To support the design of Indonesia’s R&D policy with a focus on R&D infrastructure and funding to enhance the quality of the research and innovation ecosystem.

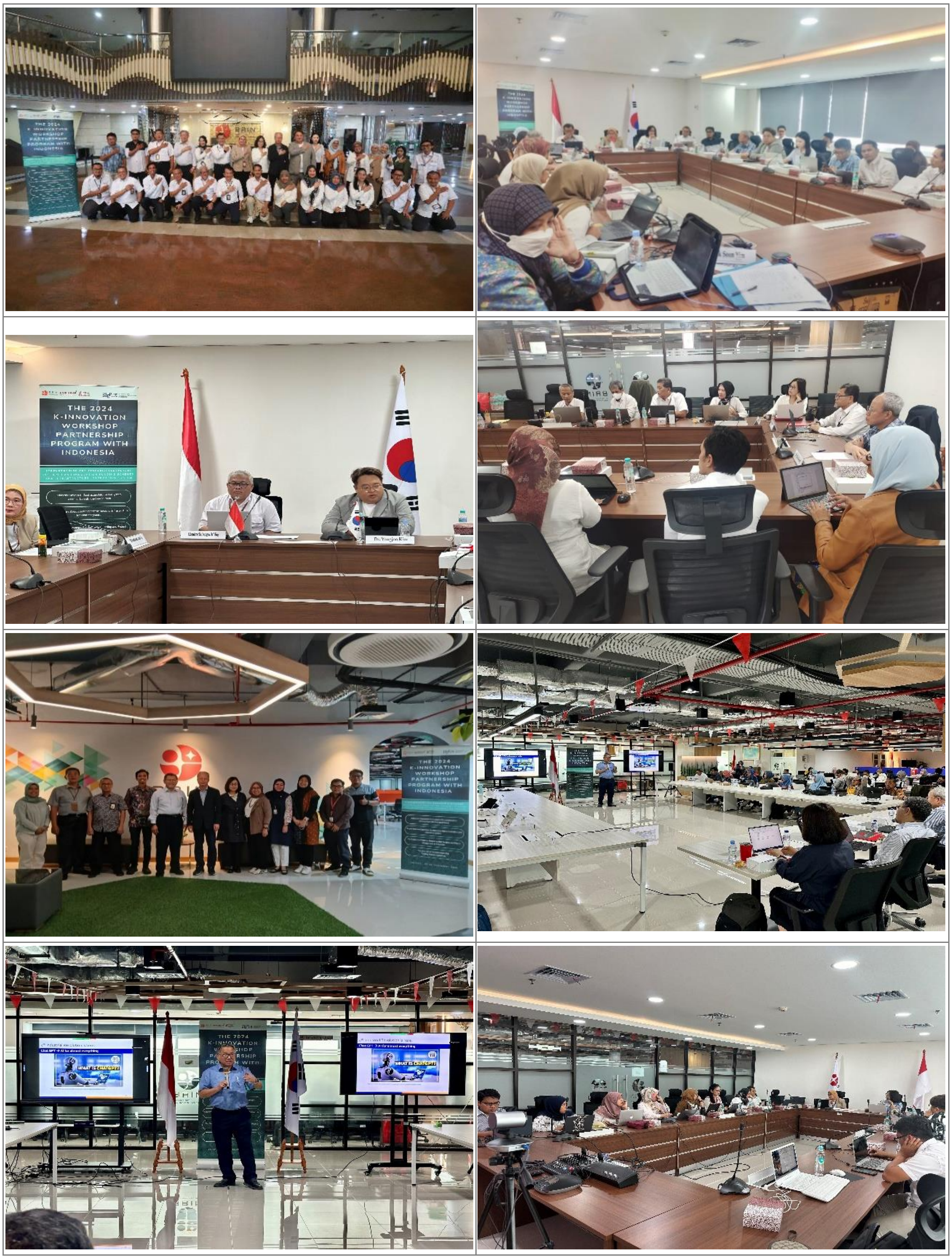
– (Kick-Off Seminar and Field Research) Determination of the 3rd year project and an Indonesian stakeholder workshop to implement the Guideline for Funding and Infrastructure. The guidelines for R&I funding and Infrastructure should be finalized by June 2025

– For the seminars and workshops, will be held two times in 2025 (concept note of the seminar and workshop should be sent by BRIN)

[Table 1-6] Joint Research Workshop Programme

<table>
<tr><th>Date</th><th>Programmes</th><th>Presenters</th></tr>
<tr><td rowspan="3">23 September
Meeting room 11</td><td>Indonesia Research Infra Information system: Comparison with Korea Case</td><td>Dr. Yongjoo Kim
Director / Office of Policy Strategy and Planning
National Research Facilities and Equipment Center</td></tr>
<tr><td>Future Vision and Objectives for Indonesia Research Infra Information system</td><td>Mr. Hendro Subagyo, M.Eng.
Head / Center for Data and Information</td></tr>
<tr><td colspan="2">Question and Answer (Q&A) and Group Discussion</td></tr>
<tr><td rowspan="3">23 September
Meeting room 12</td><td>The Status and problems related to R&I Funding and Project Management in Indonesia</td><td>Dr. Ajeng Arum Sari, S.T.
Director / Directorate of R&I Funding
- National Research and Innovation Agency</td></tr>
<tr><td>Direction of Indonesian R&I Funding Scheme Guidelines</td><td>Dr. Jiho Hwang
Director General / Senior Research Fellow
Office of Strategic Technology Planning
Korea Institute of S&T Evaluation and Planning</td></tr>
<tr><td colspan="2">Q&A and Group Discussion</td></tr>
<tr><td>24 September
Meeting room 11</td><td>STEPI-BRIN Seminar : Policy School</td><td>Dr. Deok Soon Yim
Consultant for International Cooperation</td></tr>
<tr><td>24 September
Meeting room 5</td><td>Expert Meeting: Introduction of Korea Programs and Discussion of Plans</td><td>Dr. Deok Soon Yim
Consultant for International Cooperation</td></tr>
<tr><td>25 September
Meeting room 11</td><td colspan="2">BRIN-STEPI Research Team Minutes of Meeting</td></tr>
</table>

[Figure 1-5] Photos of the Joint Research Workshop in Indonesia



6. Project Schedule

[Table 1-7] Project Schedule

Activities	2024												2025
	1	2	3	4	5	6	7	8	9	10	11	12	1
Project Planning	■	■	■										
Organizing the Project Team		■	■										
Kick-off Seminar				■									
International Joint Research				■	■	■	■	■	■	■			
Training Workshop							■						
Submission of Interim Report							■						
Joint Research Workshop									■				
Submission of Final Report											■		
Presentation of Final Report												■	
Evaluation of Final Report												■	
Publication													■

CHAPTER 2

Information System for Supporting the Research & Innovation Infrastructure Facilitation in Indonesia

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Chapter 2. Information System for Supporting the Research & Innovation Infrastructure Facilitation in Indonesia

1. Background

1.1 Background

Research infrastructure plays a critical role in driving scientific advancement, innovation, and technological progress. Effective management of research infrastructure ensures that resources are allocated efficiently, supporting high-impact research and enabling collaboration across institutions. This literature review explores key concepts and best practices from existing literature on research infrastructure management, innovation ecosystems, and technology transfer. It contextualizes the findings of the BRIN's infrastructure research survey and the comparative insights from Korea's Research Infrastructure indicators.

■ **The Role of Research Infrastructure in Innovation Ecosystem**

Research infrastructure is a foundational component of national innovation systems. According to Pillay et al. (2015), research infrastructure encompasses facilities, equipment, and services required for conducting high-quality research, including data centres, laboratories, and technology platforms. Well-maintained and accessible infrastructure is crucial for innovation, as it enables researchers to conduct experiments, analyse data, and develop new technologies. In turn, this drives economic growth and societal development through knowledge transfer and commercialization

Several studies highlight the importance of aligning infrastructure investments with national research priorities. For instance, Katz & Martin (2007) argue that prioritizing high-impact research fields and emerging technologies enhances the return on infrastructure investments. This approach has been adopted in several countries, including South Korea, where national research infrastructure policies are closely tied to strategic industries such as biotechnology and information technology.

■ Effective Management and Utilization of Research Infrastructure

The management of research infrastructure involves not only procurement and maintenance but also ensuring that facilities are utilized optimally. According to Georghiou et al. (2010), an integrated approach to infrastructure management requires robust monitoring systems, preventive maintenance, and transparent access policies. Such systems are designed to ensure that resources are available to a wide range of users, including both academia and industry, thereby fostering collaboration.

In the context of BRIN's E-Layanan Sains (ELSA) system, the importance of user feedback is emphasized in the literature on infrastructure management. Studies by Brousseau & Penard (2007) suggest that user-centric design of management systems can significantly improve the accessibility and efficiency of research infrastructure. By gathering data on user experiences, infrastructure managers can identify pain points, streamline access processes, and enhance overall satisfaction. This is consistent with the findings of the BRIN survey, which indicates areas where the ELSA system could be improved, such as reducing waiting times and simplifying access procedures.

■ Infrastructure Monitoring and Performance Metrics

Monitoring the performance of research infrastructure is crucial for ensuring its effective utilization. Several frameworks for measuring infrastructure performance have been proposed in the literature. For example, the OECD (2017) identifies key performance indicators (KPIs) for research infrastructure, including utilization rates, external partnerships, and the scientific output generated by facilities. Such indicators provide valuable data for decision-making, helping to identify underutilized resources and opportunities for collaboration.

Korea's approach to infrastructure performance measurement provides a comprehensive model for other countries. The country tracks detailed data on equipment utilization, budget allocation, and research outputs, enabling infrastructure managers to make informed decisions about resource allocation and policy adjustments. This model could be adapted to BRIN's context to improve the monitoring and management of research infrastructure.

■ Accessibility and Collaboration in Research Infrastructure

Ensuring that research infrastructure is accessible to a diverse range of users is a key challenge. According to Cressey (2011), equitable access to research infrastructure promotes inclusivity and collaboration, enabling researchers from different institutions and regions to benefit from national investments. Barriers to access, such as complex administrative processes or insufficient information, can hinder research productivity and innovation.

The survey conducted by BRIN highlights several accessibility challenges faced by researchers, including long waiting times and lack of information on equipment availability. To address these challenges, the literature suggests adopting more user-friendly systems and reducing administrative bottlenecks. For example, Ponds et al. (2007) propose the use of digital platforms that provide real-time information on infrastructure availability, making it easier for researchers to plan and conduct experiments. The implementation of such features in the ELSA system would likely improve user satisfaction and infrastructure utilization

Indonesia's research infrastructure faces challenges in access, efficiency, and management. To address these issues, BRIN has developed guidelines and implemented the ELSA system for infrastructure management. The initiative is aimed at optimizing research support, increasing accessibility for researchers, and improving the overall operational efficiency of research facilities. A survey was conducted among researchers in universities and industries to gather feedback on infrastructure usage and to identify areas for enhancement.

1.2 Objectives

The primary objectives of the infrastructure research and innovation (R&I) project are:

A. To improve the procurement, utilization, and maintenance of research infrastructure through systematic policy guidelines.

B. To ensure that research infrastructure is easily accessible to a broader range of researchers from both academia and industry, thereby fostering innovation and collaboration.

C. To align Indonesia's research infrastructure with international standards, particularly by learning from models such as Korea's R&I system.

1.3 Aims

A. Enhance the efficiency of BRIN's research infrastructure by streamlining processes, including equipment procurement, maintenance, and monitoring.

B. Increase transparency and accessibility for researchers through the ELSA system, improving user satisfaction and infrastructure utilization.

C. Foster collaboration across various research sectors by enabling joint infrastructure use and supporting national strategic research initiatives.

D. Establish robust monitoring and reporting mechanisms to ensure the sustainability of infrastructure and maintain high levels of service.

1.4 Progress

A. Infrastructure Management and Policy

The policy guidelines have been laid out, with lifecycle stages covering procurement, utilization, maintenance, and disposal. The BRIN infrastructure management system, especially the ELSA platform, has undergone improvements, including the standardization of procedures for infrastructure access.

B. Survey Findings to identify the barriers to Infrastructure Access and Utilization

The BRIN survey points to three primary barriers that impede the effective use of research infrastructure: long waiting times, complex procedures, and lack of information. Each of these challenges is widely acknowledged in the literature as a hindrance to productivity and research progress.

1) Long Waiting Times

Waiting periods for access to research infrastructure can significantly delay scientific progress, especially when high-demand facilities are involved. According to Schmidt and Teräs (2013), the allocation of infrastructure resources is often a bottleneck in national research systems, particularly when demand exceeds capacity. Efficient scheduling systems, clear prioritization criteria, and real-time monitoring of resource availability are recommended strategies to reduce waiting times.

2) Complex Access Procedures

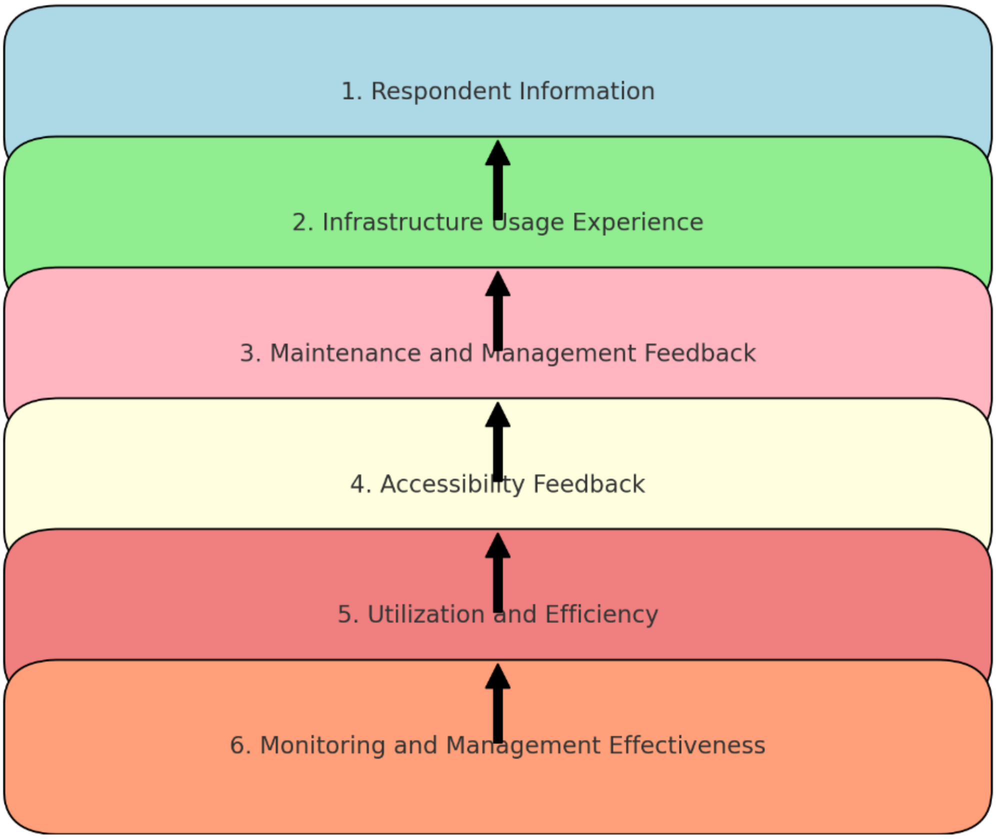
The administrative burden associated with gaining access to infrastructure is a major issue in many research environments. Katz and Martin (2007) argue that overly bureaucratic processes can discourage potential users, particularly those from industry or smaller institutions, from accessing national facilities. Simplifying access procedures and offering more transparent guidelines can help alleviate these problems, as highlighted by feedback in the BRIN survey.

3) Lack of Information

Insufficient information regarding the availability, capabilities, and scheduling of research infrastructure can make it difficult for researchers to plan and conduct their work efficiently. Cressey (2011) emphasizes the importance of providing real-time, user-friendly information systems that enable researchers to make informed decisions about infrastructure use. In the BRIN's case, improving the ELSA system to offer more detailed and accessible data on equipment availability and performance could address this barrier.

The survey reveals several challenges in accessing and utilizing research infrastructure. Issues such as long waiting times, complex procedures, and lack of information were identified as key barriers. Nevertheless, feedback from the survey provides valuable insights for future enhancements.

[Figure 2-1] Survey Structure



Source: Author’s own elaboration (2024)

Here is a flow diagram based on the survey structure, outlining the key sections from the document. It represents the flow of questions that cover:

A. Respondent information

The first section of the survey collects personal and organizational data, which is essential for analyzing how different groups (e.g., academic researchers vs. industry users) interact with research infrastructure. Literature on infrastructure usage often differentiates between the needs and behaviours of various user groups. According to Brousseau and Penard (2007), users feedback must be disaggregated to tailor infrastructure management strategies that meet the diverse needs of these groups. This foundational information helps identify trends in access and utilization across different sectors.

B. Infrastructure usage experience assessing user experience with BRIN's infrastructure

This section evaluates user experiences with BRIN's infrastructure, shedding light on operational inefficiencies. The literature highlights that negative user experiences can drastically reduce infrastructure utilization. Ponds et al. (2007) argue that user satisfaction is a key indicator of how well infrastructure systems are functioning. Survey findings show that many users find access processes cumbersome, supporting the need for simplified, user-friendly interfaces.

C. Maintenance and management feedback

Users feedback on the maintenance of research infrastructure is critical for improving system reliability. Studies by Georghiou et al. (2010) indicate that poor maintenance practices can lead to significant downtime and reduce the overall availability of infrastructure. Survey respondents noted dissatisfaction with current maintenance procedures, pointing to a need for more preventive maintenance and faster response times to repairs. This aligns with best practices in the literature, which advocate for regular, preventive maintenance schedules to maximize infrastructure uptime.

D. Accessibility feedback

Accessibility is another key challenge highlighted in the survey. Schmidt and Teräs (2013) emphasize that research infrastructure must be accessible not only to researchers in large, well-funded institutions but also to smaller universities and industry partners. The BRIN survey reveals that access is hindered by complex procedures, further supporting the literature's call for streamlining administrative processes and enhancing transparency in infrastructure allocation.

E. Utilization and efficiency

The effectiveness of infrastructure use is a central focus in the literature on research systems. According to Katz and Martin (2007), infrastructure that is not fully utilized represents a missed opportunity for innovation. The survey findings suggest that many researchers believe the BRIN’s infrastructure is underutilized, often due to inefficiencies in scheduling and access. Addressing these concerns would help maximize the return on investment in research infrastructure by ensuring that more projects can be supported with existing resources.

F. Monitoring and management effectiveness

Finally, the survey assesses how well the BRIN's monitoring systems track the use and condition of research infrastructure. Effective monitoring is essential for both ensuring optimal utilization and planning future investments. OECD (2017) advocates for real-time monitoring systems that provide actionable insights into infrastructure performance, helping to reduce downtime and inform maintenance schedules. The BRIN survey indicates a need for more robust monitoring systems that offer greater transparency and real-time data, which would help researchers to plan their use of facilities more effectively.

G. Comparative analysis with the South Korea's practice

Indonesia's infrastructure indicators are less comprehensive compared to the Korea's R&D system, which includes detailed metrics on equipment utilization, external collaborations, and output tracking.

1.5 Establishment of the Next-Generation ELSA System

The ELSA System aims to revolutionize the management of research infrastructure by delivering a comprehensive and integrated platform. The upcoming iteration of the ELSA System focuses on enhancing efficiency, transparency, and accessibility for researchers, ensuring seamless operation and utilization of scientific assets. The following key components will define the Next Generation ELSA System:

A. Procurement Management System

This module is designed to streamline the procurement of research equipment and materials, providing a robust framework for managing the end-to-end process. Key features include:

1) An integrated request platform where users can submit purchase requests with detailed specifications.
2) Real-time tracking of procurement stages, from approvals and vendor selection to delivery.
3) Comprehensive documentation of the procurement lifecycle to enhance transparency and accountability.
4) Data-driven analytics tools to support decision-making, optimize budgets, and forecast future procurement needs.

B. Maintenance Portal

The Maintenance Portal will serve as a centralized hub for managing the upkeep of research infrastructure, ensuring that equipment remains functional and reliable.

Core functionalities include:

1) Scheduling preventive maintenance to minimize downtime and extend the lifespan of assets.
2) Logging and tracking maintenance activities, including repair histories and service reports.
3) Providing real-time updates on the maintenance status to relevant stakeholders.
4) Automated reminders and notifications for upcoming service schedules or equipment checks.

C. Total Reservation System

This system will revolutionize resource allocation by providing researchers with a seamless and efficient way to reserve facilities and equipment. Its capabilities include :

1) A user-friendly interface for browsing available assets and scheduling their use.
2) Conflict management features to prevent double bookings and optimize resource allocation.
3) Integration with institutional calendars to align reservations with project timelines.
4) Automated confirmation and notification systems to ensure clear communication between users and administrators.

D. Total Monitoring System

The Total Monitoring System will enable continuous oversight of all critical aspects of research infrastructure, ensuring optimal utilization and performance. Its advanced features include:

1) Real-time monitoring of equipment usage and performance metrics.
2) Integration with Internet of Things (IoT) sensors to track environmental conditions, such as temperature and humidity, critical for certain assets.

3) Analytical dashboards providing insights into infrastructure utilization, identifying inefficiencies, and areas for improvement.
4) Alerts for any deviations or issues detected, enabling swift action to minimize disruptions.

By integrating these advanced modules, the Next Generation ELSA System will provide a cohesive ecosystem for managing research infrastructure, enhancing operational excellence, and supporting cutting-edge scientific endeavors.

[Figure 2-2] Next ELSA System

NEXT ELSA SYSTEM

	Next ELSA
Procurement management system	Detailed Laboratory Data (Accreditation, Images, Equipment, Equipment Manager) to provide data support for decision-making on equipment procurement planning and management.
Maintenance portal	Detailed Laboratory Data (Accreditation, Images, Equipment, Equipment Manager) to provide data support for decision-making on equipment maintenance planning.
Total reservation system	Elsa Transaction Dashboard containing the total reservations for the use of research and innovation equipment and laboratories.
Total monitoring system	Monitoring System Dashboard by Optimizing the Use of the Research and Innovation Facilities and Infrastructure Dashboard.

Source: BRIN (2023), processed and conceptualized by the author (2024)

Laboratory Detailed Data

This feature supports the development of ELSA related to Procurement and Maintenance. It provides data support for decision-making regarding equipment procurement and maintenance planning and management. However, the lab team must input laboratory equipment into ELSA, map the equipment to laboratories, and link them to services. Based on the features and data obtained above, the administrator can carry out the procurement and maintenance processes for laboratory equipment. This ensures efficient management by utilizing detailed data such as accreditation, equipment status, and equipment management assignments, which support informed decision-making in both procurement and maintenance activities.

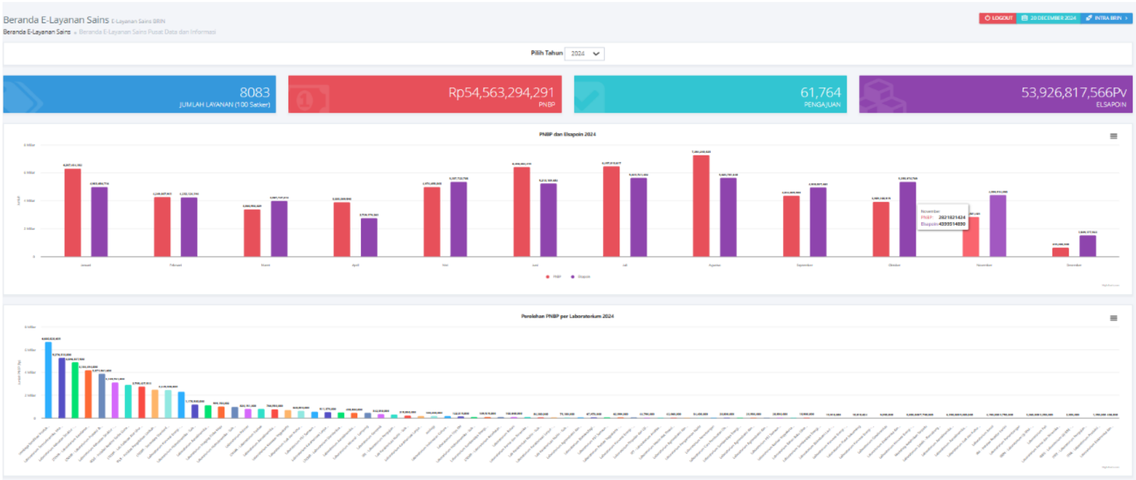
ELSA Transaction Dashboard

This feature supports the development of Elsa related to the **Total Reservation System**.

[Figure 2-3] Features of ELSA System (1)

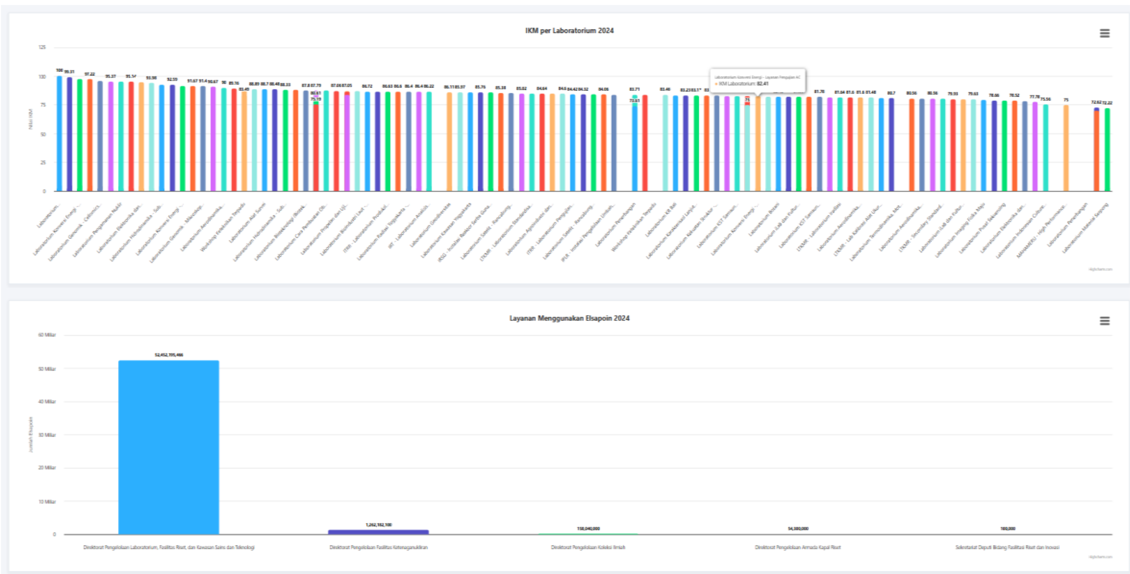
Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-4] Features of ELSA System (2)



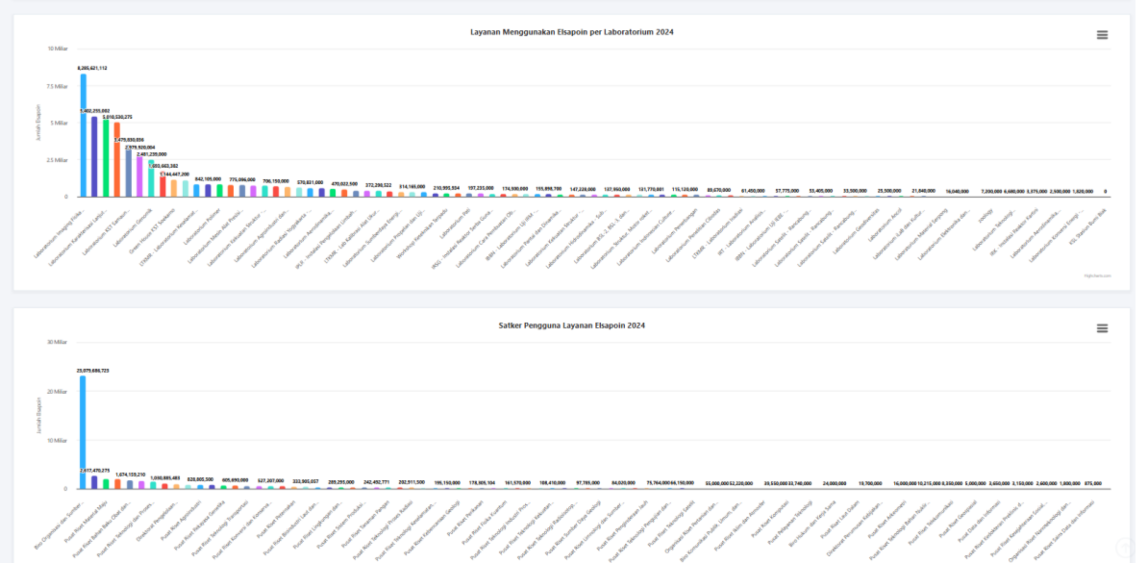
Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-5] Features of ELSA System (3)



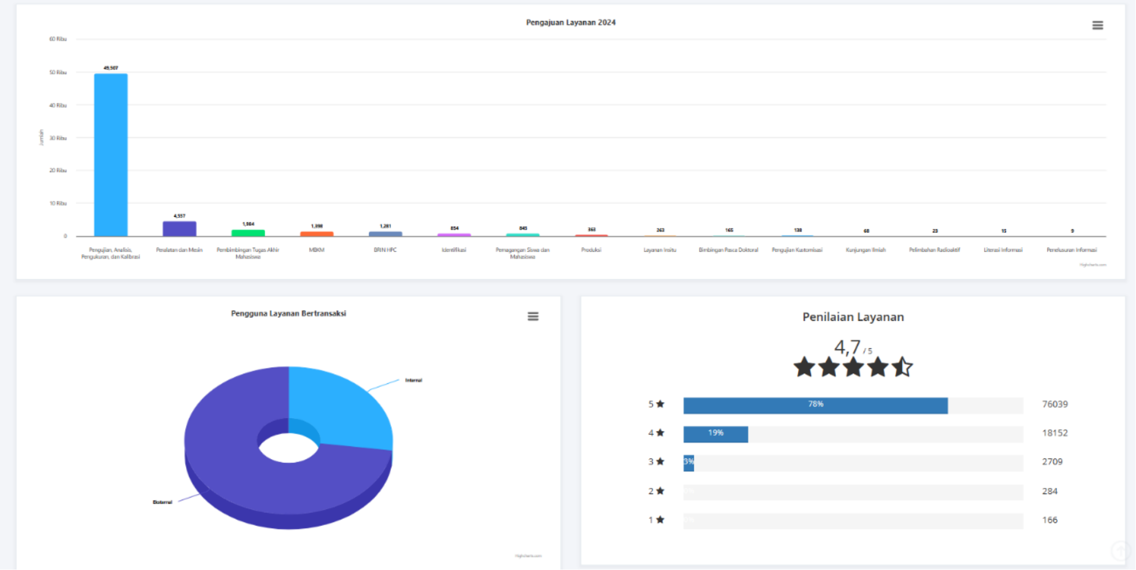
Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-6] Features of ELSA System (4)



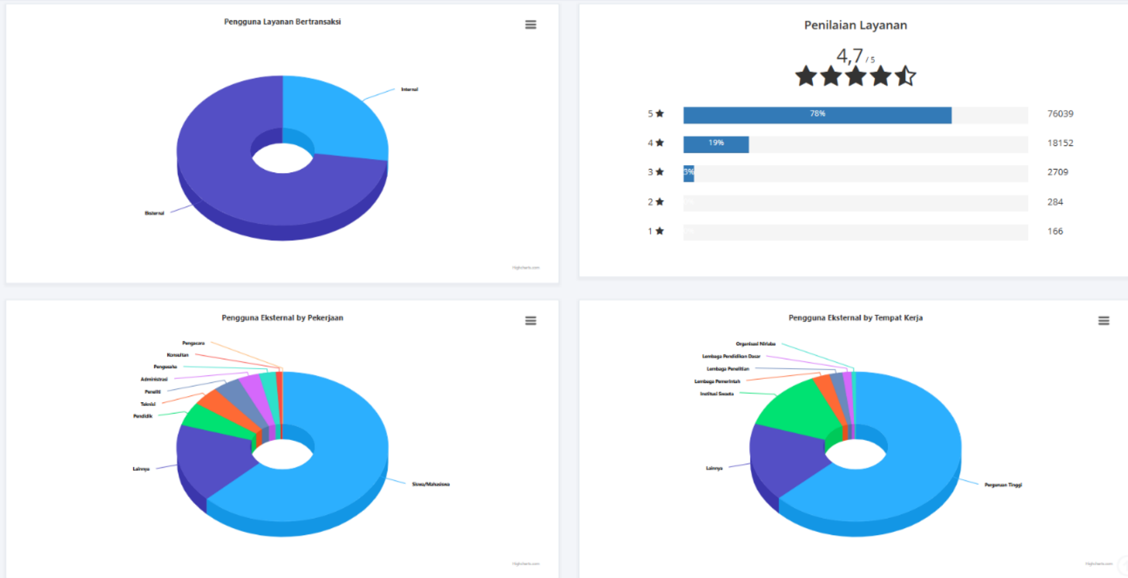
Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-7] Features of ELSA System (5)



Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-8] Features of ELSA System (6)



Source: BRIN (2023), processed and conceptualized by the author (2024)

Total Monitoring System (TMS)

Collaboration with the Data/Business Intelligence Team: Embedding dashboards in ELSA that are prepared by the Data/Business Intelligence Team related to Elsa's data.

Design of Dashboard Total Monitoring System (TMS)

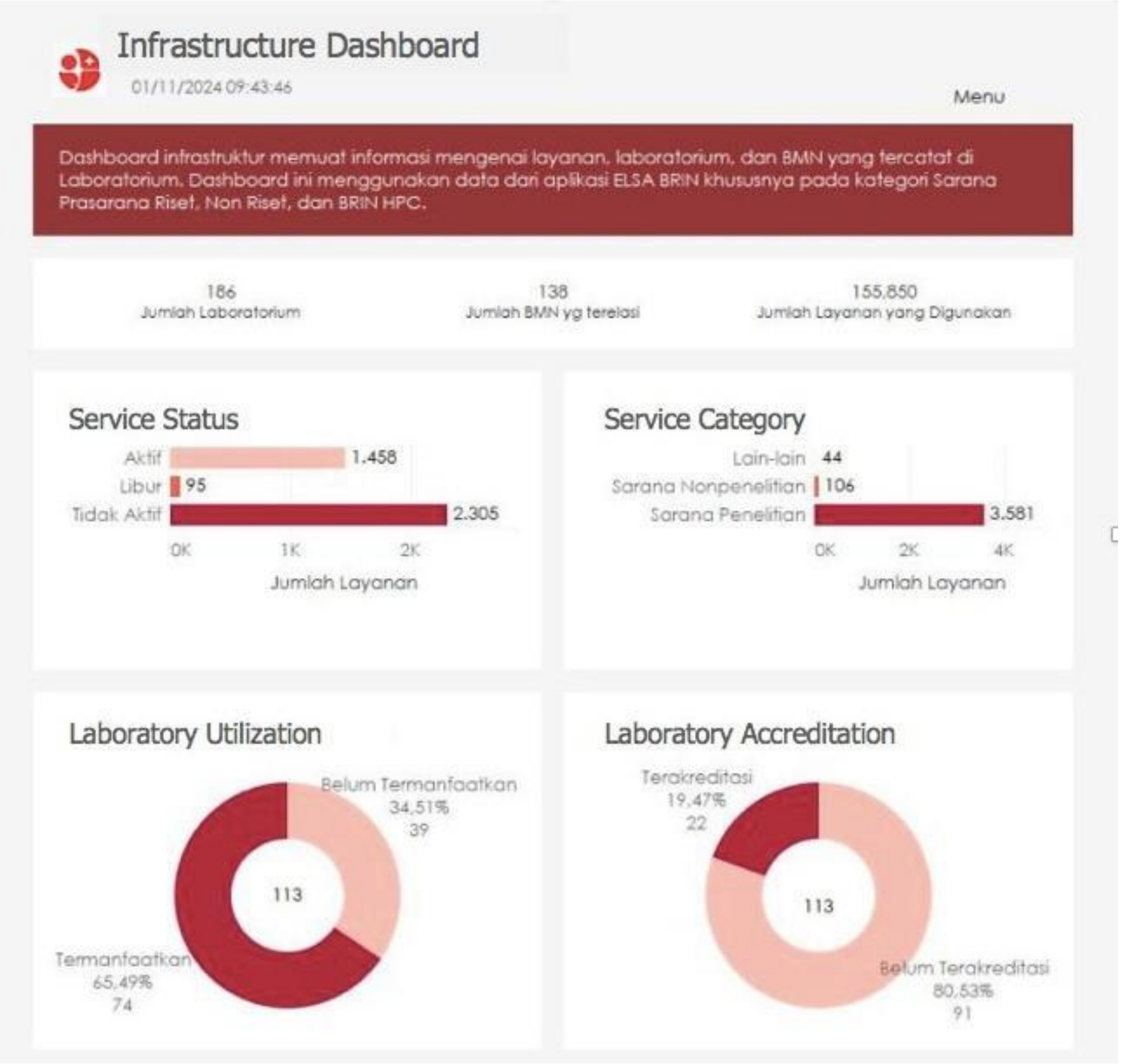
This Infrastructure Dashboard provides insights into the status, utilization, and accreditation of research facilities, laboratories, and state-owned assets (SoS) recorded in the ELSA BRIN system. It focuses on research, non-research, and the BRIN high performance computing (HPC) infrastructure.

Key Metrics:

- Total Laboratories: 186
- Total Registered SoS: 138
- Total Services Used: 155,850

Design of Dashboard Total Monitoring System :

[Figure 2-9] Infrastructure Dashboard of ELSA (1)



Source: BRIN (2023), processed and conceptualized by the author (2024)

Service Status:

- Active Services: 1,458
- On Leave: 95
- Inactive Services: 2,305

Service Categories:

- Research Facilities: 3,581
- Non-Research Facilities: 106
- Other Services: 44

Laboratory Utilization:

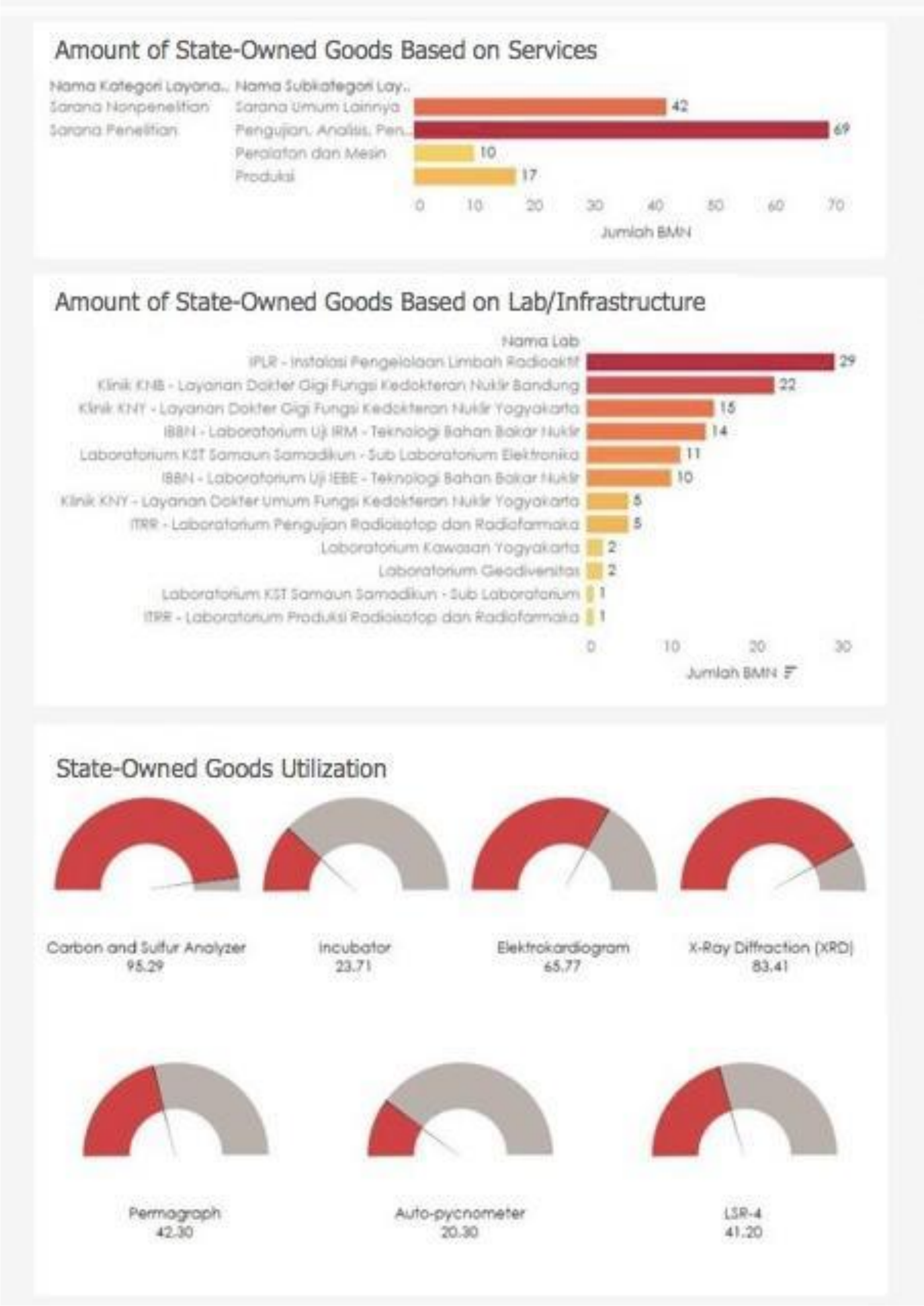
- Utilized Laboratories: 74 (65.49%)
- Not Yet Utilized: 39 (34.51%)
- Total Evaluated Laboratories: 113

Laboratory Accreditation:

- Accredited Laboratories: 22 (19.47%)
- Non-Accredited Laboratories: 91 (80.53%)
- Total Evaluated Laboratories: 113

This dashboard monitors and evaluates the availability, utilization, and accreditation of BRIN's research infrastructure. It highlights active, inactive, and categorized services while providing insights into laboratory accreditation and efficiency. The data can help in decision-making and improving infrastructure management.

[Figure 2-10] Infrastructure Dashboard of ELSA (2)



Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-11] Infrastructure Dashboard of ELSA (3)

Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-12] Infrastructure Dashboard of ELSA (4)

Dashboard Infrastruktur > Daftar BMN

01/11/2024 09:43:46

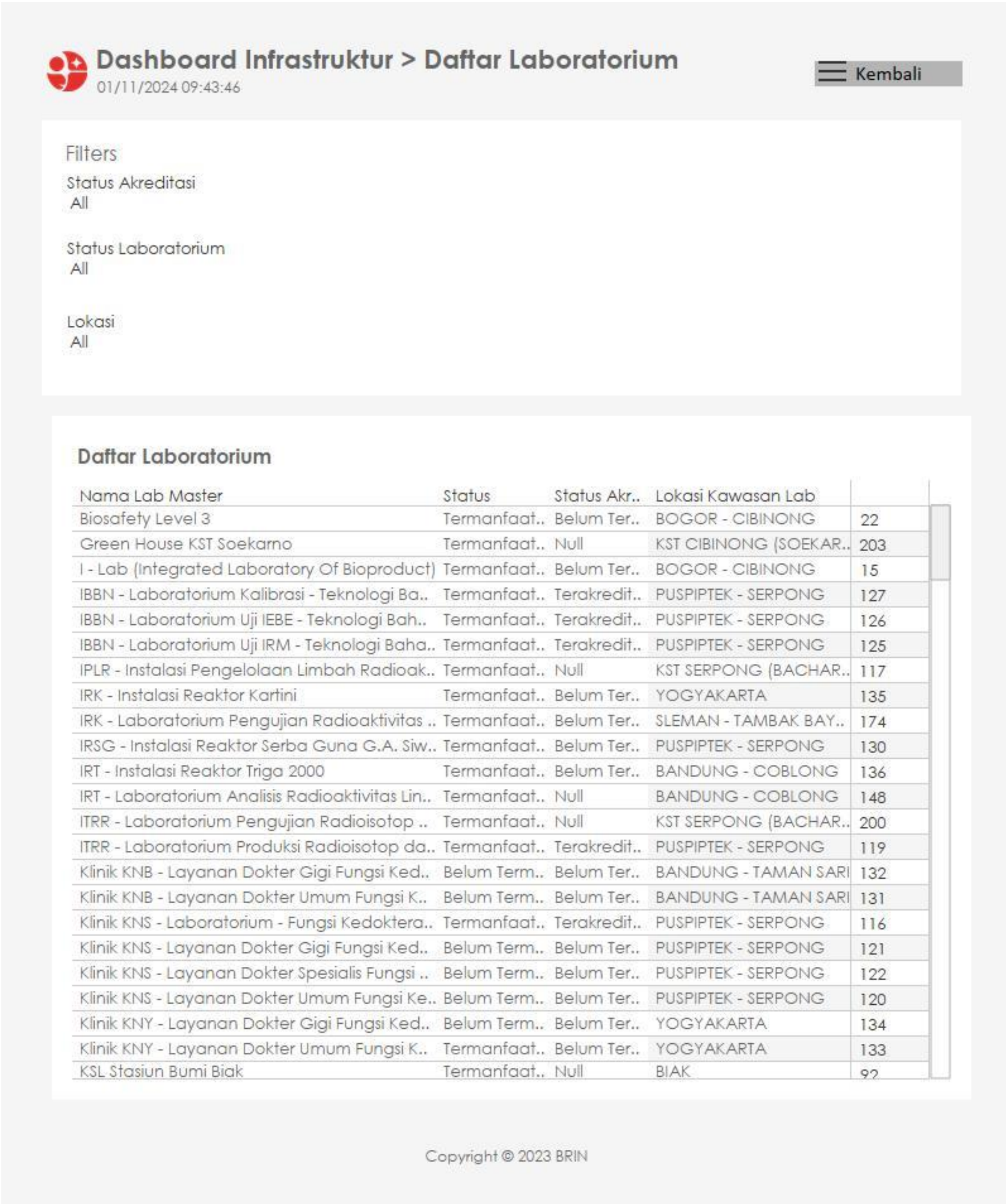
Kembali

Daftar BMN

Nama Lab	Nama Layanan Ina	Nama Barang Bmn	Tahun Per..	
IBBN - Laboratorium Uji IEBE - Teknologi Bahan Bakar Nuklir	Pengujian Densitas Nyata den..	Auto-pycnometer	2008	1
	Pengujian Karbon dan Sulfur d..	Carbon and Sulfur Analyzer	2014	1
	Pengujian Karbon/Sulfur deng..	Carbon and Sulfur Analyzer	2014	1
	Pengujian Kualitatif XRD	X-Ray Diffraction (XRD)	2013	1
	Pengujian Kuantitatif Titrasi Pot..	Titrator	2011	1
	Pengujian Kuantitatif Titrasi Pot..	Titrator	2011	1
	Pengujian Kuantitatif XRD	X-Ray Diffraction (XRD)	2013	1
	Pengujian Pola Difraksi XRD	X-Ray Diffraction (XRD)	2013	1
	Pengujian Spektrofotometer Se..	Flame Atomic Absorption Spe..	2011	1
	Pengujian Spektrofotometer Se..	Flame Atomic Absorption Spe..	2011	1
IBBN - Laboratorium Uji IRM - Teknologi Bahan Bakar Nuklir	Analisis Korosi dengan Elektroki..	Potensiostat Reference 600	2013	1
	Analisis Kualitatif Menggunaka..	Alpha-King Spectrometer	1998	1
	Analisis Kualitatif Menggunaka..	Gamma Spectrometer	2013	1
	Analisis Kuantitatif Menggunak..	Alpha-King Spectrometer	1998	1
	Analisis Kuantitatif Menggunak..	Gamma Spectrometer	2013	1
	Analisis Unsur dengan Electron ..	Scanning Electron Microscop..	2019	1
	Layanan tidak digunakan	Simultaneous Thermal Analysi..	2019	1
		X-Ray Fluoresence	2014	1
	Pengujian TG-DTA / DSC hingg..	Simultaneous Thermal Analysi..	2019	1
	Pengujian TG-DTA / DSC suhu ..	Simultaneous Thermal Analysi..	2019	1
	Pengujian XRF secara Kuantita..	X-Ray Fluoresence	2014	1
	Scanning Electron Microscope ..	Scanning Electron Microscop..	2019	1
	Uji Kekerasan Mikro Vickers	Microhardness Tester	2013	1
	Uji Tarik Pelat (Suhu Kamar)	Tensile Testing Machine	2010	1
IPLR - Instalasi Pengelolaan Limbah Radioaktif	[A] Layanan Penerimaan Limbah Radioaktif Limbah Cair..	[IPLR] Bangunan Gedung Lab..	1995	1
		Unit Resin Penukar Ion	2020	1
	[AB] Pencacahan Alpha Beta ..	[IPLR] Alpha Beta Sample Co..	2018	1
	[C] Layanan Penerimaan Limb..	[IPLR] Bangunan Gudang Tert..	1990	1
	[D] Layanan Penerimaan Limbah Radioaktif Cair Aktivita..	[IPLR] Bangunan Gedung Lab..	1995	1
		Unit Resin Penukar Ion	2020	1
	[DEMIN] Pembelian Air Demine..	[IPLR] Bangunan Gedung Lab..	2006	1
	[E] Layanan Penerimaan Limb..	[IPLR] Bangunan Gudang Tert..	2004	1
	[F] Layanan Penerimaan Limb..	[IPLR] Bangunan Gedung Lab..	1990	1
	[G1] Layanan Penerimaan Limbah Radioaktif Sumber	[IPLR] Bangunan Gudang Tert..	1990	1
		[IPLR] Bangunan Gudang Tert..	2004	1

Source: BRIN (2023), processed and conceptualized by the author (2024)

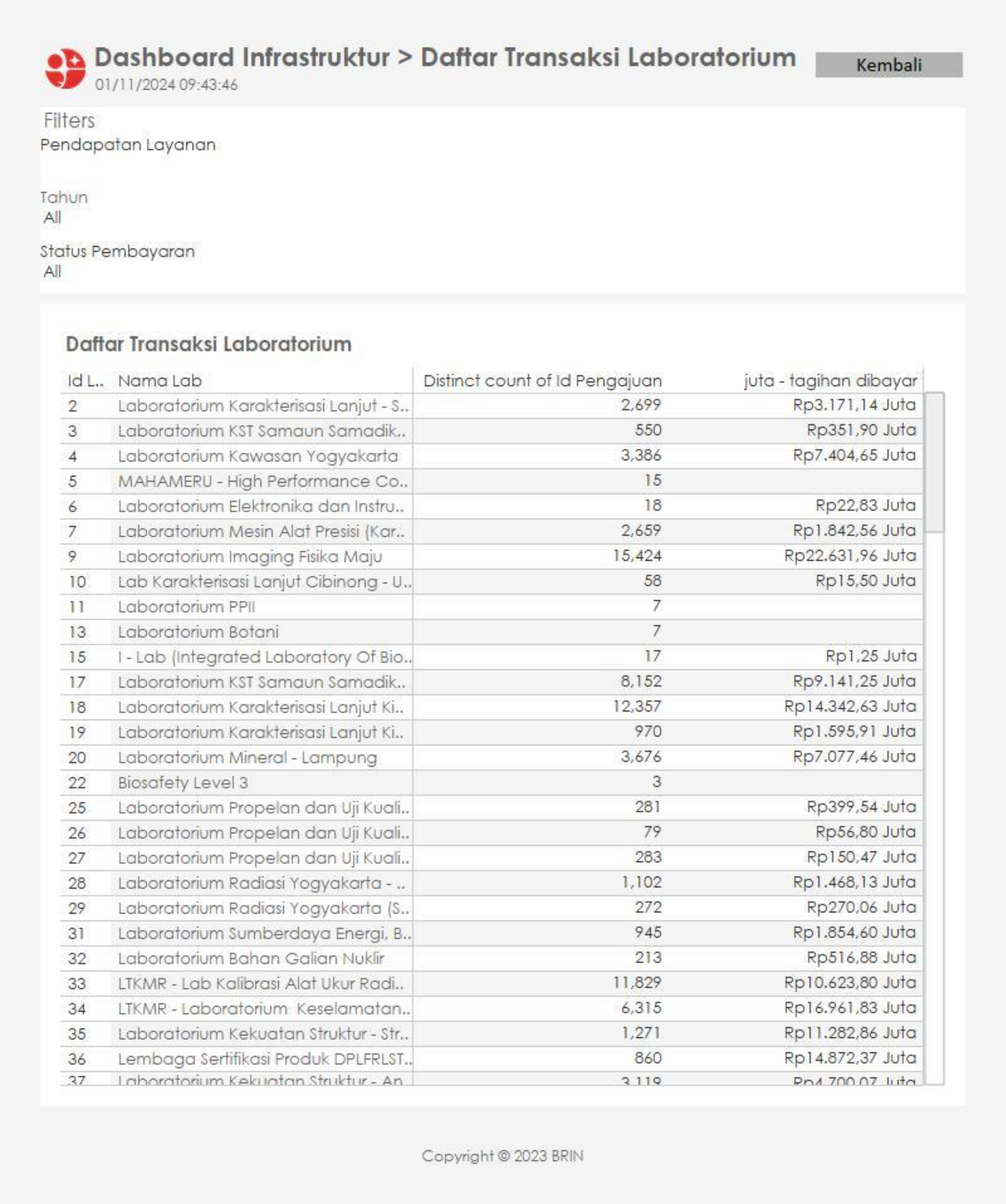
[Figure 2-13] Infrastructure Dashboard of ELSA (5)



Nama Lab Master	Status	Status Akr..	Lokasi Kawasan Lab	
Biosafety Level 3	Termanfaat..	Belum Ter..	BOGOR - CIBINONG	22
Green House KST Soekarno	Termanfaat..	Null	KST CIBINONG (SOEKAR..	203
I - Lab (Integrated Laboratory Of Bioproduct)	Termanfaat..	Belum Ter..	BOGOR - CIBINONG	15
IBBN - Laboratorium Kalibrasi - Teknologi Ba..	Termanfaat..	Terakredit..	PUSPIPTEK - SERPONG	127
IBBN - Laboratorium Uji IEBE - Teknologi Bah..	Termanfaat..	Terakredit..	PUSPIPTEK - SERPONG	126
IBBN - Laboratorium Uji IRM - Teknologi Baha..	Termanfaat..	Terakredit..	PUSPIPTEK - SERPONG	125
IPLR - Instalasi Pengelolaan Limbah Radioak..	Termanfaat..	Null	KST SERPONG (BACHAR..	117
IRK - Instalasi Reaktor Kartini	Termanfaat..	Belum Ter..	YOGYAKARTA	135
IRK - Laboratorium Pengujian Radioaktivitas ..	Termanfaat..	Belum Ter..	SLEMAN - TAMBAK BAY..	174
IRSG - Instalasi Reaktor Serba Guna G.A. Siw..	Termanfaat..	Belum Ter..	PUSPIPTEK - SERPONG	130
IRT - Instalasi Reaktor Triga 2000	Termanfaat..	Belum Ter..	BANDUNG - COBLONG	136
IRT - Laboratorium Analisis Radioaktivitas Lin..	Termanfaat..	Null	BANDUNG - COBLONG	148
ITRR - Laboratorium Pengujian Radioisotop ..	Termanfaat..	Null	KST SERPONG (BACHAR..	200
ITRR - Laboratorium Produksi Radioisotop da..	Termanfaat..	Terakredit..	PUSPIPTEK - SERPONG	119
Klinik KNB - Layanan Dokter Gigi Fungsi Ked..	Belum Term..	Belum Ter..	BANDUNG - TAMAN SARI	132
Klinik KNB - Layanan Dokter Umum Fungsi K..	Belum Term..	Belum Ter..	BANDUNG - TAMAN SARI	131
Klinik KNS - Laboratorium - Fungsi Kedoktera..	Termanfaat..	Terakredit..	PUSPIPTEK - SERPONG	116
Klinik KNS - Layanan Dokter Gigi Fungsi Ked..	Belum Term..	Belum Ter..	PUSPIPTEK - SERPONG	121
Klinik KNS - Layanan Dokter Spesialis Fungsi ..	Belum Term..	Belum Ter..	PUSPIPTEK - SERPONG	122
Klinik KNS - Layanan Dokter Umum Fungsi Ke..	Belum Term..	Belum Ter..	PUSPIPTEK - SERPONG	120
Klinik KNY - Layanan Dokter Gigi Fungsi Ked..	Belum Term..	Belum Ter..	YOGYAKARTA	134
Klinik KNY - Layanan Dokter Umum Fungsi K..	Termanfaat..	Belum Ter..	YOGYAKARTA	133
KSL Stasiun Bumi Biak	Termanfaat..	Null	BIAK	92

Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-14] Infrastructure Dashboard of ELSA (6)



Dashboard Infrastruktur > Daftar Transaksi Laboratorium

01/11/2024 09:43:46

Kembali

Filters

Pendapatan Layanan

Tahun
All

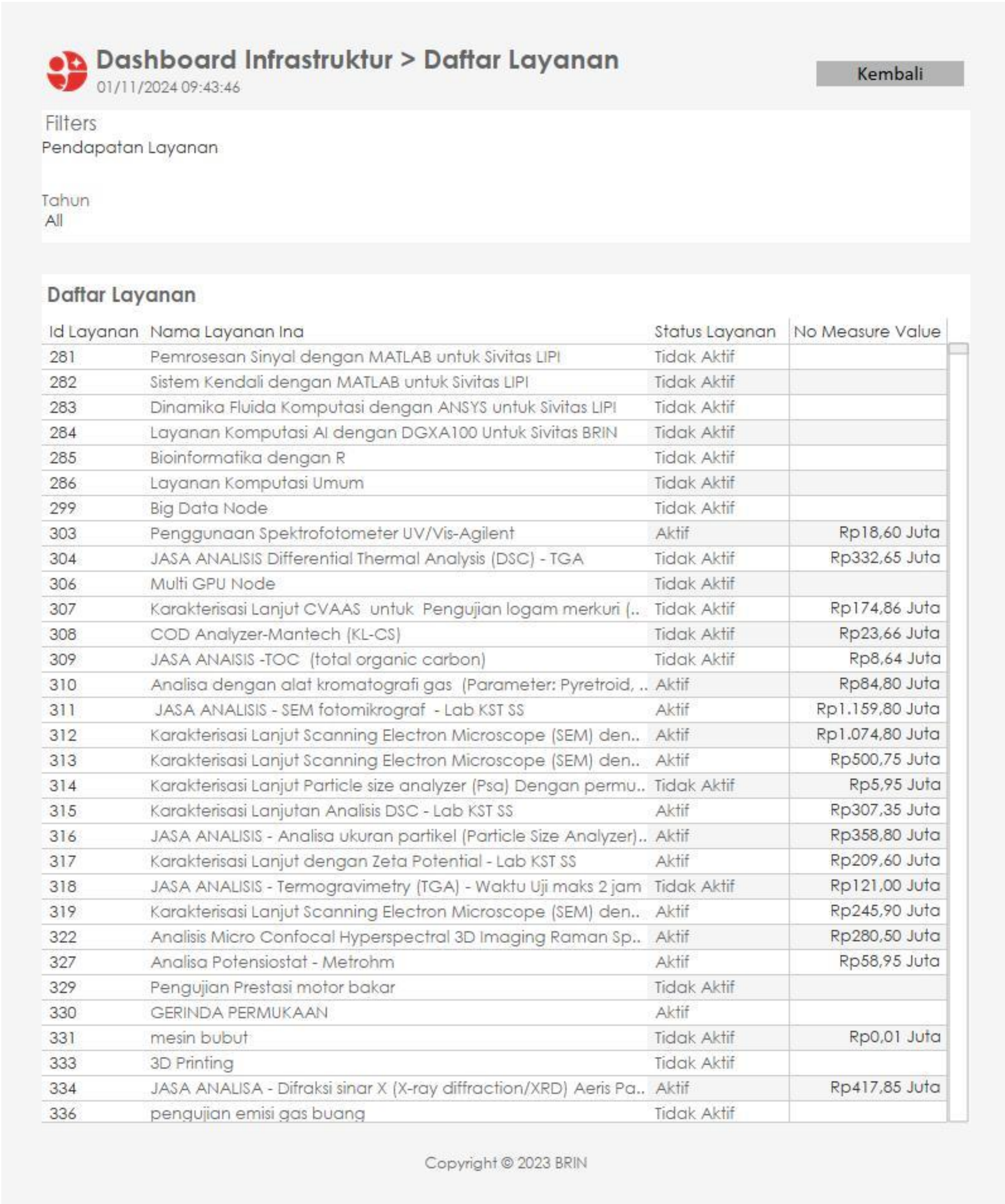
Status Pembayaran
All

Daftar Transaksi Laboratorium

Id L..	Nama Lab	Distinct count of Id Pengajuan	juta - tagihan dibayar
2	Laboratorium Karakterisasi Lanjut - S..	2,699	Rp3.171,14 Juta
3	Laboratorium KST Samaun Samadik..	550	Rp351,90 Juta
4	Laboratorium Kawasan Yogyakarta	3,386	Rp7.404,65 Juta
5	MAHAMERU - High Performance Co..	15	
6	Laboratorium Elektronika dan Instru..	18	Rp22,83 Juta
7	Laboratorium Mesin Alat Presisi (Kar..	2,659	Rp1.842,56 Juta
9	Laboratorium Imaging Fisika Maju	15,424	Rp22.631,96 Juta
10	Lab Karakterisasi Lanjut Cibinong - U..	58	Rp15,50 Juta
11	Laboratorium PPII	7	
13	Laboratorium Botani	7	
15	I - Lab (Integrated Laboratory Of Bio..	17	Rp1,25 Juta
17	Laboratorium KST Samaun Samadik..	8,152	Rp9.141,25 Juta
18	Laboratorium Karakterisasi Lanjut Ki..	12,357	Rp14.342,63 Juta
19	Laboratorium Karakterisasi Lanjut Ki..	970	Rp1.595,91 Juta
20	Laboratorium Mineral - Lampung	3,676	Rp7.077,46 Juta
22	Biosafety Level 3	3	
25	Laboratorium Propelan dan Uji Kuali..	281	Rp399,54 Juta
26	Laboratorium Propelan dan Uji Kuali..	79	Rp56,80 Juta
27	Laboratorium Propelan dan Uji Kuali..	283	Rp150,47 Juta
28	Laboratorium Radiasi Yogyakarta - ..	1,102	Rp1.468,13 Juta
29	Laboratorium Radiasi Yogyakarta (S..	272	Rp270,06 Juta
31	Laboratorium Sumberdaya Energi, B..	945	Rp1.854,60 Juta
32	Laboratorium Bahan Galian Nuklir	213	Rp516,88 Juta
33	LTKMR - Lab Kalibrasi Alat Ukur Radi..	11,829	Rp10.623,80 Juta
34	LTKMR - Laboratorium Keselamatan..	6,315	Rp16.961,83 Juta
35	Laboratorium Kekuatan Struktur - Str..	1,271	Rp11.282,86 Juta
36	Lembaga Sertifikasi Produk DPLFRLST..	860	Rp14.872,37 Juta
37	Laboratorium Kekuatan Struktur - An..	3,119	Rp4.700,07 Juta

Source: BRIN (2023), processed and conceptualized by the author (2024)

[Figure 2-15] Infrastructure Dashboard of ELSA (7)



Dashboard Infrastruktur > Daftar Layanan
01/11/2024 09:43:46
Kembali

Filters
Pendapatan Layanan

Tahun
All

Daftar Layanan

Id Layanan	Nama Layanan Ina	Status Layanan	No Measure Value
281	Pemrosesan Sinyal dengan MATLAB untuk Sivitas LIPI	Tidak Aktif	
282	Sistem Kendali dengan MATLAB untuk Sivitas LIPI	Tidak Aktif	
283	Dinamika Fluida Komputasi dengan ANSYS untuk Sivitas LIPI	Tidak Aktif	
284	Layanan Komputasi AI dengan DGXA100 Untuk Sivitas BRIN	Tidak Aktif	
285	Bioinformatika dengan R	Tidak Aktif	
286	Layanan Komputasi Umum	Tidak Aktif	
299	Big Data Node	Tidak Aktif	
303	Penggunaan Spektrofotometer UV/Vis-Agilent	Aktif	Rp18,60 Juta
304	JASA ANALISIS Differential Thermal Analysis (DSC) - TGA	Tidak Aktif	Rp332,65 Juta
306	Multi GPU Node	Tidak Aktif	
307	Karakterisasi Lanjut CVAAS untuk Pengujian logam merkuri (..	Tidak Aktif	Rp174,86 Juta
308	COD Analyzer-Mantech (KL-CS)	Tidak Aktif	Rp23,66 Juta
309	JASA ANAISIS -TOC (total organic carbon)	Tidak Aktif	Rp8,64 Juta
310	Analisa dengan alat kromatografi gas (Parameter: Pyretroid, ..	Aktif	Rp84,80 Juta
311	JASA ANALISIS - SEM fotomikrograf - Lab KST SS	Aktif	Rp1.159,80 Juta
312	Karakterisasi Lanjut Scanning Electron Microscope (SEM) den..	Aktif	Rp1.074,80 Juta
313	Karakterisasi Lanjut Scanning Electron Microscope (SEM) den..	Aktif	Rp500,75 Juta
314	Karakterisasi Lanjut Particle size analyzer (Psa) Dengan permu..	Tidak Aktif	Rp5,95 Juta
315	Karakterisasi Lanjutan Analisis DSC - Lab KST SS	Aktif	Rp307,35 Juta
316	JASA ANALISIS - Analisa ukuran partikel (Particle Size Analyzer)..	Aktif	Rp358,80 Juta
317	Karakterisasi Lanjut dengan Zeta Potential - Lab KST SS	Aktif	Rp209,60 Juta
318	JASA ANALISIS - Termogravimetry (TGA) - Waktu Uji maks 2 jam	Tidak Aktif	Rp121,00 Juta
319	Karakterisasi Lanjut Scanning Electron Microscope (SEM) den..	Aktif	Rp245,90 Juta
322	Analisis Micro Confocal Hyperspectral 3D Imaging Raman Sp..	Aktif	Rp280,50 Juta
327	Analisa Potensiostat - Metrohm	Aktif	Rp58,95 Juta
329	Pengujian Prestasi motor bakar	Tidak Aktif	
330	GERINDA PERMUKAAN	Aktif	
331	mesin bubut	Tidak Aktif	Rp0,01 Juta
333	3D Printing	Tidak Aktif	
334	JASA ANALISA - Difraksi sinar X (X-ray diffraction/XRD) Aeris Pa..	Aktif	Rp417,85 Juta
336	pengujian emisi gas buang	Tidak Aktif	

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Source: BRIN (2023), processed and conceptualized by the author (2024)

This Infrastructure Dashboard page displays a list of research services provided by BRIN, including their status (active or inactive) and revenue values (if applicable).

Key Information:

- Filters Applied: Service Revenue
- Year: All

Service List:

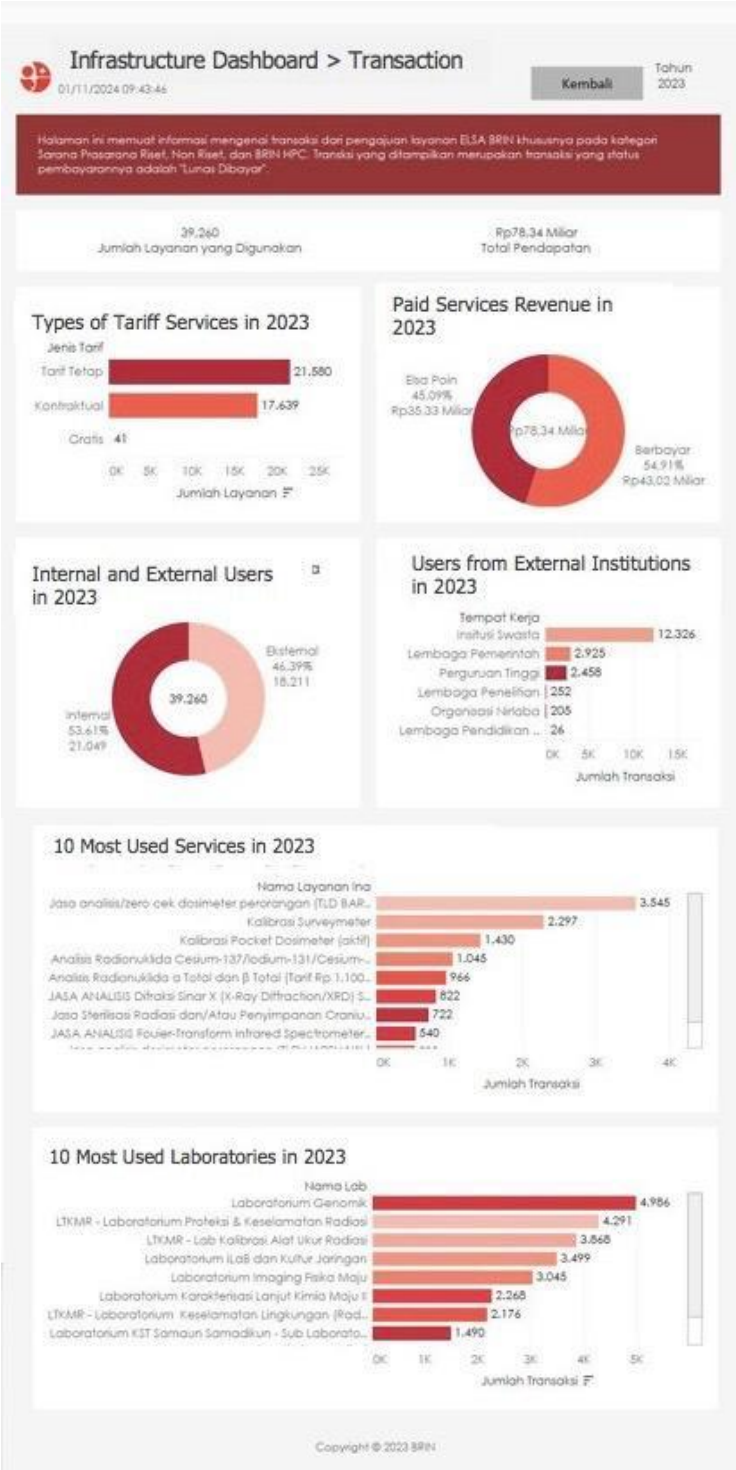
- Each service is listed with a Service ID, Name, and Status (Active or Inactive).
- Some services display revenue values in Indonesian Rupiah (IDR).

Examples of Services:

1. Inactive Services (e.g., MATLAB signal processing, AI computation, bioinformatics, GPU node)
2. Active Services with Revenue:
   - Spectrophotometer UV/Vis-Agilent – Rp18.60 million
   - Differential Thermal Analysis (DSC - TGA) – Rp332.65 million
   - Chromatographic Gas Analysis – Rp84.80 million
   - SEM Microscopy Analysis – Rp1.159 billion
   - X-Ray Diffraction Analysis (XRD) – Rp417.85 million
3. Active and Inactive Laboratory Testing Services (e.g., metal characterization, total organic carbon analysis, micro confocal hyperspectral imaging, 3D printing, fuel testing).

This dashboard provides an overview of research services availability and their financial contribution. While some services generate significant revenue, many remain inactive, potentially indicating low utilization or lack of demand.

[Figure 2-16] Infrastructure Dashboard of ELSA (8)



Source: BRIN (2023), processed and conceptualized by the author (2024)

The Infrastructure Dashboard provides information on services, laboratories, and state-owned assets (SoS) recorded in the laboratory database. The data comes from the ELSA BRIN application, specifically covering research, non-research, and the BRIN High Performance Computing (HPC) facilities.

Key Metrics:

- Total Laboratories: 186
- Total Registered SoS: 138
- Total Services Used: 155,850

Service Status:

- Active Services: 1,458
- On Leave: 95
- Inactive Services: 2,305

Service Categories:

- Research Facilities: 3,581
- Non-Research Facilities: 106
- Other: 44

Laboratory Utilization:

- Utilized Laboratories: 74 (65.49%)
- Not Yet Utilized: 39 (34.51%)
- Total Laboratories Evaluated: 113

Laboratory Accreditation:

- Accredited Laboratories: 22 (19.47%)
- Non-Accredited Laboratories: 91 (80.53%)
- Total Laboratories Evaluated: 113

This dashboard reflects the current status of BRIN's research infrastructure, highlighting utilization levels, accreditation status, and service availability.

2. Survey Method and Result

2.1 Survey Method

This survey was conducted through an online method which was distributed through an email to selected respondents spanning from public-owned research or testing laboratory units, private laboratories, and public universities. The closed and semi-opened questions were distributed to the 9 (nine) key respondents as representatives of ELSA service users. General questions captured demographic details, research field information, and user experiences of BRIN's research infrastructure and the ELSA system. Quantitative questions consisted of multiple choice and the likert-scale questions, while qualitative questions were open-ended prompts for comments or feedbacks to ELSA service improvement.

The determinant requisites of nine selected respondents who used BRIN's infrastructure through ELSA system comprised of university and industry with various research fields (e.g: natural science/engineering/medical science) with minimum experiences 1-2 years or more. The selection of those respondents in this survey was carried out through a purposive approach to ensure that participants represented the diverse profiles of ELSA system users. The respondents were selected based on their active involvement in research activities and their direct experience in utilizing BRIN's research infrastructure or ELSA services. All respondents met the minimum criterion of having at least one to two years of research experience, ensuring that their feedback was grounded in practical experience rather than general perception.

The respondents came from various institutional backgrounds, consisting of public-owned research or testing laboratories, private laboratories, and public universities. Each group accounted for roughly one-third of the total respondents, creating a balanced representation across the public, private, and academic sectors. This even distribution allowed the survey to capture perspectives from different operational environments, ranging from state-regulated laboratories and industry-driven R&D centers to academic institutions conducting both applied and fundamental research.

In terms of research disciplines, the respondents represented a wide spectrum of scientific fields. Nearly half (45%) were engaged in Natural Sciences, including agriculture, fisheries, biology, and physics. Another 33% specialized in Engineering and Technology, covering mechanical, civil, electrical, and informatics-related areas, while the remaining 22% were from Medical and Health Sciences, such as pharmacy, nursing, and clinical research. This multidisciplinary composition highlights the broad scope of ELSA's user base and its relevance across different sectors of research and innovation.

The survey participants also varied in their research experience levels. More than half (56%) had been conducting research for less than five years, indicating a strong representation of early-career researchers who are actively adapting to digital research systems like ELSA. Meanwhile, about one-third (33%) had between five and ten years of experience, and the remaining 11% had more than a decade of experience, reflecting senior-level professionals with deeper institutional knowledge and system familiarity. This combination provided a comprehensive picture of user experience, capturing both the perspectives of younger researchers who rely on ELSA for accessibility and efficiency, and seasoned experts who can identify areas for system improvement and policy refinement.

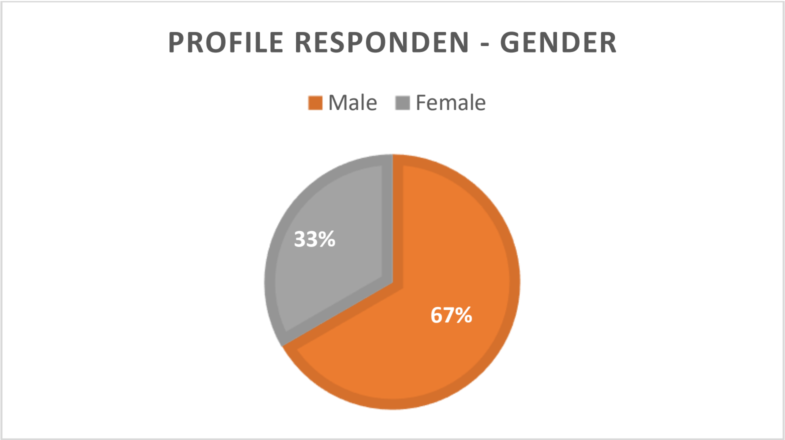
Overall, the composition of these nine respondents represents a balanced microcosm of BRIN's research ecosystem, combining voices from academia, government, and industry. Their diverse disciplinary and experiential backgrounds make them well-positioned to reflect on the accessibility, efficiency, and quality of ELSA-facilitated infrastructure services. The results drawn from this group therefore offer valuable insights into how BRIN's research infrastructure is perceived and utilized by different user segments, and how the system can be further enhanced to support Indonesia's broader research and innovation landscape.

2.2 Survey Result

Here are the survey results from the nine selected respondents who participated in the questionnaire.

Profile of Respondents

[Figure 2-17] Profile of Respondents



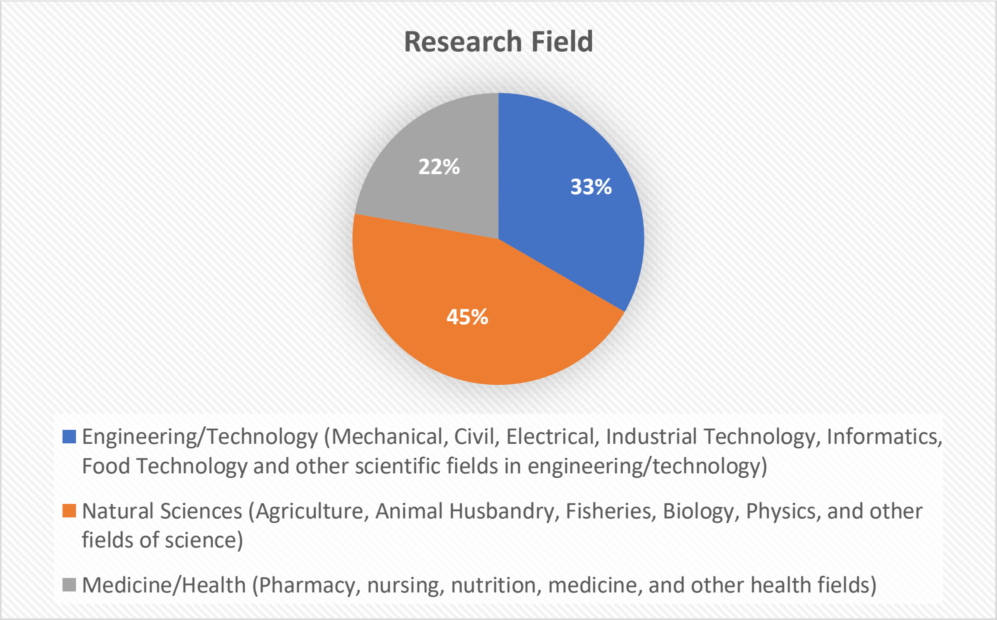
Source: Research result (2024)

This chart depicts the gender breakdown of the respondents, showing that 67% of participants identify as male, while 33% identify as female. The predominance of male respondents is notable, with roughly two-thirds of the surveyed population being men. This imbalance in representation may shape the overall perspectives captured by the survey, particularly on issues where gender can influence opinion or experience.

2.2.1 Basic Information

2.2.1.1 Research Field

[Figure 2-18] Research Field

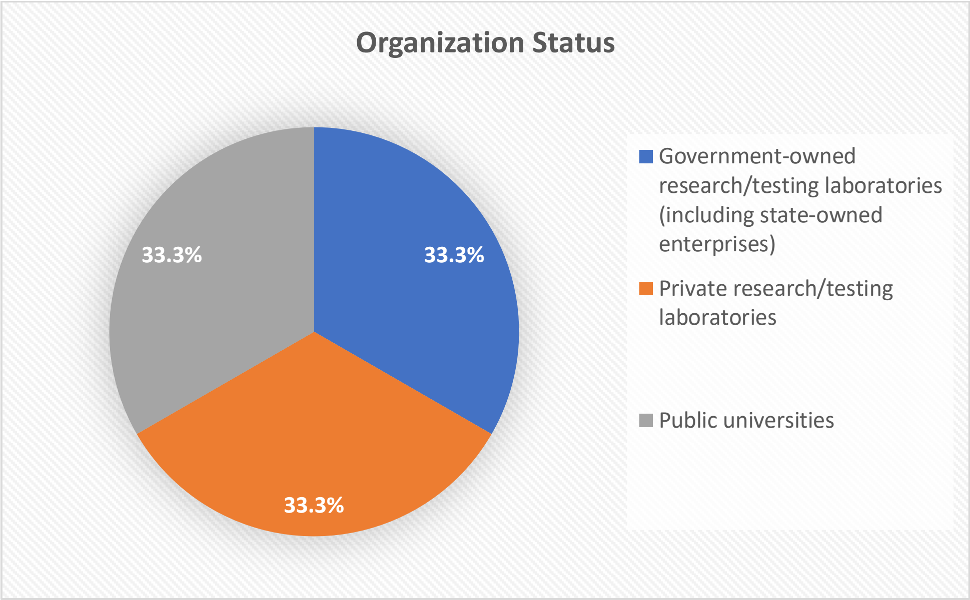


Source: Research result (2024)

This chart shows that nearly half of the respondents (45%) are from the Natural Sciences—covering fields such as agriculture, animal husbandry, fisheries, biology, and physics. Meanwhile, 33% come from Engineering and Technology disciplines, including mechanical, civil, electrical, informatics, and related areas. The remaining 22% are engaged in Medicine and Health fields, such as pharmacy, nursing, nutrition, and medicine. This distribution indicates that the majority of participants have backgrounds rooted in the broad spectrum of natural sciences, with substantial representation from engineering/technology and health-related fields as well.

2.2.1.2 Organization Status

[Figure 2-19] Organization Status



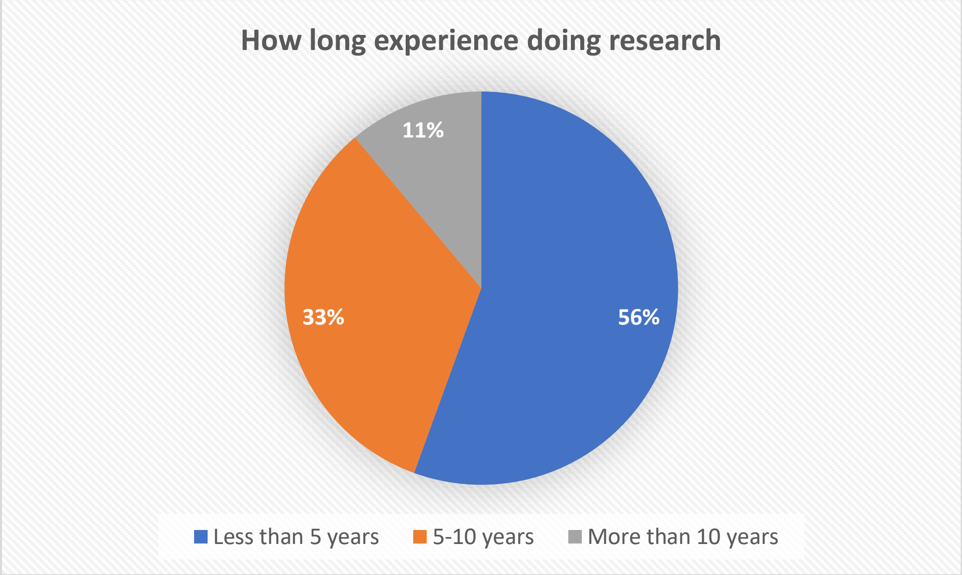
Source: Research result (2024)

This pie chart shows an evenly split sample among three types of organizations. Government-owned research/testing laboratories (including state-owned enterprises), private research/testing laboratories, and public universities each account for about 33.3% of the total respondents. This even distribution suggests a balanced representation of institutional perspectives in the survey.

2.2.1.3 How long have you been doing research

(if researcher/lecturer/lab/other laboratory personnel at the university: including participation in postdocs, doctoral programs, and master's programs)?

[Figure 2-20] Experience Doing Research

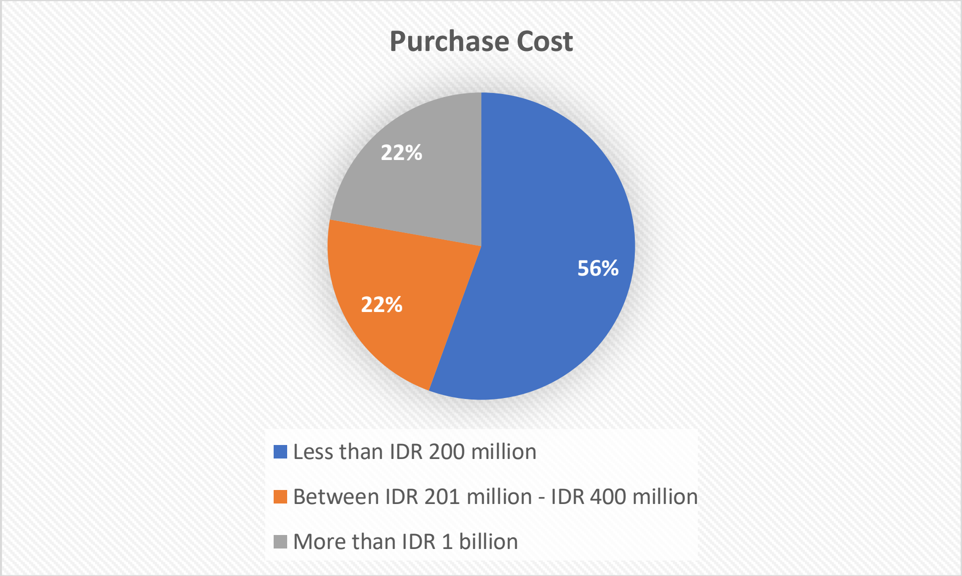


Source: Research result (2024)

According to this chart, more than half of the respondents (56%) have been conducting research for less than five years, indicating a predominance of relatively early-career researchers. Meanwhile, a significant portion (33%) report five to ten years of experience, and the remaining 11% have been active in research for over a decade. This distribution suggests a predominantly junior to mid-career demographic, with a smaller group of more seasoned researchers.

2.2.1.4 What was the purchase price/acquisition cost/purchase cost/cost of acquiring your major research infrastructure/tools?

[Figure 2-21] Purchase Cost

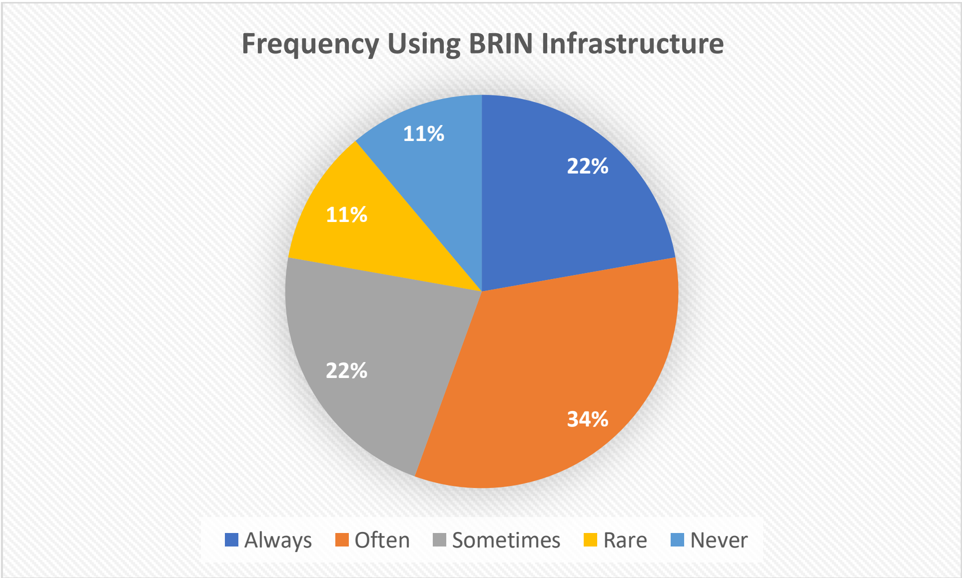


Source: Research result (2024)

This chart shows that the largest share of respondents (56%) report purchasing instruments or equipment at a cost of under IDR 200 million. Meanwhile, 22% fall within the mid-range category of IDR 201 million to IDR 400 million, and an equal share (22%) invest in instruments costing above IDR 1 billion. These findings suggest that most respondents spend within a relatively modest budget, although a sizable fraction indicates significantly higher expenditures.

2.2.1.5 Have you used BRIN infrastructure in your research?

[Figure 2-22] Frequency Using BRIN Infrastructure



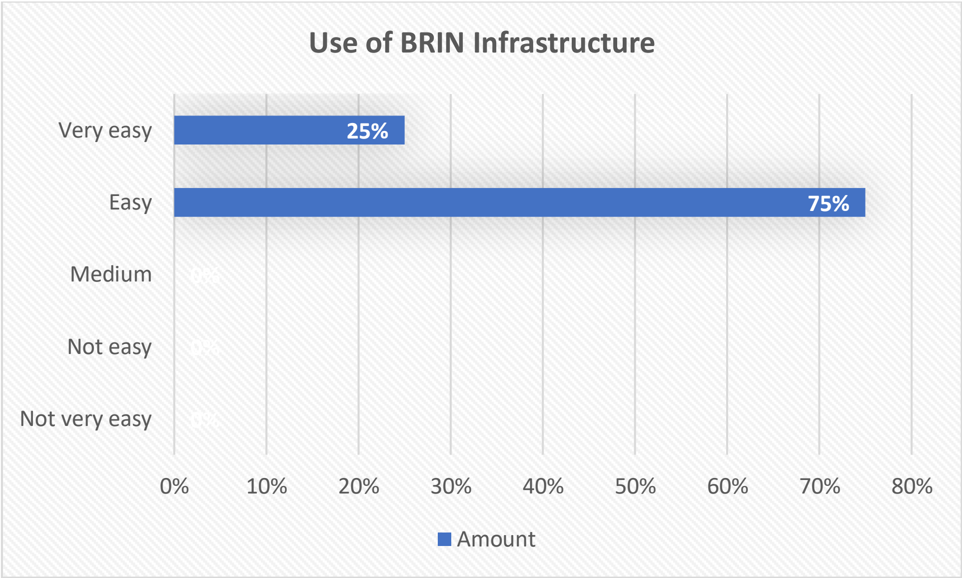
Source: Research result (2024)

From this chart, 34% of respondents report using BRIN infrastructure “often,” while 22% say they “always” use it, and another 22% use it only “sometimes.” Meanwhile, 11% rarely take advantage of BRIN infrastructure, and 11% have never used it at all. This distribution suggests that although a solid majority engage with BRIN infrastructure regularly, a smaller segment of respondents either seldom or never incorporate it into their research activities.

2.2.2 Experience in the use of research infrastructure

2.2.2.1 How easy do you find the process of using BRIN infrastructure through ELSA for your research?

[Figure 2-23] Use of BRIN Infrastructure

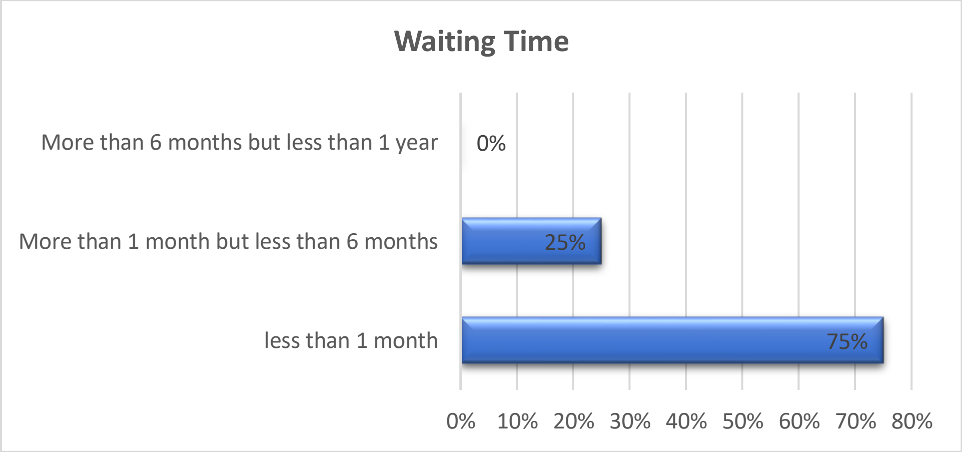


Source: Research result (2024)

Based on this chart, all respondents perceive BRIN infrastructure to be at least easy to use, with 75% rating it "easy" and the remaining 25% describing it as "very easy." None of the participants selected the medium or more difficult options, indicating a generally positive user experience in accessing or operating BRIN facilities.

2.2.2.2 How long is the waiting time to start using BRIN's research infrastructure services?

[Figure 2-24] Waiting Time

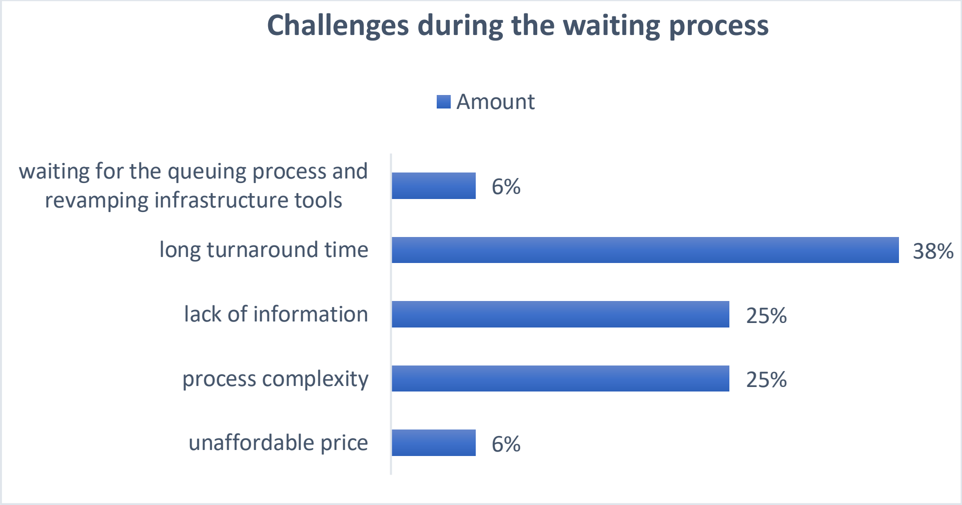


Source: Research result (2024)

According to this chart, three-quarters of respondents (75%) experience a waiting time of under one month, while the remaining quarter (25%) report waits of over one month but under six months. Notably, no one indicated a waiting period longer than six months, suggesting that most users can access BRIN infrastructure relatively quickly.

2.2.2.3 What challenges did you face while waiting to use BRIN's research infrastructure?

[Figure 2-25] Challenge During the Waiting Process

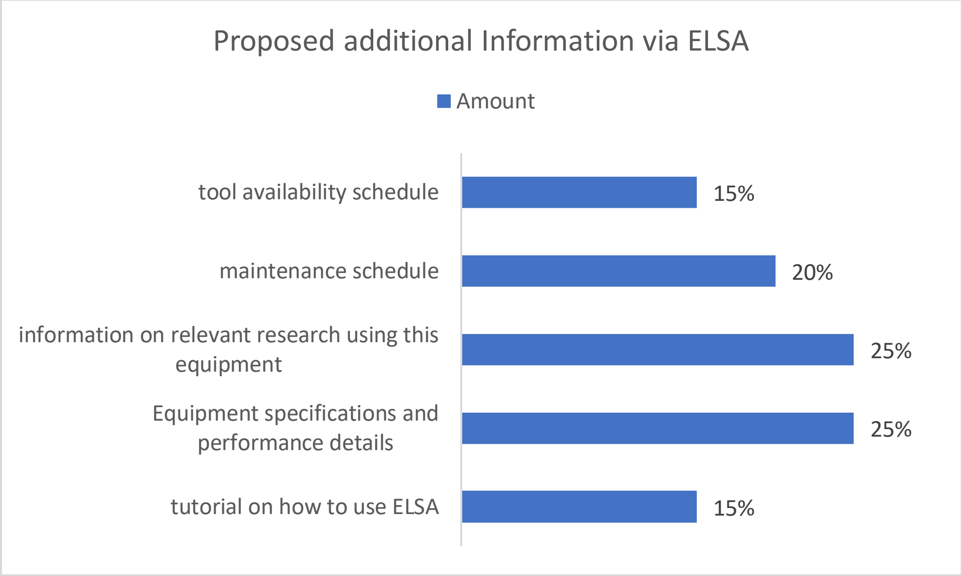


Source: Research result (2024)

According to this chart, the most frequently cited challenge (38%) is a long turnaround time, followed by two issues—lack of information and process complexity—each at 25%. Meanwhile, 6% of respondents point to waiting for the queueing process or ongoing infrastructure upgrades, and another 6% highlight unaffordable pricing. These findings indicate that most users see time-related delays as their primary concern, while informational gaps, procedural hurdles, and financial constraints also pose noteworthy difficulties.

2.2.2.4 What additional information would you like to receive when selecting and utilizing BRIN's research infrastructure through ELSA?

[Figure 2-26] Additional Information via ELSA



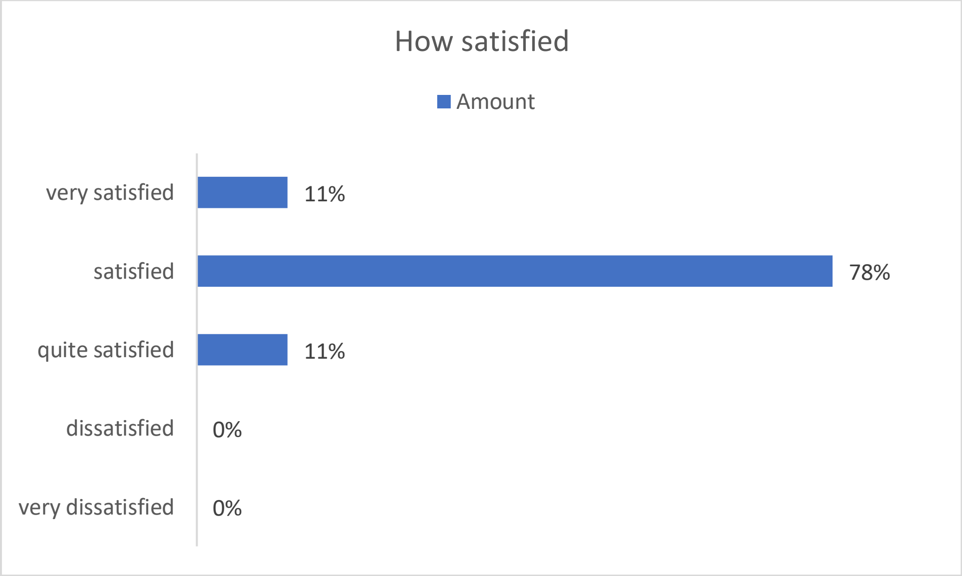
Source: Research result (2024)

This chart reveals that the most commonly suggested additions to ELSA's information offerings are details on relevant research using the equipment (25%) and comprehensive specifications/performance information (also 25%). A further 20% of respondents propose including a maintenance schedule, while 15% would like a tool availability schedule, and another 15% would find a tutorial on using ELSA beneficial. These suggestions point toward a desire for more in-depth technical and usage guidance, as well as clearer scheduling information.

2.2.3 Research Infrastructure Maintenance and Management

2.2.3.1 How satisfied are you with the BRIN research infrastructure maintenance system you use?

[Figure 2-27] Service Satisfaction

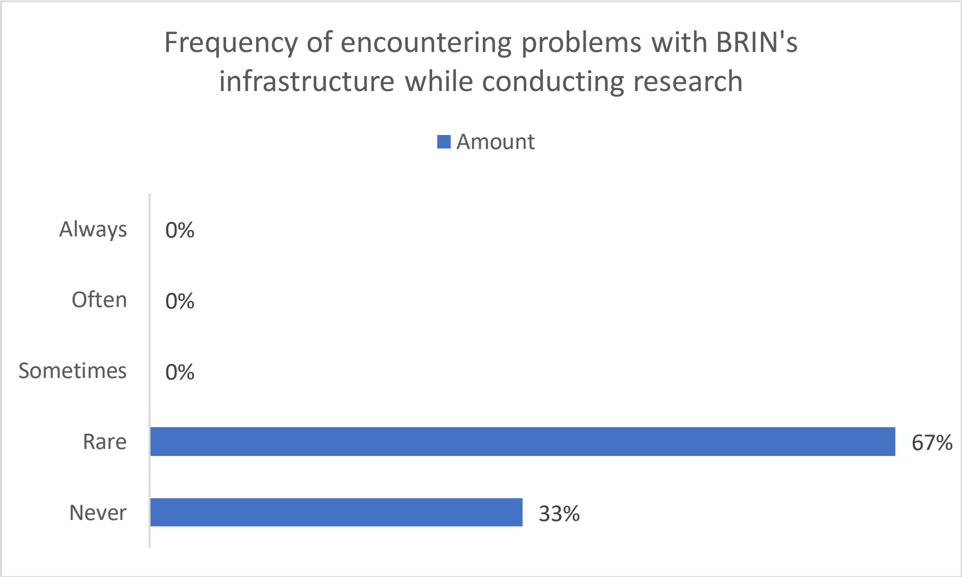


Source: Research result (2024)

The chart shows overwhelmingly positive satisfaction levels, with the vast majority (78%) indicating they are “satisfied.” Meanwhile, an additional 11% each say they are “very satisfied” or “quite satisfied,” and none of the respondent report dissatisfaction. This suggests a strongly favorable reception among users.

2.2.3.2 How often have you faced problems due to BRIN research infrastructure failures during your research?

[Figure 2-28] Frequency of Encountering Problem

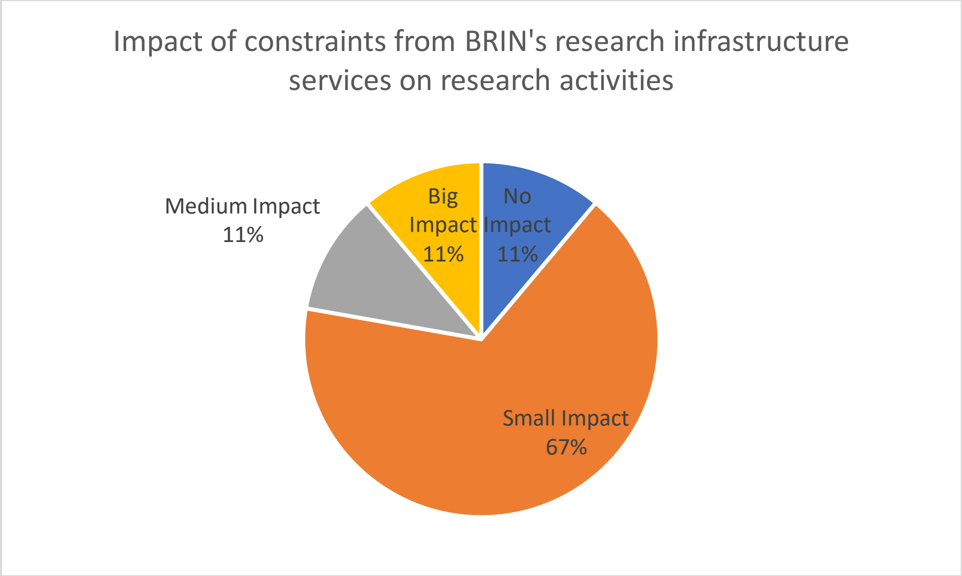


Source: Research result (2024)

According to this chart, no respondents report encountering problems “always,” “often,” or even “sometimes.” Instead, 67% experience issues only “rarely,” and the remaining 33% say they “never” have any problems. This pattern suggests that most users seldom, if ever, face infrastructure-related difficulties.

2.2.3.3 How much of an impact do the issues/constraints from BRIN's research infrastructure services have on your research activities?

[Figure 2-29] Impact of Constrains from BRIN’s Research Infrastructure

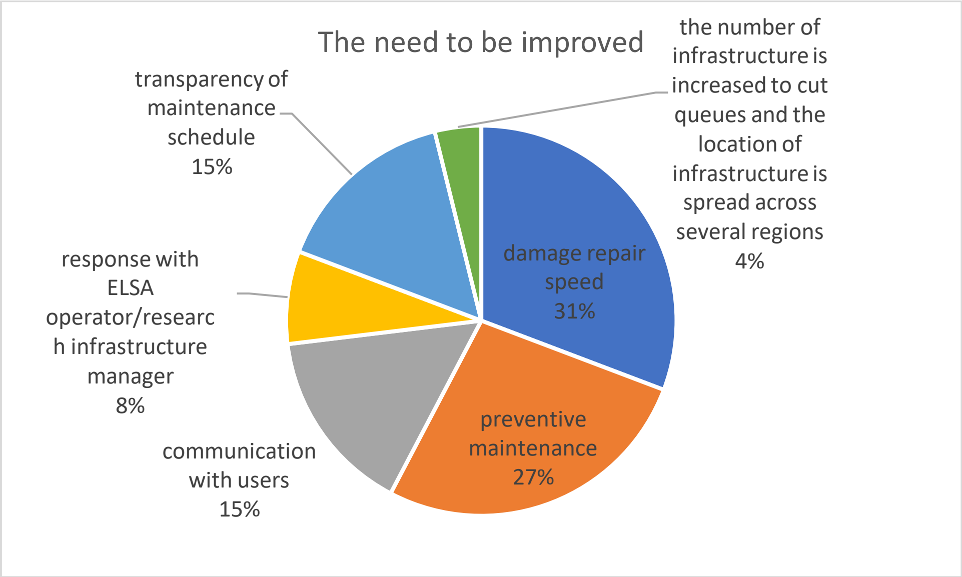


Source: Research result (2024)

A clear majority (67%) report experiencing only a small impact from BRIN’s research infrastructure constraints, while another 11% notice no impact at all. Meanwhile, 11% categorize the effect as medium and an additional 11% view it as significant, indicating that although most users are minimally affected, a smaller subset encounters more pronounced challenges.

2.2.3.4 What needs to be improved and enhanced regarding BRIN's research infrastructure services?

[Figure 2-30] The Need to be Improved

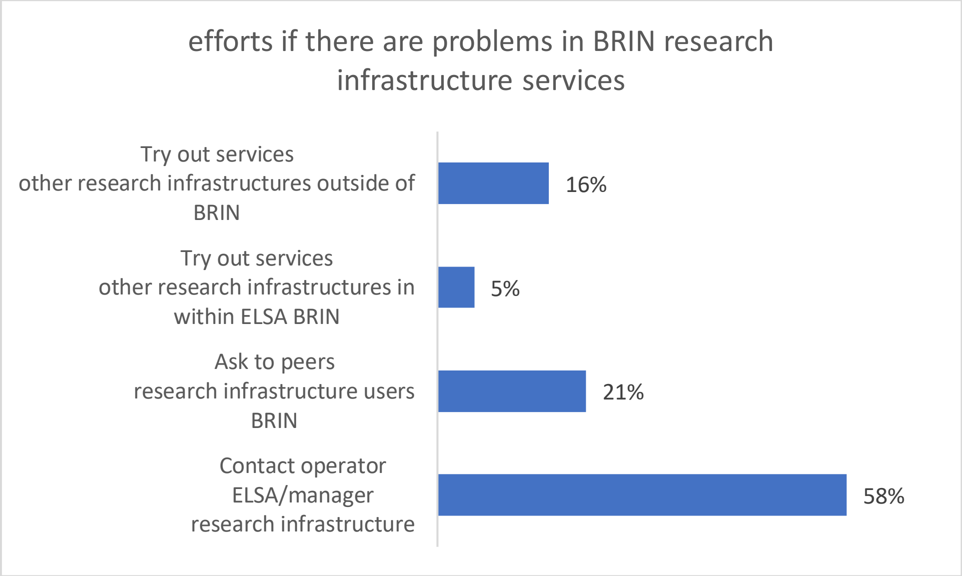


Source: Research result (2024)

From this chart, the largest share of respondents (31%) see speeding up damage repairs as the top priority for improvement, followed by better preventive maintenance (27%). Communication with users and greater transparency about maintenance schedules each garner 15%, while 8% emphasize more responsive support from ELSA operators or research infrastructure managers. A small portion also suggests expanding and decentralizing the infrastructure itself (e.g., increasing the number of sites) to reduce queues and improve overall access.

2.2.3.5 What efforts will you take if there are obstacles in BRIN's research infrastructure services?

[Figure 2-31] Efforts to Address Problems in Using Infrastructure



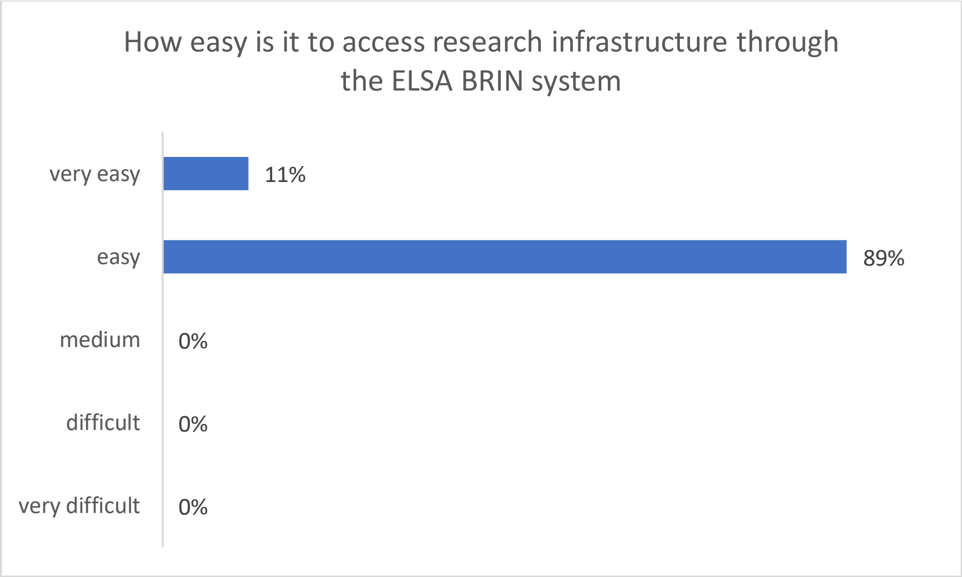
Source: Research result (2024)

Respondents respond if there are problems in BRIN research and infrastructure services with the first strategy are to contact operator ELSA/manager research infrastructure (58%) as the top oriorities, to ask to peers research infrastructure users of BRIN (21%), to try out services other research infrastructures outside of BRIN (16%), and to try out services other research infrastructures in within ELSA BRIN (5) as the lowest priority strategy.

2.2.4 Accessibility of research infrastructure

2.2.4.1 How easy do you find it to access research infrastructure through the ELSA BRIN system?

[Figure 2-32] Easy Access in Using ELSA

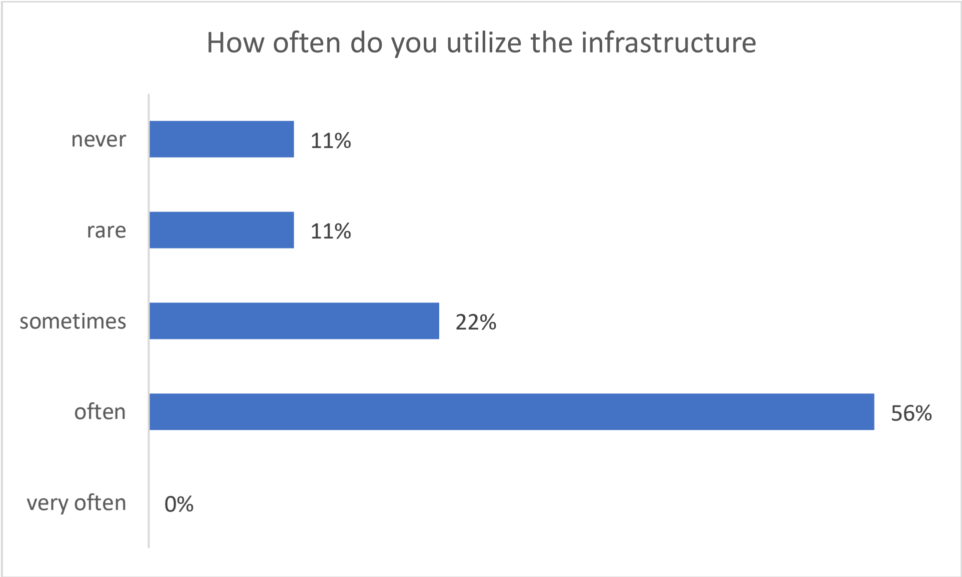


Source: Research result (2024)

Respondents respond “easy” (89%) and “very easy” (11%) in using and accessing research infrastructure through the ELSA BRIN system. No one respond “medium”, “difficult”, and “very difficult” to use BRIN infrastructure through ELSA system. It means that the respondents have positive opinion and experience in using BRIN infrastructure by using ELSA system.

2.2.4.2 How often do you utilize the infrastructure?

[Figure 2-33] Frequency to utilize BRIN infrastructure

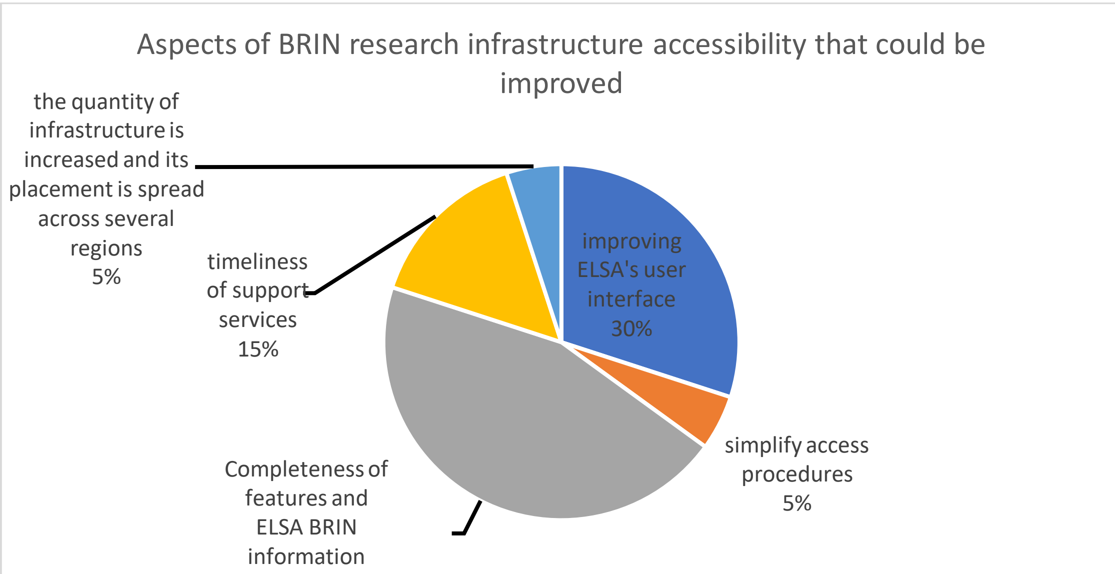


Source: Research result (2024)

A little over half of respondents (56%) report using the infrastructure “often,” while 22% do so “sometimes.” Meanwhile, 11% use it only “rarely,” and another 11% “never” use it at all. Notably, no one selected “very often,” suggesting that while most respondents do engage with the infrastructure regularly, there is still a segment that makes little or no use of it.

2.2.4.3 What aspects of BRIN's research infrastructure accessibility could be improved?

[Figure 2-34] Aspects of BRIN Research Infrastructure



Source: Research result (2024)

This chart indicates that most respondents (45%) want more comprehensive features and information from ELSA BRIN, followed by 30% who focus on improving ELSA's user interface. Another 15% highlight the timeliness of support services, while 5% suggest simplifying access procedures and an equal 5% call for increasing and spreading out infrastructure across different regions. Overall, the biggest priorities center on content completeness and user-friendliness, with smaller but still noteworthy attention given to service responsiveness, ease of access, and expanded infrastructure.

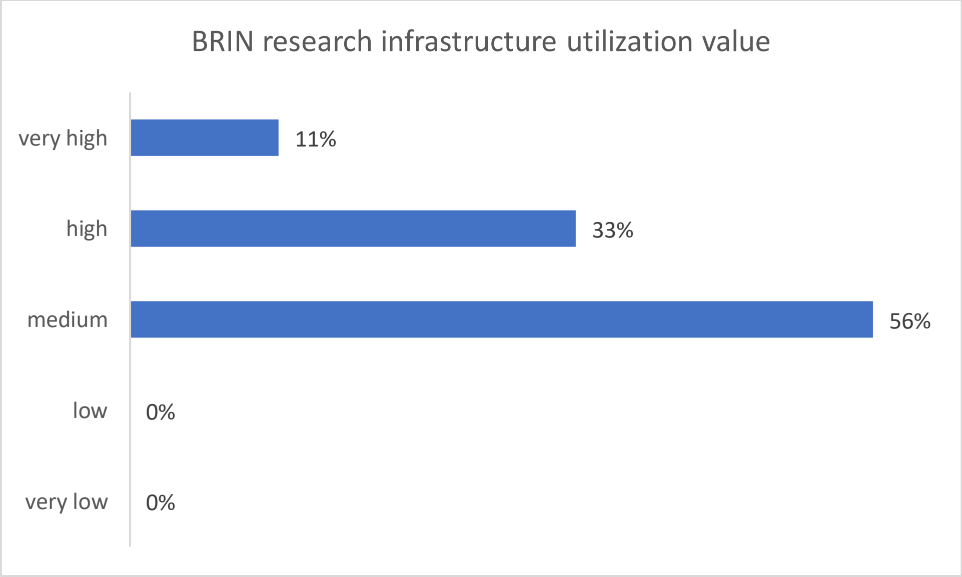
2.2.4.4 What aspects of BRIN's research infrastructure accessibility are good or do you like?

- At the time of testing if there is no long queue the work will be fast
- Online registration is very helpful so there is no need for physical presence for administrative matters
- Digitization
- No access discrimination between researchers in BRIN and outside BRIN
- Completeness of ELSA BRIN features and information
- Ease of Use of ELSA Application
- Easy to reach because it is online
- Service is quite fast & easy
- ease of access
- Web service

2.2.5 Research Infrastructure Utilization and Efficiency

2.2.5.1 How would you rate your current utilization of BRIN's research infrastructure?

[Figure 2-35] BRIN Research Infrastructure Utilization Value

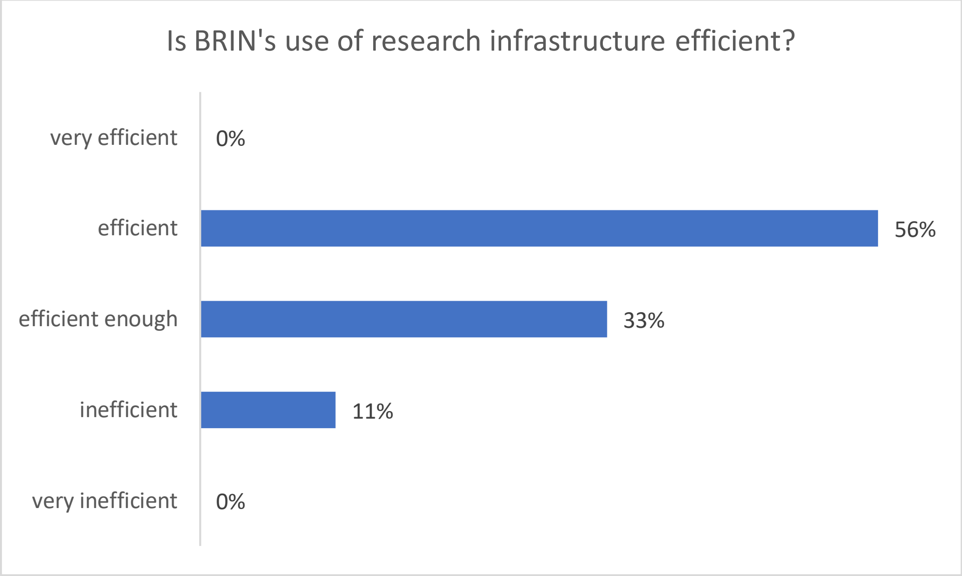


Source: Research result (2024)

In this chart, the majority of respondents (56%) rate the utilization value of BRIN's research infrastructure at a "medium" level, followed by 33% who consider it "high," and 11% who deem it "very high." Notably, no one reports a "low" or "very low" value, suggesting that most users perceive at least a moderate to high benefit from the infrastructure.

2.2.5.2 Do you think BRIN's use of research infrastructure is efficient?

[Figure 2-36] Efficiency of BRIN Research Infrastructure Use



Source: Research result (2024)

According to this chart, more than half of respondents (56%) describe BRIN's use of research infrastructure as "efficient," while a further 33% consider it "efficient enough." Only 11% perceive it as "inefficient," and no one selected "very efficient" or "very inefficient." These findings indicate that overall, respondents view BRIN's use of infrastructure positively, although a minority sees room for improvement.

2.2.5.3 explanation of answer 5.2

Efficient

- Customers have the flexibility to choose their preferred testing time.
- The work is carried out efficiently and promptly.
- The service has been consistently reliable and satisfactory.
- However, there is room for improvement, particularly in providing complete information about the tools and samples used for each test.

Efficient enough

- The existing tools are quite efficient, though they often operate at overcapacity.
- Some equipment is damaged, and repairs take a long time. If we may provide input, we suggest that BRIN continue to offer technical training, such as those provided at the Polymer Technology Center for Injection Molding.
- The use of research tools is not widely known within my academic community.

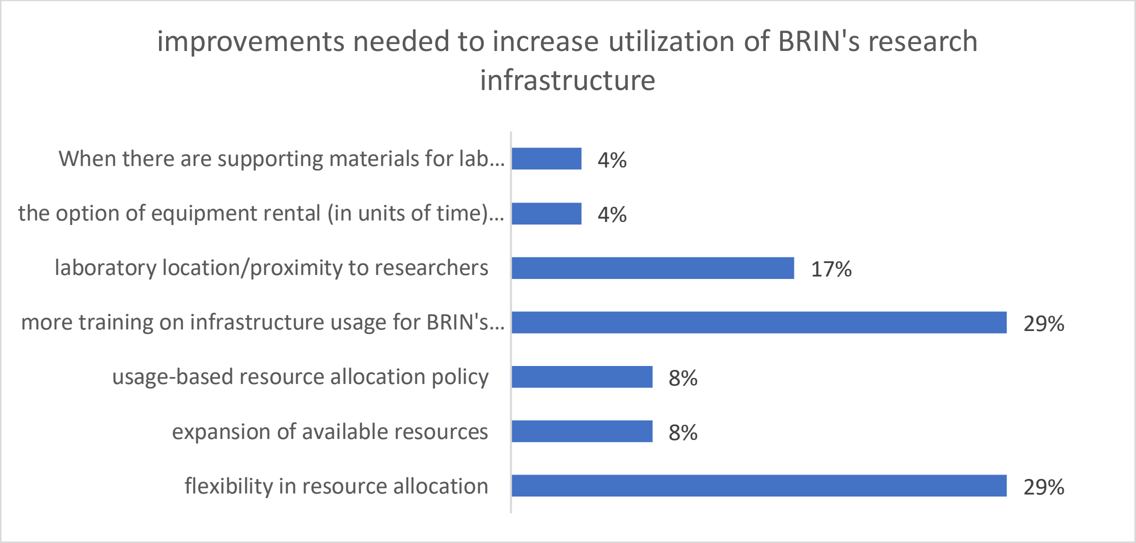
Inefficient

- Specifically for the use of horizontal tensile testing equipment in the Structural Strength Lab, the tool has a high capacity (4000 kN). However, due to its limitations, testing can only be conducted on specimens with loads below 1000 kN.

- Despite the lack of similar equipment with this capacity in Indonesia, BRIN is expected to accommodate such tests, particularly for government projects that require third-party testing approval.

2.2.5.4 What improvements do you think are needed to improve the utilization of BRIN's research infrastructure?

[Figure 2-37] Improvement of Utilization of BRIN Research Infrastructure



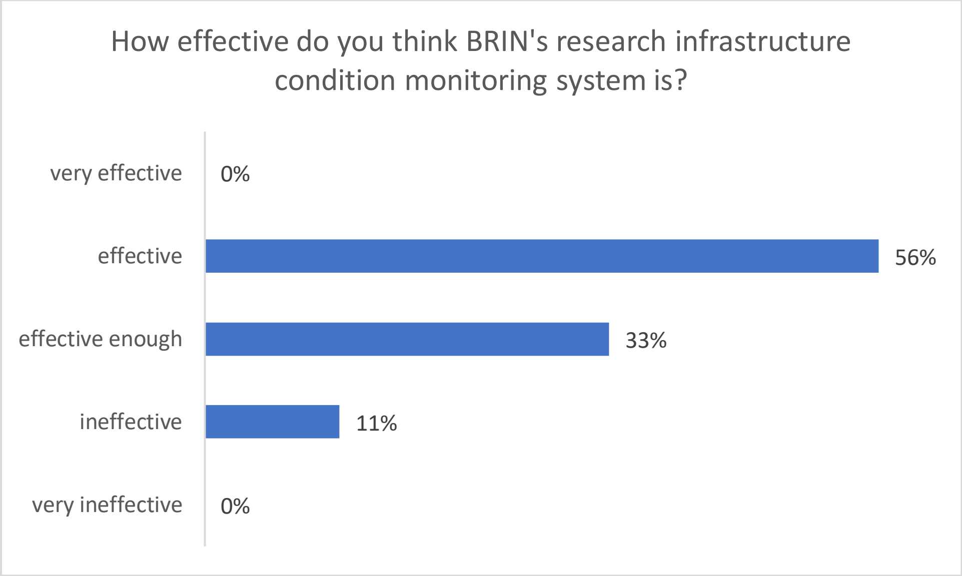
Source: Research result (2024)

Referring to the chart above, respondents focus on the improvement of BRIN's research infrastructure through management aspects. and technical aspects. Management aspects are priorities in this context. More training on infrastructure usage for BRIN's external users (29%), flexibility in resource allocation (29%), laboratory location/proximity to researchers (17%), usage-based resource allocation policy (8%), and expansion of available resources (8%) are managerial aspects to be improved by BRIN management. While, the options of equipment rental (4%) and the supporting materials for laboratory operations including the purchase acceleration (4%) are technical aspects to be revamped by BRIN.

2.2.6 Integrated Monitoring and Effective Management

2.2.6.1 How effective do you think BRIN's research infrastructure condition monitoring system is?

[Figure 2-38] Effectiveness of BRIN Research Infrastructure Monitoring System



Source: Research result (2024)

Based on this chart, the two most frequently cited improvements (each at 29%) are increasing flexibility in resource allocation and providing more training on infrastructure usage for external users. Next, 17% emphasize enhancing laboratory location/proximity to researchers, while 8% would like a usage-based resource allocation policy and another 8% call for expanding available resources. Lastly, smaller groups (4% each) suggest streamlining purchasing processes for supporting materials and offering short-term equipment rental options. Overall, these results highlight a strong desire for greater adaptability, training, and accessibility within BRIN's research infrastructure system.

2.2.6.2 explanation of answer 6.1

Effective

- Many tools still lack regular maintenance.
- Efforts should be made to increase awareness of BRIN's research infrastructure.

Effective enough

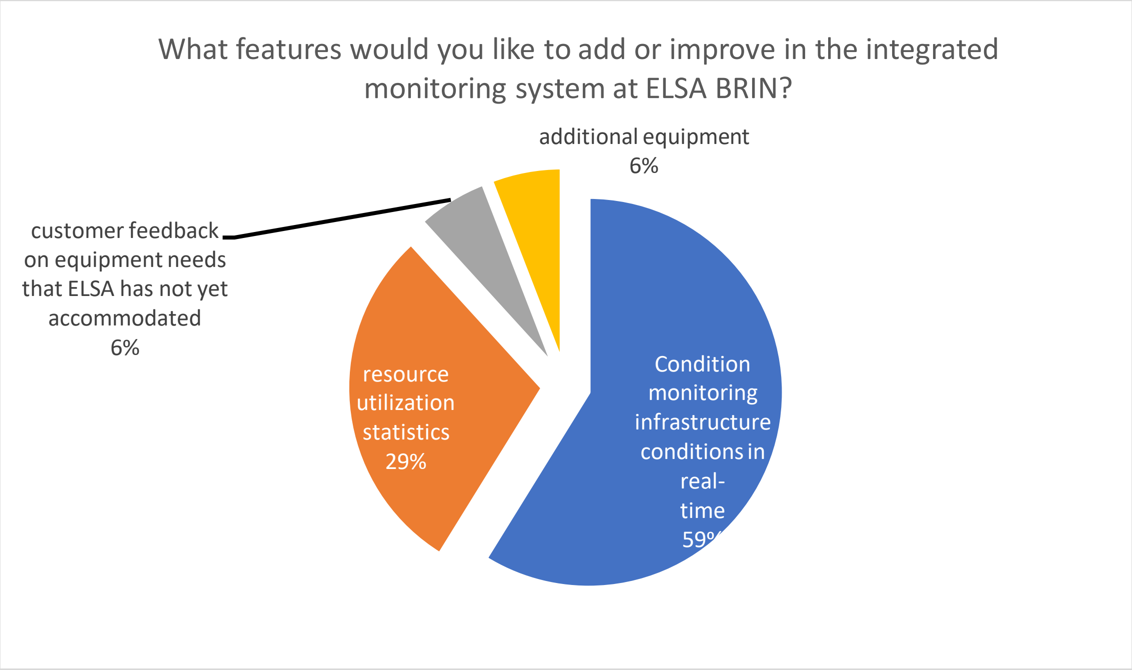
- The available tools are quite effective.
- Accessible through the ELSA app.
- Some equipment lacks a standard operating procedure (SOP) for usage.

Ineffective

- We are uncertain about how to monitor the effectiveness of BRIN's research infrastructure.

2.2.6.3 What features would you like to add or improve in the integrated monitoring system at ELSA BRIN?

[Figure 2-39] The Necessity of Added Features of ELSA



Source: Research result (2024)

Based on this chart, the largest proportion (59%) of respondents would like to see real-time monitoring of infrastructure conditions. Another 29% emphasize the need for resource utilization statistics. Meanwhile, 6% suggest adding more equipment and an equal 6% propose incorporating a mechanism for customer feedback on unaddressed equipment needs. This feedback highlights a strong desire for better system transparency, combined with improved tracking and support for future equipment demands.

2.2.7 Additional Comments

2.2.7.1 Constraints/problem in using BRIN's ELSA system for research infrastructure facilitation:

- The ELSA system only supports basic measurements, while many other measurable parameters are essential for researchers.
- Long loading times.
- The addition of human resources is needed.

2.2.7.2 Additional advice and input on the use of BRIN's ELSA system for research infrastructure facilitation:

- Additional human resources and reduced calibration costs.
- Option to use equipment beyond the offered measurement parameters, along with time-based equipment rental options.
- ELSA BRIN provides good web services; however, developing an Android version and enhancing features related to publications and BRIN infrastructure facilities would be highly beneficial.

2.2.7.3 Do you use other institutions' infrastructure besides BRIN's in your research?

- Yes, Saraswati Lab

To optimize their infrastructure research facility, BRIN should focus on enhancing user accessibility and expanding the breadth of information available on its platform. The majority of respondents suggest increasing flexibility in resource allocation, offering more comprehensive training for external users, and improving the transparency of infrastructure conditions with real-time monitoring. Additionally, addressing long turnaround times, improving preventive maintenance, and offering better communication regarding maintenance schedules are critical areas for improvement. Expanding the infrastructure network and boosting the current level of service responsiveness, as well as simplifying access procedures, will further enhance the user experience.

CHAPTER 3

Current Status of BRIN's Research and Innovation Funding Scheme and Project Management

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Chapter 3. Current Status of BRIN's Research and Innovation Funding Scheme and Project Management

1. BRIN's Roles & Functions in National S&T Policy

1.1 Background

Research and innovation are essential drivers for economic growth, societal progress, and global competitiveness. In Indonesia, the National Research and Innovation Agency (BRIN) plays a vital role in promoting these activities. To improve funding mechanisms for research and innovation, BRIN needs a comprehensive and strategic approach.

Drawing inspiration from South Korea's successful model, several key factors have been identified. Analyses conducted between the Science and Technology Policy Institute (STEPI) and BRIN last year revealed that BRIN should adopt strategic measures, including efficient governance, to ensure transparent fund allocation and reduce bureaucratic hurdles. Collaboration with academia, industry, and global partners is crucial for creating a synergistic research and innovation ecosystem, while investments in human capital enhance the quality of research outputs.

Simplifying R&I funding processes and fostering a culture that encourages innovation and calculated risk-taking are vital for achieving transformative breakthroughs. Diversifying funding sources through public-private partnerships and international collaborations strengthens financial resilience and supports research and development (R&D) initiatives. Robust monitoring and evaluation frameworks are essential for measuring project impact and informing future strategies.

Additionally, public engagement campaigns can highlight the societal benefits of research, garnering support for sustainable funding. By implementing these measures, BRIN can drive impactful research and innovation, significantly contributing to Indonesia's development.

To improve research and innovation funding in Indonesia, insights from South Korea highlight the importance of establishing medium- and long-term plans and aligning strategies with Indonesia’s national science and technology framework, as outlined in its law on the national science and technology system. BRIN should develop a clear five-year national R&D investment strategy to enable the strategic allocation of the national R&D budget. BRIN’s responsibilities include allocating and coordinating budgets for major national R&D programs. Priority research areas should receive dedicated funding rather than competing in open selection processes. Additionally, BRIN should establish a funding committee to coordinate and review R&D budget proposals in collaboration with the Indonesian Ministry of Finance. This approach ensures that science and technology considerations are integrated into national funding decisions.

Sustainable development increasingly relies on effective research and innovation policies, as demonstrated by developed nations where science and technology have become key drivers of economic growth. In Indonesia, exploring alternative policies to strengthen research and innovation funding is crucial for achieving similar success.

To enhance research and innovation funding, BRIN can implement several forward-thinking strategies that leverage collaboration, incentives, and inclusivity. By collaborating with venture capital firms, BRIN can attract private investments that combine financial backing with industry expertise. Public R&I fund managed by BRIN could be deployed to facilitate this to materialize. Establishing Public-Private Partnerships (PPPs) can also allow BRIN to share risks and rewards with private entities, aligning research objectives with market needs and diversifying funding sources. These approaches strengthen the BRIN’s capacity to drive impactful innovation while fostering stronger ties with the private sector.

BRIN has implemented a key strategy by introducing tax incentives for corporations engaged in R&I activities. Offering tax breaks or credits to companies participating in collaborative research projects with BRIN can stimulate private-sector investment in innovation. This policy reduces the financial burden on companies while fostering mutually beneficial partnerships between the public and private sectors.

To further encourage innovation, modernizing intellectual property (IP) regulations is essential. Updated IP laws would provide stronger protections for researchers and innovators, creating an environment conducive to private funding and partnerships. With secure intellectual assets, businesses are more likely to invest confidently in R&D collaborations.

BRIN should also adopt agile funding mechanisms, such as challenge-based funding, to quickly address emerging research priorities and opportunities. Flexible funding approaches ensure that critical projects receive timely support, enhancing the adaptability of the research landscape. Simultaneously, establishing an open innovation platforms would enable collaboration among academia, industry, and Non-Government Organizations (NGOs). These platforms could serve as bridges, connecting stakeholders and fostering a more cohesive and resource-rich ecosystem for innovation.

Performance-based funding offers another avenue for improvement. By linking funding to key performance indicators (KPIs) and project milestones, BRIN can ensure accountability and focus resources on high-impact projects. Similarly, implementing innovation procurement policies would allow the government to prioritize products and services developed through advanced research, encouraging further investment in innovation.

BRIN can significantly benefit from accessing global resources and expertise. By strengthening ties with international research consortia, engaging in joint projects, and participating in global funding programs, BRIN can enhance its capabilities and broaden its impact.

Inclusive innovation policies are equally important to ensure that funding opportunities are accessible to underrepresented groups. By promoting diversity, BRIN can tap into a wider talent pool and address a broader range of societal challenges.

Strategic communication and advocacy are crucial for raising awareness about the societal benefits of research and innovation. By engaging the public, policymakers, and industry stakeholders through targeted campaigns, BRIN can foster greater support for funding initiatives.

By adopting these diverse policies, BRIN can build a more dynamic, inclusive, and adaptive innovation ecosystem. These measures will not only diversify funding sources and stimulate private-sector involvement but also position BRIN as a key driver of national development in Indonesia. All of the R&I policy ideas explained above would be the basic ingredients of R&I funding guidelines.

1.2 BRIN's Role and Function in National S&T Policy

The National Research and Innovation Agency (BRIN), as the government's primary R&D organization, plays a crucial role in the 2020–2024 National Medium-Term Development Plan (RPJMN). BRIN is tasked with actively addressing Indonesia's low critical mass issue, in alignment with the National Research Master Plan (RIRN) 2017-2045. Without tackling this fundamental challenge, the national research sector would struggle to contribute and to propel the development of science and technology towards an advanced Indonesia.

BRIN's strategies and initiatives to address these challenges are guided by Law No. 11/2019 on the National System of Science and Technology. This legislation aims to enhance the use of science and technology for sustainable national development, improved quality of life, and public welfare. Moreover, the National System of Science and Technology forms the foundation and a vital component of the national development planning system, which includes long-term development plans (RPJP), medium-term development plans (RPJM), and annual strategic development plans (Renstra).

In the 2020-2024 RPJMN, the President has outlined five main directives as strategies to implement the Nawacita mission and achieve the goals of Indonesia's Vision 2045. These directives include Human Resource Development, Infrastructure Development, Regulatory Simplification, Bureaucratic Simplification, and Economic Transformation. Consequently, BRIN supports these visions and missions by:

- Providing technical and administrative support, along with rapid, accurate, and responsive analysis to the President and Vice President. This includes conducting research, development, assessment, and application, as well as invention and innovation, national nuclear energy management, and national space management in an integrated manner, while also monitoring, controlling, and evaluating the execution of BRIDA's tasks and functions.
- Enhancing the quality of human resources and infrastructure for research and innovation in national nuclear energy and space management in an integrated manner, and fostering the execution of BRIDA's tasks and functions.
- Delivering effective and efficient services in the fields of supervision, general administration, information, and institutional relations.

To fulfill BRIN's mission, the policy direction focuses on several key areas:

- **Increasing National Competitiveness**: BRIN aims to boost its contribution to enhancing the nation's competitiveness on a global scale.
- **Improving Environmental Quality and Disaster Resilience**: Efforts are directed towards better environmental management and strengthening resilience against disasters.
- **Fostering a Research and Innovation Climate**: Creating an environment that thrives on research and innovation results is a priority.
- **Strengthening Research and Innovation Resources**: BRIN is committed to enhancing its resources dedicated to research and innovation.

BRIN has initiated various efforts to reform research management in accordance with global norms and standards. This includes accelerating the enhancement of research capacity and competence by improving human resource qualifications. Additionally, BRIN aims to strengthen its role as a provider of national research infrastructure, encompassing both human resources and technological resources (hardware/software). It also serves as a hub for collaborative activities in science and technology, open to all sectors, including academia, students, and industry.

Furthermore, under Presidential Regulation Number 78 of 2021, BRIN will facilitate research and development through several funding schemes, such as: Research and Innovation for Advanced Indonesia (RIfAI) or in Indonesia is called RIIM Competition consists of RIIM Invitation, RIIM Collaboration, RIIM Expedition, RIIM Research-Based Startup Companies, Testing of Health Innovation Products, Testing of Agricultural Innovation Products, Testing of Technology Innovation Products, and many more.

BRIN recognizes the importance of supporting all parties interested in actively participating in the establishment of a research and innovation ecosystem. It also emphasizes developing research and innovation connectivity for the advancement of science and technology with partners.

National research spending is a primary indicator for comparing research activities among countries. Each country compiles research funding sources across five sectors: government, higher education, business/industry, non-profit private sector, and international institutions. In this context, national research spending includes the total expenditure in the government, higher education, and business/industry sectors. The distribution and percentage of national research spending by sector for 2022 and 2023 are detailed in the following Table 3-1.

[Table 3-1] The Distribution and Percentage of National Research Spending by Sector in 2022-2023

Sectors	Research Spending (Rupiahs)/% of total	
	2022	2023
Government (BRIN, Local Goverment, and LPDP*)	11.743.096.134.752 (66%)	6.611.817.610.505 (32,42%)
Higher education	3.029.724.677.000 (17%)	9.385.943.837.278 (46,02%)
Business/industry	2.963.369.254.279 (17%)	4.397.217.936.223 (21,56%)
Total	17.736.190.066.031 (100%)	20.394.979.384.006 (100%)

Note: LPDP is the national research and education fund sourced from the Indonesian Ministry of Finance

Source: Book of Indicators of Science, Technology, Research, and Innovation of Indonesia, BRIN, 2023 & 2024

A country's dedication to research activities can be assessed by analyzing its budget allocation and expenditure. In developing nations, the government sector plays a crucial role in distributing funds for research initiatives. Government Budget Allocations for Research and Development (GBARD) include the total funding provided by the government across various economic sectors, such as higher education, business, public sector, and international research programs.

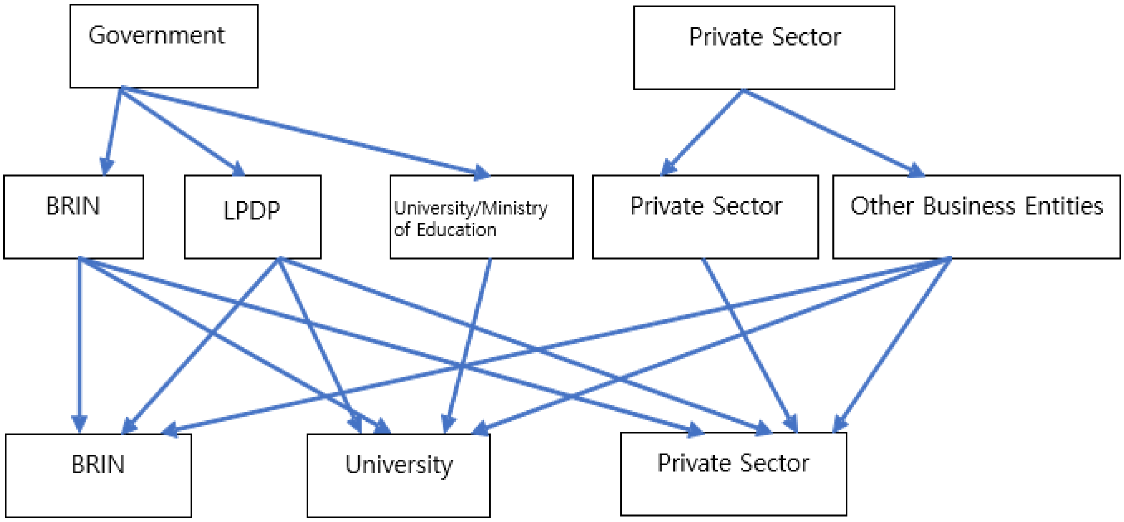
The government sector's research budget covers expenses related to personnel, capital, equipment, and services. These are managed by the central government, including entities like the National Research and Innovation Agency (BRIN), local government agencies (Pemda), and the Public Service Agency of Indonesia Endowment Fund for Education Agency (Badan Layanan Umum Lembaga Pengelolaan Dana Pendidikan/ BLU-LPDP). The following sections detail the government sector research budget and its percentage of the Gross Domestic Product (GDP).

[Table 3-2] Government Sector Research Budget and Percentage of GDP (Gross Domestic Product) 2022-2023

Sectors	**Research Spending (Rupiahs)**	
	2022	**2023**
Central Government (BRIN)	10.513.489.517.466	6.356.162.008.000
Local Governments	982.365.233.377	1.163.443.825.814
Indonesia Endowment Fund for Education Agency (LPDP)	247.241.383.909	409.485.000.000
Total Government Research Budget	11.743.096.134.752	7.929.090.833.814
GDP (Gross Domestic Product)	19.588.400.000.000.000	20.892.400.000.000.000
Government Research Budget as a Percentage of GDP	0,06%	0,04%

Source: Book of Indicators of Science, Technology, Research, and Innovation of Indonesia, BRIN, 2023 & 2024

[Figure 3-1] Key Features of Current R&D Investment



Source: BRIN (2024)

In Indonesia, the research budget is sourced from the National Revenue and Expenditure Budget (APBN) through BRIN and the Ministry of Finance in the name of LPDP as well as contributions from the private sector (industries and private R&D institutes). This budget is allocated to researchers through institutions such as the National Research and Innovation Agency (BRIN), the Indonesia Endowment Fund for Education (LPDP), and various universities. Additionally, the private sector provides funding for research projects that cater to industry needs. However, the contribution from the private sector remains relatively low compared to the government's research budget. Finally, funding supports from venture capital both to R&I in public and private sector are still limited at best.

2. Optimization of the BRIN's R&I Funding Scheme and Process

2.1 BRIN's R&I Funding Scheme and Progress

BRIN is implementing strategic measures to foster a research and innovation-driven environment by enhancing collaboration in the development and utilization of scientific products. This strategy aims to boost the involvement of the community, government, universities, and industry/business entities in research and innovation activities. Key efforts include:

- **Increasing Innovation Outputs**: BRIN focuses on elevating the production of innovative products from Research-Based Startup Companies it nurtures. This involves increasing the number of innovations adopted by the community and industry, as well as boosting patent filings. Innovation is a critical pillar for global competitiveness, with many countries leveraging domestic innovations to drive economic growth and sectoral advancements.
- **Promoting Collaborations**: The realization of innovation products serves as a benchmark for the success of the research and development process, transitioning from initial invention to stakeholder acceptance. Collaboration is essential for increasing participation from the community, universities, and industry. By sharing BRIN's research facilities, stakeholders can achieve optimal research outcomes. Both domestic and international research collaborations are vital for strengthening Indonesia's research ecosystem through effective facilitation mechanisms.
- **Balancing Public and Private Sector Roles**: Currently, government involvement in Indonesian research is more dominant compared to the private sector, which contrasts with the global research landscape. BRIN aims to shift this balance by increasing private sector involvement. This begins with initiating funding and facilitation schemes that are accessible to all parties, including researchers from research institutions, universities, and industry. These schemes are available year-round and are implemented through self-

management and contracts, with each undergoing a rigorous selection process.

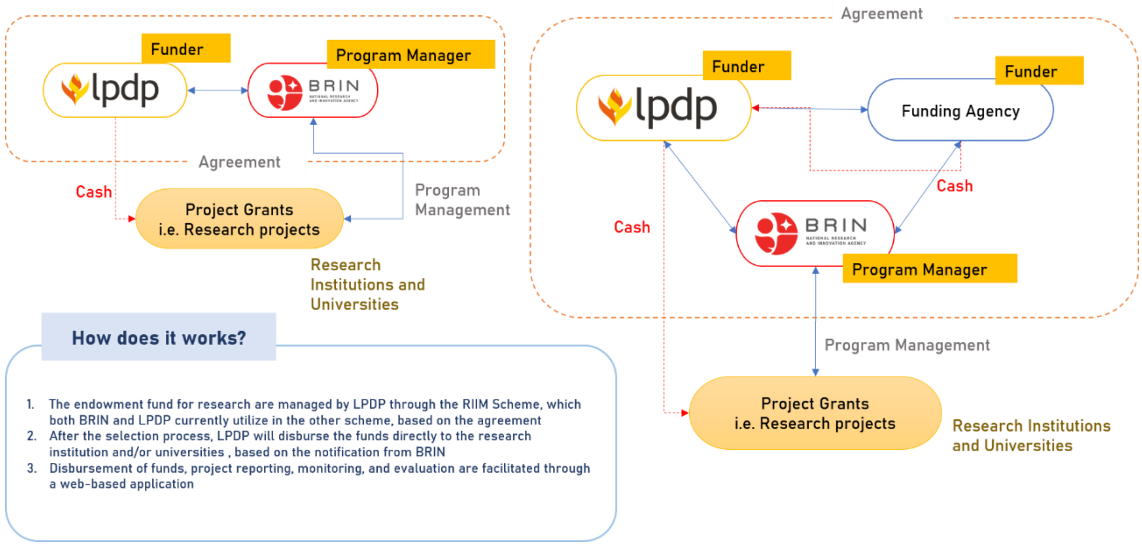
- **Structured Implementation Process**: The funding schemes involve a structured process that includes calls for proposals, selection, determination of funding recipients, execution of research activities, and monitoring and evaluation. This comprehensive approach ensures that research activities are well-supported and aligned with BRIN's strategic objectives.

It is important to note that most of key efforts above are basically funded by conventional R&I grant-base. This opens up opportunities to expand the existing R&I funding programs to a more sophisticated and complex funding scheme, such as vouchers, loan and loan guarantees.

To strengthen the research and innovation ecosystem, it is essential to optimize funding for research and innovation activities and to enhance the system for discoveries. Several funding sources within the facilitation scheme include the research endowment fund and the national budget (APBN) managed by BRIN (National Research and Innovation Agency). The research endowment fund, overseen by the Indonesia Endowment Fund for Education (LPDP), can be allocated for research activities through BRIN, universities, and even businesses. The government actively encourages the private sector to engage in research activities.

However, the participation of businesses or the private sector in Indonesia in contributing to the research budget remains quite low. This is evident from the research budget data from 2018, which shows that funding provided by businesses accounted for only 4.33% of the total research budget in Indonesia (Katadata, 2018).

[Figure 3-2] Current Funding Mechanism



Source: BRIN (2023)

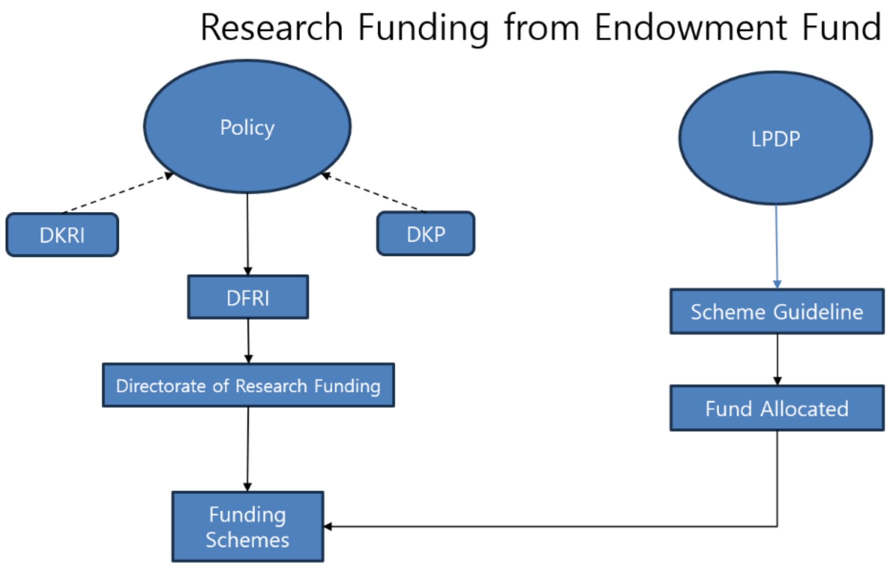
In accordance with Presidential Regulation 111/2021, the Indonesia Endowment Fund for Education (LPDP) is now responsible for managing all endowment funds related to education. This includes the Education Endowment Fund, Research Endowment Fund, Higher Education Endowment Fund, and Cultural Endowment Fund. To effectively implement these programs, LPDP collaborates with key technical ministries and agencies, such as the Ministry of Education, Culture, Research and Technology (Kemendikbud Ristek), the Ministry of Religious Affairs (Kemenag), and the National Research and Innovation Agency (BRIN).

- **Role of BRIN in Research & Innovation Funding**: As a pivotal research institution in Indonesia, BRIN manages the Research Endowment Fund. This fund is allocated to researchers through various funding schemes, supporting research, development, studies, and applications aimed at fostering inventions and innovations. These funds are accessible to researchers from universities, industries, and research institutions across Indonesia, ensuring a broad reach and impact.

- **Complementary Funding Institutions**: In addition to BRIN, Indonesia has other funding bodies like the Indonesia Science Fund (ISF) (or Indonesian is DIPI) and the Palm Oil Plantation Fund Management Agency (POPFMA) (or Indonesian is BPDPKS). DIPI operates under the Indonesian Academy of Sciences (IAoS) (or Indonesian is AIPI) and focuses on enhancing the national research ecosystem through competitive and sustainable funding. Its mission is to elevate the quality of fundamental research, thereby boosting Indonesia's global competitiveness.
- **BPDPKS and Sustainable Development**: The Palm Oil Plantation Fund Management Agency (BPDPKS) is tasked with managing the Palm Oil Plantation Fund, operating under the Ministry of Finance. It adheres to policies set by a steering committee and aligns with government programs. Established under Article 93 of Law No. 39 of 2014 on Plantations, BPDPKS collects funds from plantation business operators, known as the CPO Supporting Fund (CSF), to support sustainable practices in the palm oil sector.

These coordinated efforts and diverse funding mechanisms are crucial for advancing Indonesia's research and innovation landscape, promoting sustainable development, and enhancing the country's competitive edge on the global stage.

[Figure 3-3] Procedures related to BRIN's R&D Funding



Source: BRIN and LPDP (2023)

Notes:

DKRI: The Deputy for Research and Innovation Policy – BRIN

DKP: The Deputy for Development Policy – BRIN

DFRI: The Deputy of Facilitation for Research and Innovation – BRIN

LPDP: The Indonesia Endowment Fund for Education (LPDP) – the Indonesian Ministry of Finance

The Deputy of Facilitation for Research and Innovation (DFRI) is a unit within the National Research and Innovation Agency (BRIN) tasked with the development and implementation of policies that facilitate research and innovation. DFRI operates under the guidance of policies and regulations established by the Deputy for Research and Innovation Policy (DKRI) and the Deputy for Development Policy (DKP). These policies provide the strategic framework necessary for DFRI to effectively allocate research funds.

- **Funding Allocation by DFRI**: The Directorate of Research and Innovation Funding, under DFRI, is responsible for distributing research funds. These funds are primarily sourced from the Research Endowment Fund, which is managed by the Indonesian Endowment Fund for Education (LPDP). This endowment fund is a critical resource for supporting a wide range of research activities across Indonesia, ensuring that researchers have the financial backing needed to pursue innovative projects.

- **Strategic Role of DFRI**: By leveraging the Research Endowment Fund, DFRI plays a strategic role in enhancing Indonesia's research capabilities. The allocation of funds is designed to support projects that align with national priorities and foster innovation across various sectors. This approach not only strengthens the research ecosystem but also contributes to the country's long-term economic and technological development.

Through these efforts, DFRI ensures that Indonesia's research and innovation landscape is well-supported, promoting a culture of innovation and discovery that is essential for national progress.

■ Specific issues and challenges with BRIN’s R&I funding

- **Administrative Challenges**: From an administrative standpoint, securing funding from the Indonesian Ministry of Education, Culture, Research, and Technology (Kemendikbusristek) is perceived as more straightforward and flexible compared to BRIN, where the process feels more bureaucratic and less focused on research needs.

  A. **Need for Financial Standardization**: There is a necessity to standardize financial procedures across Research Organizations, as each organization has distinct business processes related to fund utilization.

  B. **Micro-Level Technicalities**: Attention to detail at the micro level is crucial. For instance, the information provided by each Commitment Making Official varies, particularly concerning the use of research funds for purchasing materials.

  C. **Sense of Belonging in Financial Management**: The financial implementers within Research Organizations are often from the Planning and Finance Bureau rather than the Research Organization itself, leading to a lack of ownership. This disconnect also poses challenges for the head of the Research Organization in assigning tasks effectively.

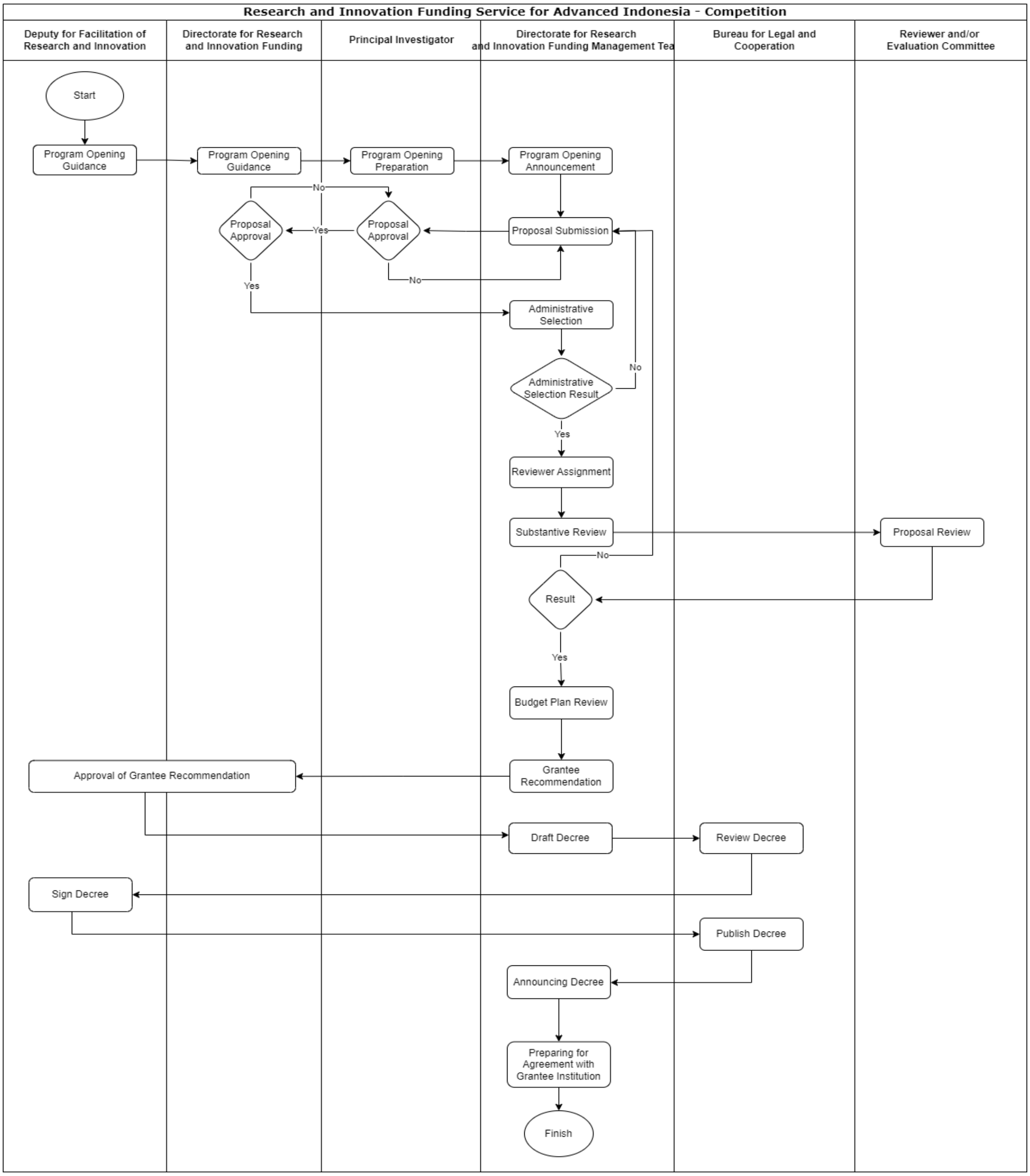
  D. **Complexity of Administrative Matters for Researchers**: Researchers often find administrative tasks more complex and cumbersome compared to their research activities.

E. **Lack of R&D Project Management**: Currently, there is no established R&D project management system or annual R&D planning in place, which hampers effective project execution.

F. **Centralization of Laboratory Tools**: The centralization of laboratory tools requires researchers to learn how to manage lab environments, such as the layout and tool usage for specific samples. For example, in genomics, labs should maintain sterility and enforce strict personal protective equipment protocols. However, current practices deviate from these standards, posing contamination risks. Additionally, equipment like ovens are sometimes used interchangeably for food testing and other tests, such as soil analysis, which can lead to cross-contamination.

3. Optimization of the BRIN’s R&I Project Management System

[Figure 3-4] Standard Operating Procedure for RIIM Competition



Source: BRIN (2024)

The Directorate of Research and Innovation Funding announces a call for proposals for funding schemes, which are managed according to a predetermined schedule. This call aims to gather research proposals from researchers across Indonesia on specific themes. The submitted proposals will undergo both administrative and substantive selection processes.

- **Administrative Selection**: This involves verifying documents, checking the proposal's format, and ensuring all required files are complete.
- **Substantive Evaluation**: Conducted by reviewers and/or an assessment committee appointed by the RIIM (Research and Innovation Management) program organizer, this evaluation assesses the novelty of the research theme, its scientific merit, and the proposed methods for addressing the problem. The evaluation covers the following aspects:

  A. **Research Planning**: This includes evaluating the quality of the research team's track record, qualifications, reputation, and consistency of experience in the relevant field, such as their education, work, and research activities. It also considers the background of the research, the problem being addressed, and the research roadmap.

  B. **Research Framework**: This involves assessing data collection methods, testing, measurement, and analysis of research results.

  C. **Outputs**: The quantity and quality of outputs each year should be clearly stated and measurable both quantitatively and qualitatively, along with the potential for research sustainability.

Following this, the budget proposal evaluation is conducted by an evaluation team from the Directorate of Research and Innovation Funding at the National Research and Innovation Agency. This assessment aims to recommend the amount of funding to be allocated to each proposal. The final recommendations are reviewed in a plenary leadership meeting to determine the recipients of the RIIM funding. The recipients are then formally announced through the issuance of a Decision Letter.

3.1 Specific Issues and Challenges with BRIN's project management system

To foster an optimal research and innovation ecosystem that yields usable results, synergy among various elements is essential. This includes conducting research, facilitating research, and utilizing research outcomes. The utilization of research marks the culmination of downstream research and development, aimed at creating innovation. This process necessitates the involvement of industrial partners as production units that employ technology. The government and industry collaborate to support the development of research and innovation as a shared agenda, focusing on social and environmental aspects to achieve sustainable economic growth.

As the sole government-owned institution for research, development, and implementation, BRIN requires a strategy to address the challenge of limited research funds. BRIN plans to collaborate with funding institutions, particularly industrial partners, who provide budget support both in cash and in kind. However, BRIN encounters several challenges in facilitating research funding:

1. **Policy**: As previously mentioned, policies regarding science and technology are outlined in Law Number 11 of 2019 concerning the National System for Science and Technology. This law includes the research concept in Article 48, which states that BRIN was established to conduct integrated research, development, study, and implementation activities, as well as inventions and innovations, under the National Research and Innovation Agency. Implicitly, research encompasses research, development, assessment, and implementation activities. The implementation of this law was further detailed in Presidential Regulation Number 78 of 2021 concerning the National Research and Innovation Agency, which mandates the integration of all research institutions in Indonesia.

   Currently, there is no specific policy or regulatory framework governing the management of research and innovation funding. In 2023, BRIN, through the Directorate of Research and Innovation Funding, is in the process of drafting regulations of the BRIN's Chairman regarding research and innovation funding. It is anticipated that this regulation will serve as a regulatory foundation for national research and innovation management mechanisms.

2. Management

BRIN manages research and innovation funding through the Directorate of Research and Innovation Funding. In 2023, the directorate has 43 personnel responsible for overseeing eight funding schemes. This number is insufficient for managing schemes with large budgets, and there is an uneven distribution of skills among the staff. The management of these funding schemes is conducted via a website-based or online system, requiring all management personnel to possess equal proficiency in using the system. Additionally, the centralized decision-making process employed in managing these schemes slows down the distribution of research funds.

3. Infrastructure

The Directorate of Research and Innovation Funding utilizes two systems to manage funding schemes: the research and innovation funding system (www.pendanaan-risnov.go.id) and the Education Fund Management Institution system (https://risprolpdp.kemenkeu.go.id). Effective management requires smooth and secure interoperability between these two systems. However, a significant challenge is the lengthy synchronization or data interoperability process between them. BRIN, through its Data and Information Center, is actively working on developing these research and innovation funding systems. However, due to limited human resources, progress in this development is slow.

4. External Funds

The Directorate of Research and Innovation Funding, through the RIIM Collaboration scheme, aims to partner with both domestic and international funding agencies to support research activities, particularly those managed by BRIN. Despite these efforts, the proportion of research funding from external sources remains lower than anticipated. In 2023, international collaborations included partnerships with the Bill & Melinda Gates Foundation (BMGF) for banana-themed research proposals, as well as collaborations with the Science and Technology Research Partnership for Sustainable Development (SATREPS) and the New Energy and Industrial Technology Development Organization (NEDO) funding from Japan. The RIIM Competition scheme, which manages over 1,000 research proposals, has yet to achieve the desired level of collaboration with industrial partners. Of these proposals, less than 1% involve collaboration with industrial partners in their research activities.

5. Monitoring and Evaluation Processes

The implementation of the funding scheme involves reviewers during both the proposal selection and the monitoring and evaluation stages of research activities. However, there is a limitation in the number and expertise of reviewers available at these stages. This shortage results in prolonged proposal selection, as well as extended monitoring and evaluation processes. As the volume of research proposals increases, the time required for evaluation also grows. The limited number of reviewers poses significant challenges to the evaluation process, which needs to be conducted promptly at the conclusion of the research activities.

4. Engaging Indonesian Companies in R&I

4.1 Current participation of Indonesian companies in R&I

To foster an optimal research and innovation ecosystem that yields practical results, it is essential to create synergy among various elements involved in conducting research, facilitating it, and utilizing its outcomes. The utilization of research marks the culmination of the research and development process, leading to innovation. This stage necessitates the involvement of industrial partners who act as production units employing the developed technology.

To stimulate investment in research within the business sector, the National Research and Innovation Agency (BRIN) actively supports the development of Indonesia's research ecosystem. This is achieved by providing industry partners with opportunities to maximize the use of BRIN-managed infrastructure and research facilities. Such initiatives aim to remove barriers that hinder corporate sector research.

Despite these efforts, the research and innovation funding scheme managed by the Directorate of Research and Innovation Funding has not yet fully collaborated with industry or business entities for research partnerships. To address this, the government has introduced a super tax deduction of up to 300% for companies contributing to research budgets. Additionally, the Directorate continues to develop funding schemes that engage the industrial sector, such as the Industrial Research Collaboration Center and joint funding schemes.

The Industrial Research Collaboration Center, In Indonesia known as Type II, is a research collaboration initiative proposed by industries. It involves researchers from the National Research and Innovation Agency and may also include universities. This center focuses on developing products and services derived from research and innovation, ensuring they are effectively utilized by the industry.

However, more efforts are needed to boost business participation in R&I and fortunately funding scheme can be an effective instrument to do that. For example, voucher scheme might offer partial funding support for small medium enterprises (SMEs) to initiate its own R&I with collaboration with research institute, such as universities. Another example is loan guarantee that might share the risks of financial institutions so they are more prudent in providing credit for R&I projects run by businesses.

BRIN has initiated partnerships with both domestic and international funding institutions to foster research and innovation through the Collaborative Research and Innovation for the Advanced Indonesia Scheme, also known as the RIIM Collaboration Scheme. This scheme is specifically designed to facilitate cooperation between BRIN and partner countries or funding institutions, whether they are based within Indonesia or abroad.

The RIIM Collaboration Scheme aligns with BRIN's goal of serving as a platform for inclusive and collaborative national and global cooperation. It focuses on research and innovation in critical areas such as food, health, and energy, which are vital for national independence. Additionally, it encompasses other fields like nuclear power, aviation and space, biology and environment, electronics and informatics, earth and maritime sciences, nanotechnology and materials, archaeology, language and literature, social humanities, as well as governance, economics, and social welfare.

Through this scheme, BRIN aims to provide open opportunities for both BRIN and non-BRIN researchers, including State Civil Apparatus and non- State Civil Apparatus individuals, to enhance their capabilities. This is achieved through national research collaborations among Indonesian researchers and international collaborations with researchers from other countries.

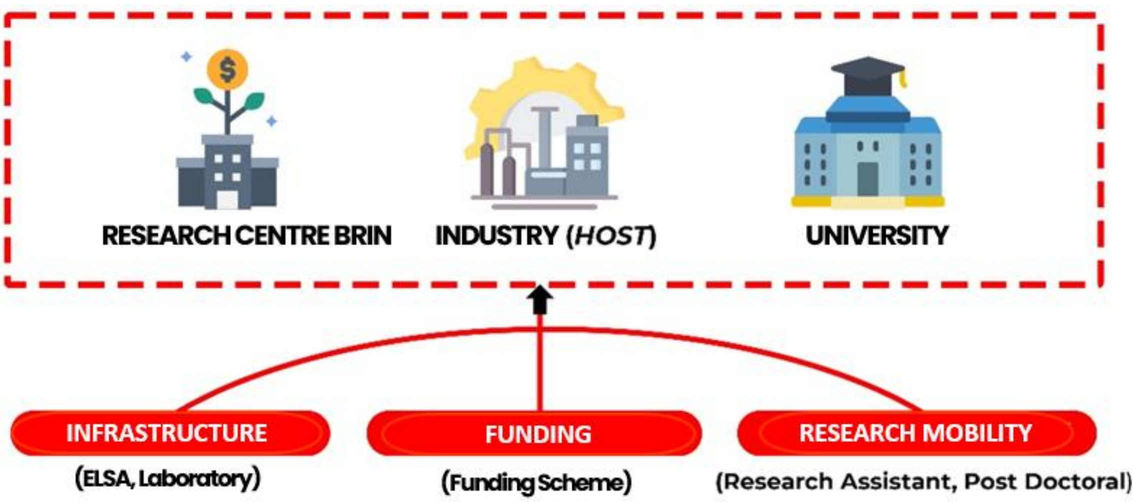
The expected outcomes of these research collaborations include an increase in the volume of research leading to novel findings or new technologies. Additionally, it aims to improve the quantity and quality of human resources by developing researchers who can compete globally. This is facilitated by enhancing their ability to publish scientific articles in international journals with medium or high global reputations and optimizing the use of research infrastructure owned and managed by BRIN.

■ BRIN's expectation for Indonesian companies to get more involved in R&I

BRIN aspires for greater involvement of Indonesian companies in research and innovation (R&I). According to Law No. 11 of 2019 concerning the National Science and Technology System, science and technology are pivotal for national development. This implies that building a robust research and innovation ecosystem must consider the technological needs essential for national progress.

Bappenas (2024) emphasizes that strengthening science, technology, and innovation requires improvements not only in the supply ecosystem, which includes universities, research institutions, and industrial R&D, but also in the demand ecosystem. This demand ecosystem comprises industries, community businesses, the general public, and both central and local governments. Enhancing both ecosystems is crucial to ensure that the outcomes of science, technology, and innovation are optimally utilized, thereby increasing the added value of industrial products.

[Figure 3-5] Encourage Industry Engagement in Funding Schemes: Industry Research Collaboration Centre (PKR Industry)



Source: BRIN (2023)

Industry Research Collaboration Centre (IRCC) or In the term of Indonesia is called PKR. The diagram illustrates a collaborative funding framework designed to strengthen national research and innovation capacity through the integration of multiple key stakeholders. In this model, BRIN research centers, industry as host institutions, and universities form a synergistic partnership to optimize the use of national R&I resources. The central aim is to bridge the gap between research excellence, industrial needs, and academic advancement, enabling a more coordinated and outcome-driven innovation ecosystem in Indonesia. This approach reflects a shift from fragmented and sectoral R&D activities toward a unified and mission-oriented model of national innovation governance.

The funding scheme supports the collaboration through three main mechanisms: research infrastructure access, financial support for research activities, and research mobility programs. Infrastructure support—via platforms such as ELSA and national laboratory networks—ensures that advanced research facilities are utilized efficiently and equitably. The structured funding scheme encourages co-financing and enhances accountability in project selection and execution. Meanwhile, research mobility supports the placement of research assistants and postdoctoral researchers across institutions, enhancing knowledge transfer and human capital development. Together, these mechanisms reinforce a holistic system that accelerates innovation and fosters stronger linkages among government, industry, and academia.

■ **Benefit of PKR Industry**

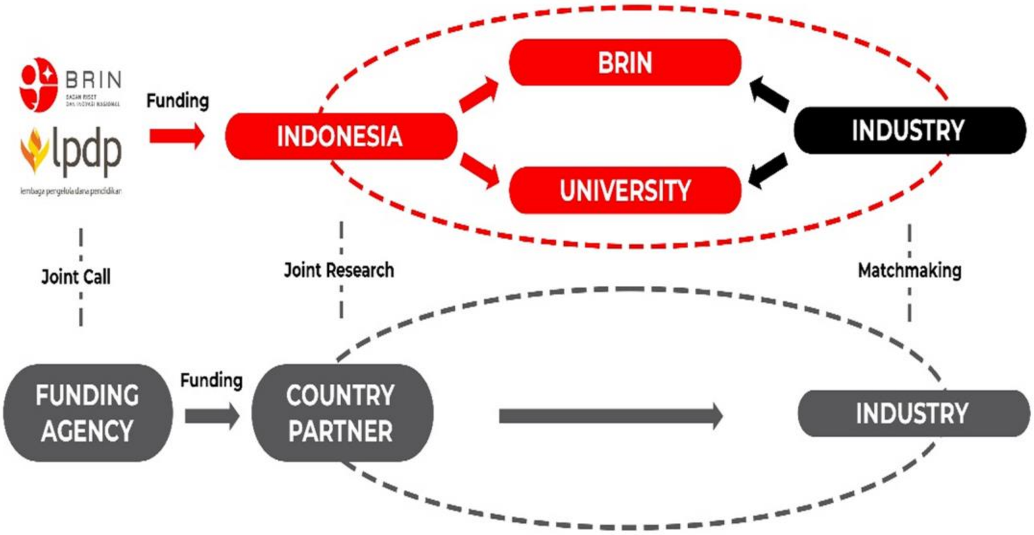
1. Survey of the interests of Heads of Research Centers and researchers regarding Industrial PKR guidelines: 20 Research Centers and 2 Directorates with a total of 570 data.
2. Survey of the usefulness of Change Projects in the form of managing funding schemes related to industry
3. Acquisition of external budget from the commitment of funding agencies in the joint funding scheme amounting to Rp 242,859,445,000

■ Development of Funding Program Strategy to Support Industry Engagement in R&D

Memorandum of Understanding (MoU) and Implementation Guidelines for Collaboration with Funding Agency

1. MoU TUBITAK Turki
2. MoU MIGHT Malaysia
3. Call Text BRIN-TUBITAK
4. Guidelines Joint Call BRIN-KONEKSI (Australia)

[Figure 3-6] Encourage Industry Engagement in Funding Schemes
Joint Funding: Program 2 + 2



Source: BRIN (2023)

The diagram illustrates the concept of a **Joint Funding 2+2 Program**, designed to enhance industry participation in national research and innovation initiatives. In this scheme, BRIN and LPDP provide funding support for collaboration between Indonesian research institutions—such as BRIN research centers and universities—and domestic industry partners through co-designed research, technology development, and matchmaking activities. Simultaneously, collaboration with international partners is facilitated through joint calls and aligned funding from foreign agencies, enabling cross-border industrial engagement to accelerate commercialization and innovation deployment. This model ensures a more balanced contribution from government and industry, thereby fostering a more sustainable and market-oriented R&D ecosystem in Indonesia.

[Figure 3-7] Business Matching between Industry

Business Matching between Industry and Researchers in BJ Habibie Building, Jakarta (30 May 2024)

Source: BRIN (2024)

The images depict a Business Matching event held at the BJ Habibie Building in Jakarta on 30 May 2024, which served as a strategic platform to connect industry stakeholders with researchers in health and life science–related fields. This activity aimed to facilitate collaboration opportunities through the presentation of research outcomes that align with industrial needs, as well as through direct discussion and potential project negotiation. By fostering communication and partnership development, the event supports the strengthening of innovation ecosystems and accelerates the translation of research outputs into commercially viable products and services. This initiative demonstrates BRIN's commitment to enhancing industry engagement and advancing mission-driven R&I collaboration in Indonesia.

[Table 3-3] Economic Benefits of Change Projects

No.	Partner Institution	Form of Collaboration	Funding Commitment
1	BMGF	Co-funding	USD 5,429,490 or equivalent to IDR 89.600.000.000
2	e-ASIA Joint Research Program (JRP)	Joint Call	IDR 81.122.515.000
3	Joint Funding Scheme (JFS) Southeast Asia-Europe (SEA-EU)	Joint Call	IDR 32.336.930.000
4	DFAT Australia –KONEKSI	Joint Call	AUD 3,000,000 or equivalent to IDR 32.000.000.000
5	NEDO	Collaborative program	IDR 3.400.000.000
6	SATREPS	Joint Call	Equipment and capacity enhancement
7	JST	Joint Call	1) Director expenses: JPY 24,000,000 2) Indirect expenses: 30% dari direct expenses or equivalent to IDR 3.200.000.000
8	TUBITAK	Joint Call	TRY 2,500,000 or equivalent to IDR 1.200.000.000
9	MIGHT	Joint Call	MYR 150.000 or equivalent to IDR 518.200.000
10	Belmont Forum Collaborative Research Action (CRA) on Tropical Forest		

Source: BRIN (2023)

The table presents an overview of collaboration commitments established between BRIN and a range of institutional partners from industry, academia, and funding organizations. Each partnership is categorized based on the specific form of collaboration—such as joint research, technology development, infrastructure access, commercialization support, or research personnel mobility—and is linked to a corresponding level of funding commitment. This structure enables transparent documentation of how each partner contributes to the advancement of R&D initiatives in Indonesia.

Funding commitments from funding agencies in the joint funding scheme amounted to IDR 242,859,445,000. It is exceeding DFRI's 2024 state budget of IDR 39 billion.

The national research budget ratio managed by BRIN (Deputy for Research and Innovation Facilitation with LPDP) comes from 3% rupiah from BRIN (DFRI), 70% from APBN managed by LPDP, and 27% from collaboration with foreign funding agencies.

■ **Problems/Issues/ Challenges related to research with industry are:**

1. **Engagement with Industry Associations**: It is crucial to engage in discussions with the Chamber of Commerce and Industry (CoCI) (In Indonesian is Kadin) and relevant industry associations to enhance the attractiveness and participation in the PKR Program.
2. **Differing Priorities**: Meetings between industry representatives and researchers reveal a disparity in priorities. Researchers often focus on novelty, while industries prioritize increasing local content requirements. Industries typically have limited time for research due to its time-consuming nature. The profit-driven nature of industries makes establishing effective collaboration challenging, as their priorities significantly differ from those of researchers. Therefore, aligning researchers' emphasis on innovation with the industry's practical needs is essential, as industries are more concerned with product development than research or novelty.
3. **Funding Alignment**: Feedback from the Directorate of Research and Innovation Utilization in Industry, BRIN, indicates that industry stakeholders are willing to allocate funding if it aligns with their needs.
4. **Collaboration’s challange with BRIN**: From industry’s point of view, collaboration with BRIN presents challenges, especially when industries seek assistance. For example, the phytopharmaceutical sector requires BRIN's support and infrastructure from the Deputy for Research and Innovation Infrastructure, but relevant researchers may be located in different divisions, leading to managerial issues and bottlenecks for the industry.

5. **Research on High-Demand Products**: There is a significant need for increased research on high-demand products, such as herbal remedies. However, BRIN's requirement for researchers to hold a doctoral degree has resulted in many researchers pursuing further education, reducing the availability of qualified personnel for immediate research initiatives.

6. **Underutilized Tax Incentives**: Although tax deductions are available to encourage industry participation in research, they are rarely used. A significant barrier is the complexity of the procurement process. When companies submit procurement requests for research activities, they often fall under different catalog systems. For example, a research procurement request might be managed by the Ministry of Health, which lacks a mechanism to accommodate tax deductions. This misalignment creates obstacles for industries seeking to benefit from research incentives.

7. **Fragmented Policy Integration**: At a macro level, Indonesia's science and technology policies lack integration. Unlike South Korea, where Science, Technology, and Innovation (STI) policy is a core part of industrial policy, Indonesia's National Long-Term Development Plan is separated from BRIN and the industrial sector. This separation results in a fragmented approach to innovation and research.

8. **Technology Readiness Level Discrepancies**: There is often a mismatch between how researchers and industries assess technology readiness level (TRL). Researchers may rate their innovations as highly ready, such as Level 7, but industries often assign lower ratings due to different assessment standards. Aligning these assessments with industry needs is crucial to support industrial development.

9. **Industrial Research Collaboration Center (IRCC) or PKR in Indonesian**: Increasing industry engagement in research can be achieved by establishing the Industrial Research Collaboration Center. The Research and Innovation for Advanced Indonesia (RIfAI) or RIIM (In Indonesian) Invitation Scheme could also complement this effort. However, this scheme has not yet been fully developed or implemented to support industry-research collaboration.

10. **Centralized Research Infrastructure**: From an industrial perspective, the centralization of research infrastructure poses a challenge. Regardless of whether a cooperation agreement exists, industries must pay fees and wait in queues to access research infrastructure, which limits the efficiency and accessibility of these resources for industrial innovation.

CHAPTER 4

Draft Guidelines for Improving Information Systems of Research & Development Infrastructure in Indonesia (Information Systems Master Plan)

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Chapter 4. Draft Guidelines for Improving Information Systems of Research & Development Infrastructure in Indonesia (Information Systems Master Plan)

1. Information System Master Plan Overview

Research infrastructure (research facilities and equipment) is a key component of science and technology research infrastructure, and its role in creating innovative research outputs is gradually expanding. In Indonesia, BRIN was established in April 2021 as a cabinet agency directly under the President to consolidate existing research institutes into one and to operate and manage the research infrastructure owned by those institutes.

Establishment of the Indonesian Research and Innovation Agency (BRIN): Indonesia established BRIN through Presidential Decree No. 78/2021, which integrates government operations in the fields of research, development, evaluation, and application, as well as invention, innovation, nuclear energy management, and national space management. Integrating research functions distributed across five research institutes and 78 government ministries has created management demands on various research infrastructures, and a management organization has been established to manage them efficiently. BRIN has been operating the E-Layanan Sains (ELSA) system for the efficient utilization and management of integrated research infrastructure, which supports integrated reservation, consulting, tutoring, and collaboration through research infrastructure. However, it provides basic research infrastructure reservation and consulting functions, and it is necessary to expand the functions of ELSA and reorganize the management structure in order to efficiently manage research infrastructure.

[Table 4-1] Status of BRIN’s Research Infrastructure

	Type of facility	Number
1	Laboratories	41
2	Botanical Gardens	5
3	Field Stations	25
4	Observation Stations	7
5	Research Vessels	5
6	Science and Technology Parks	4

Source: BRIN’s internal data

BRIN intends to promote efficient management of research infrastructure by expanding the functions of ELSA and, accordingly, to establish an informatization strategic plan for next-generation information systems. To this Object, the BRIN-STEPI research team discussed the following:

- September 2023 Consultation on refining the scope of research from building a research infrastructure management system to improving information systems for research infrastructure management utilization.
- In March 2024, the BRIN-STEPI research team meeting (Jakarta, Indonesia) introduced a mutual research infrastructure information system and revised the main content of policy advice to research infrastructure information system improvement.
- In April 2024, Korea explained the operation status and specific functions of the ZEUS system and the flow of collected information during the visit and agreed to prepare a plan through subsequent meetings.

An Information System Master Plan (ISMP) is necessary for efficient improvement and systematic promotion of information systems, and specific research is needed. At the BRIN-STEPI research team meeting in September 2024 (Jakarta, Indonesia), the status of BRIN's system and future development plans were discussed. The ZEUS system in Korea was explained as an example of a target system. The process of establishing an ISMP for Indonesia's BRIN was explained using the Korean ISMP establishment process. In 2024, we will advise on the specific method for drafting the ISMP and publish the final ISMP in 2025.

The following issues are found in the information system of research infrastructure in Indonesia.

- (Procurement) Indonesia has its own procurement process for research infrastructure, but there is no system to utilize data from existing infrastructure, such as redundancy analysis. In order to build a research infrastructure that excludes redundancies, it is necessary to consider the utilization of existing research infrastructure (e.g., which infrastructure is heavily utilized and there is an urgent need to introduce additional infrastructure, and which infrastructure is not being utilized and a request to introduce additional infrastructure is made), but there is no process and information system in place for this.
- (Maintenance) There does not appear to be a business process for identifying which research infrastructure needs maintenance and how to handle it, nor a system to support it. Research infrastructure requires continuous monitoring for failures and immediate response to failures, but no monitoring process and information system is in place. In addition, there are no processes and information systems for regular research infrastructure status checks and data management.

- (Accessibility) Utilization is managed at the same level, from small equipment to huge research infrastructure. The general method of utilizing research infrastructure is to conduct preliminary work using small research infrastructure and then use medium and large infrastructure to obtain research data. Currently, small research infrastructures require exclusive use by a single researcher; BRIN is that small, medium, and large infrastructures are centrally managed, causing great dissatisfaction among researchers. (Utilisation) BRIN does not have a procedure to measure the utilization of its research infrastructure. Utilization can be measured by deriving the available usage of the research infrastructure (time, number) and having a certain amount of usage. However, all research infrastructures are measured only by the number of uses, making it difficult to identify quantitative utilization. Of course, there are aspects that make it difficult to apply quantitative utilization figures to all research infrastructures, but it is necessary to establish a system to classify and manage infrastructures that are not utilized to at least some extent.
- (Integrated monitoring) There is no system in place to monitor what new research infrastructure will be acquired, where it will be deployed, which researchers will use it, and what results will be achieved. Logbooks are maintained on the utilization of research infrastructure, which does not allow for systematic and integrated monitoring.
- (Effective management) No systematic management system for research infrastructure is in place. In order to achieve efficient investment in research infrastructure within limited financial resources, a system should be in place to deploy research infrastructure at the right place and utilize it well.

As a solution to these problems, the following goals and improvement measures are proposed.

(Improvement Direction 1) Establishment of research infrastructure information collection system

- (Proposal 1) Establish a management system and standards for each stage of research infrastructure information generation and life cycle, and establish an information linkage system and utilization system.
- (Proposal 2) Establishment of information actualization and statistical calculation methods such as survey, survey analysis, etc.
- (Action 3) Developing a reservation system and information utilization system from the user's perspective and preparing an informatization strategic plan

(Improvement Direction 2) Establish an efficient joint utilization system

- (Proposal 1) Establish a separate management standard and management system for income generated through the use of research equipment and establish a reinvestment system through it.
- (Proposal 2) Establish a separate reward system in addition to point rewards for voluntary co-utilization, and establish support projects so that it can be actively utilized.
- (Proposal 3) Designate joint utilization bases in accordance with Indonesia's strategic goals for science and technology, and prepare necessary research infrastructure for each base, strategic investment plans, and staffing plans to foster them.

2. Current Status of Research Infrastructure Information System in Indonesia

Since the establishment of BRIN in 2021, there has been a lack of integrated information monitoring of research infrastructure, resulting in duplication of investment, poor maintenance, and limited utilization due to limited access to information.

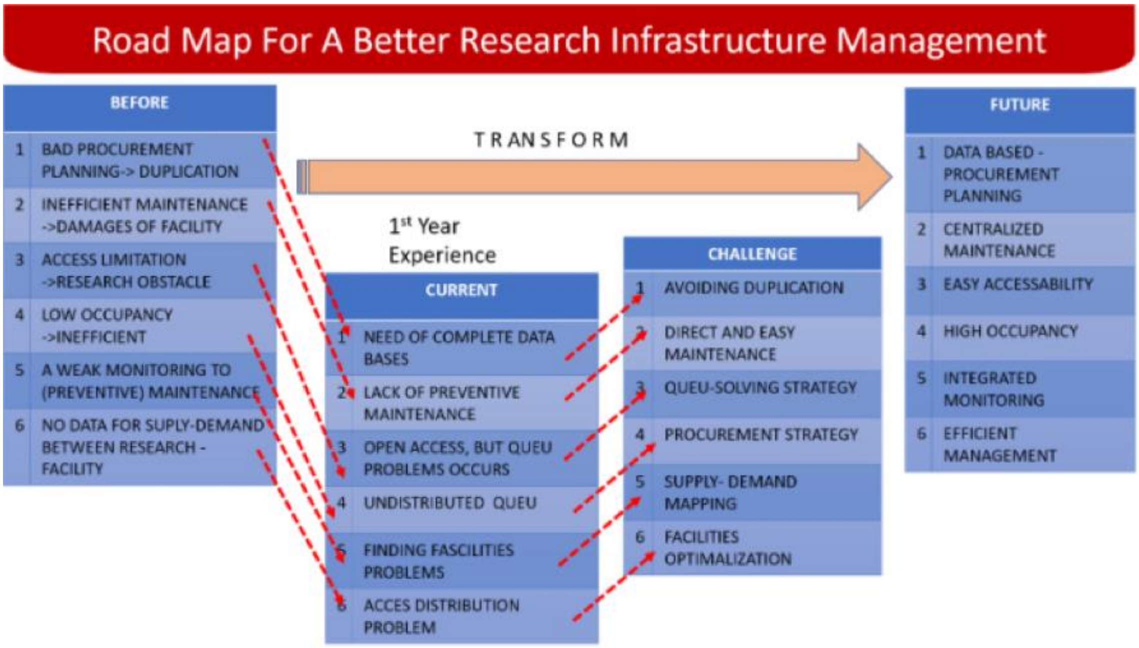
[Table 4-2] Analysing Indonesia's Research Infrastructure Problems

NO	ISSUE	ANALYSIS
1	Poor procurement planning	Procurement duplication between agencies due to lack of procurement plan coordination ⇨ Procurement duplication, unequal procurement skills (inter-agency skills gap)
2	Inefficient maintenance	Non-standardized maintenance services ⇨ Inefficient budgeting and substandard maintenance services
3	Restricted access	Disconnected information between institutions Research Infrastructure is dominated by only a few research groups, resulting in unauthorized ownership by those researchers Therefore Other researchers are unaware of the existence of research infrastructure relevant to their research And, Non-standardized rules and regulations prevent them from being utilized, and there are problems in gaining access
4	Low usage	Low utilization of equipment as a direct result of access restrictions
5	Weak monitoring	Equipment usage hours are not monitored, resulting in breakdowns ⇨ Difficult to implement preventive maintenance
6	No supply and demand data between research facilities	Supply and demand data on procurement and researcher demand is unavailable ⇨ No control to achieve optimized equipment usage based on supply and demand.

Source: Author´s elaboration

In response to this, a roadmap for policy improvement has been established and the following strategies are being implemented.

[Figure 4-1] Indonesia Research Infrastructure Problem Analysis and Solution



Source: BRIN’s internal data

BRIN's ELSA system was established in 2021 to efficiently utilize BRIN's research infrastructure and supports various functions for its utilization. As of 2024, 98 research facilities provided 7,633 services, and the cumulative usage amounted to 183,160 cases.

The ELSA system mainly provides reservation functions, expert consulting, and functions for research infrastructure cooperation and provides a total of 18 services in detail.

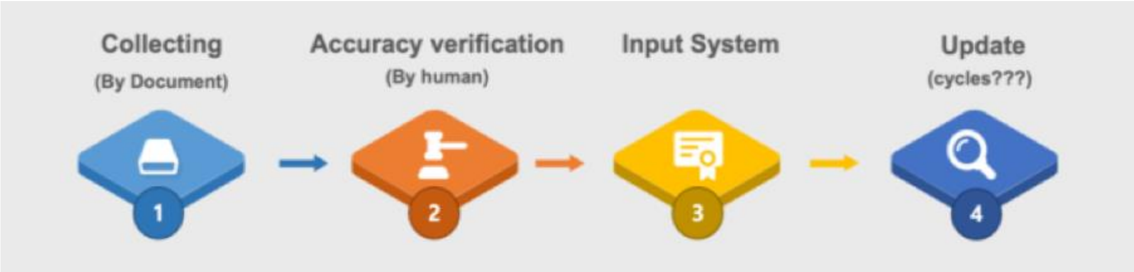
[Table 4-3] Analysis of ELSA System Provided Function

Menu(level 1)	Menu(level 2)	Menu(level 3)	Category
Facilities	research	Testing, Analysis, Measurement and Calibration	① Reservation
Facilities	research	Equipment and Machinery	① Reservation
Facilities	research	Production	① Reservation
Facilities	research	Onsite Service	① Reservation
Facilities	research	Customization Testing	① Reservation
Facilities	research	Radioactive Waste	① Reservation
Facilities	non-research	building	① Reservation
Facilities	non-research	room	① Reservation
Facilities	non-research	Other Non-Research Facilities	① Reservation
Facilities	non-research	Software Rental	① Reservation
Consultation	Research Expertise	-	② Expert Consulting
	Scientific Consultation	-	② Expert Consulting
	Scientific Speaker	-	② Expert Consulting
ETC	BRIN HPC	-	① Reservation
	Specimen	-	① Reservation
Tutoring	Training in Research Methodology and Scientific Writing	-	② Expert Consulting
Cooperation	Domestic Cooperation	-	③ Cooperation
	foreign cooperation	-	③ Cooperation

Source: BRIN’s internal data

ELSA has procedures in place to collect, database, and verify the consistency of BRIN-owned research infrastructure information that needs to be shared to provide these services. The research infrastructure information is collected in the form of documents, and a database, but the procedure for updating the information on a regular basis could not be identified. In addition, it was found that there is no specific information on how the research infrastructure is decided to be built and the specific information items for this.

[Figure 4-2] ELSA System Information Collection Procedure



Source: BRIN (2024)

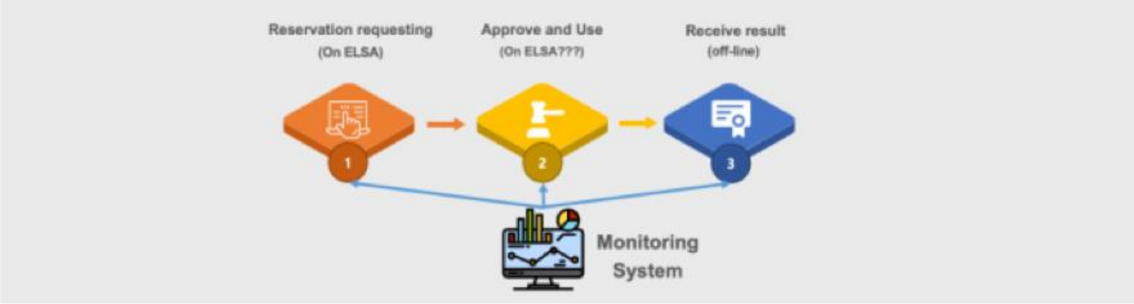
ELSA's service users are BRIN researchers, students, small medium enterprise (SME) researchers, and foreign researchers. - The reservation service provides services for BRIN researchers. It provides analytical services, direct equipment utilization services, and research prototyping.

- It also provides expert consulting services for students and SMEs, and international cooperation services for overseas researchers.
- To prevent duplication of research infrastructure, research infrastructure procurement users need to research information, and for policy making, various statistical services should be provided to policymakers and ministry officials, but these functions are not identified

■ **ELSA's research infrastructure utilization monitoring is understood to collect information on the research infrastructure's reservation, utilization, and satisfaction**

- ELSA has a system that supports the reservation and utilisation of research infrastructure, but it is not known whether it monitors what kind of research infrastructure is being built, what kind of users are accessing ELSA and utilising ELSA's research infrastructure, and how the research infrastructure currently registered with ELSA is operated and managed.

[Figure 4-3] ELSA System Monitoring Scope



Source: BRIN (2024)

To improve ELSA's system, it is necessary to analyze which users' data should be collected and how to use it to provide the system.

- Comparative analysis of Korea's ZEUS system with the current situation to derive complementary points for ELSA

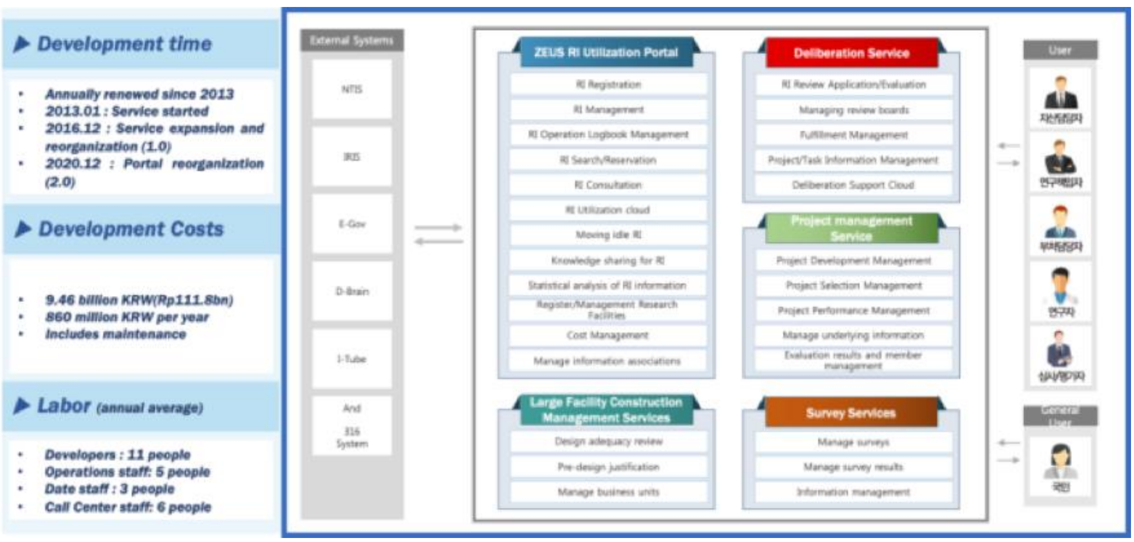
Korea's Zone of Research Equipment Utilisation Service (ZEUS) System, a research infrastructure information system, is being built and operated to promote efficient investment and systematic utilization of research infrastructure.

- ZEUS has been under construction and operation since 2013. It began operating to efficiently utilize the research infrastructure owned by the Korean government. It started with an initial 2,000 equipment reservations and currently provides more than 70,000 equipment reservations.
- In 2015, a deliberation service was established for efficient procurement of research infrastructure, and it was expanded and reorganized into a service that provides

information for reference when procuring research infrastructure and transformed into a comprehensive portal.

- In 2016, a survey system for checking the information status and utilization status of research infrastructure was established and applied to the system, and an online logbook and survey service was opened. In 2020, a construction management service for large-scale research infrastructure was opened, and in 22, a service for project management of large-scale research infrastructure was developed and provided.
- ZEUS provides services for a wide range of users, from simple users of research infrastructure to research directors who promote the construction of new research infrastructure, officials from various government departments involved in research infrastructure policy, and budget managers. It also promotes the utilization of excellent research infrastructure to the general public.
- It also provides research infrastructure-based performance information inquiry and management services by establishing a linkage system with various science and technology-related information systems.

[Figure 4-4] ZEUS system configuration in Korea



Source: ZEUS (2024)

The ZEUS system ultimately aims at the efficient execution of research infrastructure-related budgets and the efficient utilization of existing research infrastructure. It provides various functions that can be used as a model for the improvement of the ELSA system.

Comparative analysis of the goals of Indonesia's research infrastructure policy and the status of the ELSA system. The goals of the Indonesian research infrastructure policy can be summarised into six goals, which are consistent with the direction pursued by the Korean ZEUS system.

- The first policy objective of BRIN's research infrastructure policy is data-based procurement management, which aims to suppress duplication by using existing research infrastructure information as reference information for procuring new research infrastructure. To this end, ELSA must collect specific information on the existing research infrastructure and calculate quantitative statistical indicators, which can be used as reference information for the procurement of new research infrastructure. Therefore, the first ultimate policy direction of ELSA is to ensure that the annual research infrastructure procurement budget is efficiently spent.
- The second policy objective is to ensure the systematic maintenance of existing research infrastructure through centralized maintenance. This requires ELSA to continuously monitor the status of the research infrastructure operating in BRINs and to provide information that allows for immediate response to maintenance needs or breakdowns. Therefore, the second ultimate policy direction of ELSA is to reduce the downtime of the research infrastructure so that it can be utilized efficiently.
- The third policy objective is easy access to information, which is achieved by collecting comprehensive information on existing research infrastructure and making it openly available. To this end, ELSA should be able to systematically identify and provide information on research infrastructure to be registered and make it available through a booking system. Therefore, the third ultimate policy direction is to increase the openness of research infrastructure to increase its utilization.
- The fourth policy objective is to provide reservation and utilization services to maximize the use of open research infrastructure with high usage. To this end, ELSA should provide continuous and updated information on the utilization and availability of

available research infrastructure. Therefore, the fourth ultimate policy direction is to increase the frequency of use of the target equipment based on the high degree of openness of the research infrastructure.

- The fifth policy objective is to promote the procurement of preventive maintenance and response equipment through continuous monitoring of the utilisation of research infrastructure through integrated monitoring. To this end, ELSA should be able to identify the utilization of its research infrastructure in real-time, identify the specific uptime and usage against capacity, and provide information when the utilization limit is reached. Therefore, the fifth ultimate policy direction should be to continuously monitor the utilization of research infrastructure and provide results.
- The sixth policy objective is to identify new demand and supply for research infrastructure through efficient management. To do this, ELSA needs to identify the research infrastructure currently being procured and match it with the existing research infrastructure. Therefore, the sixth and ultimate policy direction is to ensure that the budget is appropriately matched to the demand and supply of research infrastructure.

The six policy objectives of ELSA are in line with the policy directions of ZEUS, and we aim to derive ways to improve ELSA through a functional comparison analysis with ZEUS.

[Table 4-4] Comparison of ELSA and ZEUS Goals

Policy Goal of ELSA	Ultimate policy direction	ZEUS Policy Goals
① Data-based procurement management	Efficient execution of procurement budgets	Efficient budget execution
② Centralised maintenance	Efficient utilization of research infrastructure by reducing downtime	Systematic utilisation
③ Easy access to information	Enhance utilizations by increasing the openness of research infrastructure	Systematic utilisation
④ High usage	Increase the frequency of use based on the high openness of research infrastructure	Systematic utilisation
⑤ Integrated monitoring	Continuous monitoring of research infrastructure utilizations	Efficient budget execution
⑥ Efficient management	Appropriately matching supply and demand to execute the budget	Efficient budget execution

Source: Author’s elaboration

Therefore, the gap analysis between the Korean ZEUS system and ELSA is as follows:

- (In terms of service functions) ZEUS has a total of 12 functions, of which 6 functions are related to the systemic goals of ELSA, providing a total of 21 services, and 3 services are matched with ELSA services.

[Table 4-5] Comparison of ELSA and ZEUS function

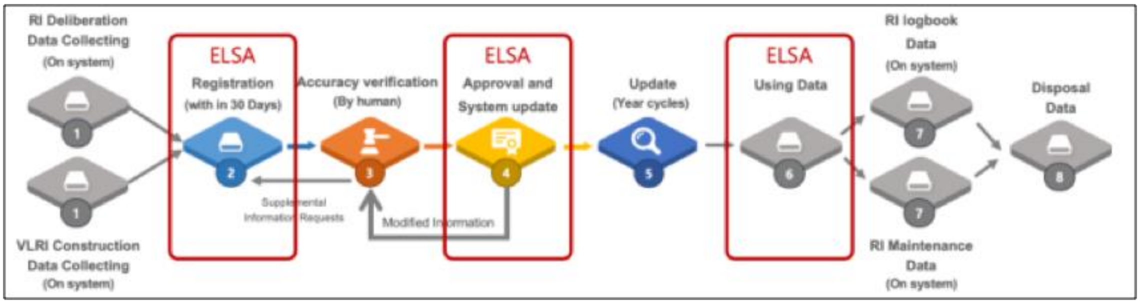
ZEUS - LEVEL1	ZEUS - LEVEL2	ELSA
RI Procurement Review	Online Deliberation Review Management	
	Managing review boards	
	Review Committee Management	
	Deliberation Fulfillment Check	
	Deliberation Process Management	
	Redundancy review	
Large Facility Construction Management Design Appropriateness	Design adequacy review	
	Managing professional committees	
	PM training	
	Customized Consulting	
RI Registration Management	About RI Online Registration	
	Registered Equipment Management	
	Equipment labour management	
	RI Reservation Search	Reservation
	SME RI reservation Service	
	Regional RI reservation Service	
	Institutional RI reservation Service	
	Cooperation Management	Cooperation
	Logbook management	
	Maintenance management	
RI Disposal	RI Sharing management	
	Disposal procedures management	
RI Training	Regular training	
	RI knowledge sharing	**Expert Consulting**
RI Statistics	Searching Statistics	
	Detailed Statistics Search	
	Theme statistics	
	Request statistical information	
Research Facilities Management	Research Facility Registration	
	Reservation management	
	Best practices for research facility	
	Overseas Research Facilities	
	National Research Facilities	
RI Operational Expense Management	Operating Expense Reserve	
	Manage operating expenses	
RI survey	Online surveys	
	Managing surveys	
	View survey statistics	
Research Facility Operations Support	Business management	
	Manage Booking Services	
	Facilities Operations Support	

Source: Author’s elaboration

(Information Collection) ZEUS collects all research infrastructure information from the procurement stage and provides research infrastructure information that can be jointly utilized as a reservation service and is continuously monitored.

- (STEP 1) All research infrastructure information built with the national research and development project budget is collected from the procurement stage, and based on this, all research institutions register the information when the construction is completed. At this time, we directly verify the integrity of the information and continuously request corrections for incorrect information.
- (STEP 2) Information that can be jointly used externally is provided to external users through the reservation service, and utilization information is collected and monitored in real-time.
- (STEP 3) We support the creation of a real-time logbook to collect information that is not used through the reservation service, and through this, we check the current operation status and identify whether it can be used in real-time.
- (STEP 4) Once a year, the entire data is checked and the actual availability is checked to identify the difference between the registered information and the actual information, and the registered information is continuously updated.
- (STEP 5) At the end of use, the information is registered and transferred so that it can be used elsewhere.

[Figure 4-5] ZEUS Information Collection System



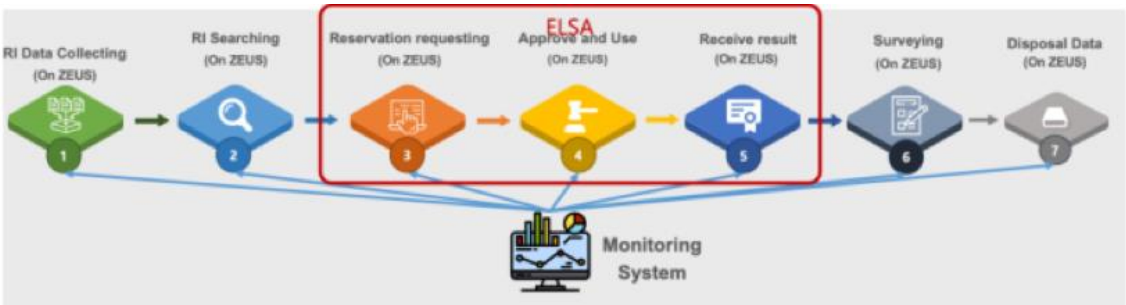
Source: Author’s elaboration

Among these ZEUS information collection procedures, information registration, information approval, and utilization information collection are functions provided by ELSA.

- (User side) ZEUS service users include general researchers, project managers of research institutes, research infrastructure operators, policy experts, and government department officials, and various services are provided for them.
- (Service monitoring aspect) ZEUS collects all information related to the entire life cycle, from real-time registration of research infrastructure information to the utilization of information, reservation status, utilization results, real-time information modification, and performance management.

The collection scope of ELSA is reservation application and utilization results, which are part of the ZEUS monitoring scope.

[Figure 4-6] ZEUS Monitoring System



Source: Author’s elaboration

The Korean ZEUS system systematically conducts efficient procurement, utilization, and monitoring of research infrastructure, which provides the following improvement implications for Indonesia's ELSA system.

In terms of improving the first information system function

- Strengthening the reservation system, ELSA's research infrastructure reservation function needs to be extended to researchers outside BRIN, small and medium enterprises (SMEs), and regional-level services to improve accessibility to a wider range of users.

- In terms of the logbook system, ZEUS supports real-time logging of equipment usage records to understand equipment status and availability, and ELSA needs to introduce a real-time logging function to enhance monitoring of equipment usage history and availability.
- As a maintenance management system, it is necessary to establish a preventive maintenance and immediate response system and minimize downtime by adding a function to provide information quickly in case of equipment failure.

The second area of improvement is the information collection and management system.

- The ZEUS information collection procedure collects information from the procurement stage so that all research infrastructure information is collected and verified from the beginning. Through this, information that can be jointly used is provided to external users as a reservation service, and real-time utilization information is collected to check operation and usage.
- In addition, we check the difference between the registered information and the actual information every year to update the data.
- Therefore, to improve the information system, ELSA also needs to collect all data from the procurement stage of research infrastructure and build a centralized database.
- As a regular data check process, it is necessary to add a process to check the registered information and actual utilization information once a year. In addition, it is necessary to establish a data transfer process at the end of use and introduce a transfer support function so that the equipment can be recycled by other organizations.

As the third part of strengthening the monitoring and integrated management function

- ZEUS monitors all information from reservation application to utilization status and performance in real-time and provides a function that warns when utilization data and equipment status reach a threshold.

- Therefore, to strengthen monitoring and integrated management, ELSA currently collects only reservation and utilization results, so it is necessary to establish a full-cycle monitoring system like ZEUS to comprehensively manage performance, uptime, etc.
- In addition, preventive maintenance scheduling is introduced by analyzing equipment usage patterns.

Fourth, expanding services and improving accessibility for users

- ZEUS provides tailored services to a wide range of users, including researchers, project managers, policymakers, and the public.
- Therefore, it is necessary for ELSA to expand the scope of services not only to researchers and government officials but also to industry and the general public to promote its usability and improve service convenience by providing dashboards tailored to the needs of each user (researchers: reservation service, policymakers: policy reports, etc.

Fifth, in terms of supporting the establishment, development, and performance evaluation of research infrastructure.

- There is a need to introduce a project performance evaluation system in ELSA to monitor the performance of infrastructure utilization and to streamline the development and evaluation of new projects.

Sixth, to streamline procurement and budgeting.

- ZEUS seeks to curb duplication of investment and improve the efficiency of research infrastructure budget execution through integrated management and utilization of infrastructure information.
- Therefore, it is necessary for ELSA to establish budget efficiency functions, such as strengthening the ability to prevent duplicate construction in new procurement through utilization analysis of existing equipment and budget execution management tools that effectively match supply and demand.

The objectives and strategies to improve the Indonesian ELSA system through policy recommendations and comparative analysis of systems are as follows :

First, Efficient research infrastructure procurement and budget execution

- (Goal) Prevent duplication and optimize the budget.
- (Detailed strategy) Provide new procurement information based on existing research infrastructure data and introduce a demand-supply matching system.

Second, Maximizing the utilization of research infrastructure

- (Goal) Maximize the availability and frequency of use of research infrastructure.
- (Detailed strategy) Expansion of SME and regional reservation systems and introduction of real-time utilization monitoring.

Third, Establishing a preventive maintenance system

- (Goal) Minimize downtime and improve equipment utilization.
- (Specific strategies) Introduce real-time logbook management and predictive maintenance systems.

Fourth, Data-based performance management and evaluation

- (Goal) Performance analysis and policy use of infrastructure utilization results.
- (Specific strategies) Introduce performance evaluation tools and automate reports related to research infrastructure.

Fifth, User accessibility and service expansion

- (Goal) Provide customized services for researchers, government officials, and the public.
- (Specific strategies) Provide customized dashboards and develop performance communication services for the public.

3. Information System Master Plan Configuration

Information System Master Plan (ISMP) refers to a planning activity that analyses the current situation and requirements for business and information technology, describes functional and technical requirements in detail to the extent that functional scores can be derived, and establishes an implementation strategy and implementation plan to prepare a detailed analysis and Request for Proposal (RFP) for the development of a specific information system.

The purpose of ISMP is to increase the likelihood of success of information system development projects through systematic planning, clarify the scope of the project, enable efficient resource management by predicting the required resources, minimize business risks by identifying potential problems in advance, and build consensus on business goals through clear communication between stakeholders.

The main contents of the ISMP are as follows

- (Identify the current situation) Analyze the current policy and information system environment and identify problems and improvements
- (Define requirements) Specify the requirements for the new information system in terms of functional and non-functional requirements.
- (Establishment of implementation strategy) Decide how to build the information system, development methodology, system architecture, etc.
- (Establishment of implementation plan) Establish project schedule, budget, manpower plan, etc.

■ **The main considerations for establishing an ISMP are as follows**

- (Alignment with business goals) The ISMP should be clearly linked to the business goals to be achieved through the construction of the information system.
- (Stakeholders) The opinions of relevant stakeholders should be actively reflected in the establishment process.
- (Latest technology trends) Design the information system in consideration of the latest IT technology trends.
- (Maintenance:) Consider how to maintain and operate the system after it is built.

In general, ISMP goes through 7 steps

- (Step 1) Define the Scope and Direction of the Information System
- (Step 2) Business and IT Status Analysis
- (Step 3) Business Requirements Analysis
- (Step 4) Information Technology Requirements Analysis
- (Step 5) Define Information System Architecture
- (Step 6) Analyze the Implementation Linkage between Requirements
- (Step 7) Preparation of Information System Requirement Documentation

Based on this, a framework for BRIN is proposed.

Information System Guidelines for S&T Research Infrastructure (ISMP) Framework

1. Introduction
    A. Purpose and necessity of guidelines

2. Purpose and Objectives of Research Infrastructure Management and Utilizations Policy
    A. Goals and Status of Research Infrastructure Management and Utilizations Policy
        I. Definition of Indonesian Research Infrastructure
        II. Definition of Life Cycle of Research Infrastructure in Indonesia
        iii. Status of policy targets
    B. Objectives of BRIN
        i. Procurement System
        ii. Utilizations system
        iii. Maintenance system
        iv. Monitoring system

3. Information System Improvement Direction to Improve Research Infrastructure Management Utilizations System in Indonesia
    A. Analyze the current status of information system and derive improvement directions
    B. Establish ISMP for optimization
        i. Scope and direction of the next ELSA system
        ii. Current ELSA system status
        iii. Mission requirement analysis
        iv. Information technology requirements analysis
        v. Next ELSA System Architecture Definition
        vi. Requirements Correlation
        vii. Next System Requirements Statement

4. future plans

By this year, the status and problem analysis of the current information system has been completed through field interviews and researchers' workshops, and based on this, the information system improvement plan for the next generation ELSA will be derived in the third year.

CHAPTER 5

Draft Guideline for Indonesia's Research & Innovation Funding Scheme

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Chapter 5. Draft Guideline for Indonesia’s R&I Funding Scheme

1. Introduction

BRIN, which was established in April 2021 as a cabinet-level independent government agency directly under the President, is responsible for policy formulation, research implementation, and research management of S&T and national innovation in Indonesia. However, there is no comprehensive guideline to serve as a basis for BRIN ’s mission of research and innovation (R&I) funding and there is an urgent need to develop guideline.

Therefore, at the March 2024 BRIN-STEPI Study Team Meeting (held at BRIN, Jakarta, Indonesia), it was decided to develop guideline for a research funding scheme that reflects the needs and characteristics of Indonesian R&I, including measures to increase industry participation in R&D. We plan to develop draft guideline in 2024 and final guideline in 2025.

To develop the guideline, we first surveyed the main areas of interest of Directorate of R&I Funding in BRIN. Directorate of R&I Funding proposed three areas: improvement of BRIN's R&I funding scheme, improvement of BRIN's R&D project management system, and establishment of measures to encourage Indonesian industry's participation in R&D.

Considering the three areas proposed by Directorate of R&I Funding, the main contents of the R&I Funding Guideline are organized as follows.

- Summarize laws and policies related to R&I funding and BRIN's mission based on them
- Summarize the current status and operation system of R&D funding
- Identify issues and challenges related to BRIN's R&I funding system and suggest directions for improvement based on Korea's advanced R&D funding experience

- Identify issues and challenges related to BRIN's R&D project management system and suggest directions for improvement based on Korea's advanced R&D funding experience
- Summarize the current status and issues of Indonesian industry's participation in R&D and suggest future directions for activating industry's participation in national R&D.

This guideline is provided to BRIN, the national R&D funding, R&D project management, and research implementation agency, and can be applied to industry, academia, and research institutes and researchers whose R&D is supported and managed by BRIN.

2. Draft Guideline for R&I Funding Scheme

2.1 Key laws and policies related to R&I funding and BRIN'S mission based on them

The main laws and policies related to R&I funding in Indonesia and their highlights are as follows.

"The 2017-2045 National Research Master Plan (RIRN)" assigns BRIN the responsibility to actively respond to the issue of low critical mass in the Indonesian R&I sector.

"Law Number 11 of 2019 concerning the national Science and Technology system (The Sisnas lptek law)" aims to increase the utilization of science and technology for sustainable national development, quality of life, and public welfare. It states that science and technology is developed through the national science and technology as the foundation and integral part of the national development planning system.

In "The Presidential Regulation Number 78 of 2021 on the National Research and Innovation Agency", BRIN is required to promote R&D by implementing a number of funding schemes, including the RIIM program. The RIIM Program is an R&I program for Advanced Indonesia. In addition, BRIN should provide support to all groups that want to actively participate in the process of building the R&D ecosystem and recognize the importance of creating R&D

collaborations with partners to advance science and technology.

"The 2020-2024 National Medium-Term Development Plan (RPJMN)" states that BRIN should provide technical and administrative support as well as rapid, accurate, and responsive analysis to the President and Vice President in conducting research, development, assessment, and application, as well as invention and innovation, national nuclear energy management, national space management in an integrated manner, while also monitoring, controlling, and evaluating the execution of BRIDA's tasks and functions. And BRIN should enhance the quality of human resources and infrastructure for research and innovation in national nuclear energy and space management in an integrated manner, and foster the execution of BRIDA's tasks and functions. In addition, BRIN shall deliver efficient and effective services in the fields of supervision, general administration, information, and institutional relations.

The "National Research and Innovation Agency Regulation No. 1 of 2021" comprehensively lays out the functions of BRIN.

As a national agency directly under the President and responsible for research, planners, policy makers, implementers, and facilitators of research, including infrastructure and funding (funding agency) for the implementation of the National Development Plan, BRIN must perform the following tasks. First, it should formulate and promote Indonesia's R&I policy and conduct integrated research, development, evaluation and application, invention and innovation, nuclear energy management, and space management. Then, it should integrate planning system development, programs, budgets, institutional frameworks, and resources for research, development, evaluation, application, invention, and innovation. It should also provide facilitation, technical guidance, mentoring, supervision, monitoring, and evaluation in the areas of research, development, evaluation, application, invention, and innovation.

A key role of BRIN should be to actively address the issue of low critical mass in the Indonesian R&I sector, and aim to change the composition of private sector participation in Indonesian research from the current government-dominated situation to align with the global research landscape. BRIN should undertake various efforts to reform research management in line with global norms and standards, improve the quality of human resources to accelerate research capacity and competitiveness, and strengthen its role as a provider of national research infrastructure (HW/SW) and a hub for S&T collaborative activities open to all sectors, including industry, academia, and research institutes. It should also support the optimization of funding for R&D activities and strengthen the recognition of discoveries to improve the R&D ecosystem. BRIN should aim to increase the number of innovative products from research-based start-up companies fostered by BRINs, the number of innovations exploited by the industry-academia community, and the number of patents, and increase collaboration on the development and exploitation of scientific outputs based on these research and innovation results.

2.2 R&D Funding Status and Structure

The current status and composition of R&D funding in Indonesia is as follows. In 2021, the national R&D budget was IDR 41.43T, accounting for 0.28% of Indonesia's GDP. Of the national R&D budget, the public sector contributed IDR 36.85T (88.95%) and the private sector contributed IDR 4.57T (11.03%).

In 2021, the government R&D budget was IDR 34.69T, accounting for 0.23% of the total government expenditure. According to KSI (2019), the government R&D budget was composed of 81.58% by the central government and 2.16% by local governments. According to Katadata (2018), the government R&D budget was composed of 80.97% by the central government, 2.91% by local governments, 2.65% by universities, 9.15% by manufacturing industries, and 4.33% by private companies.

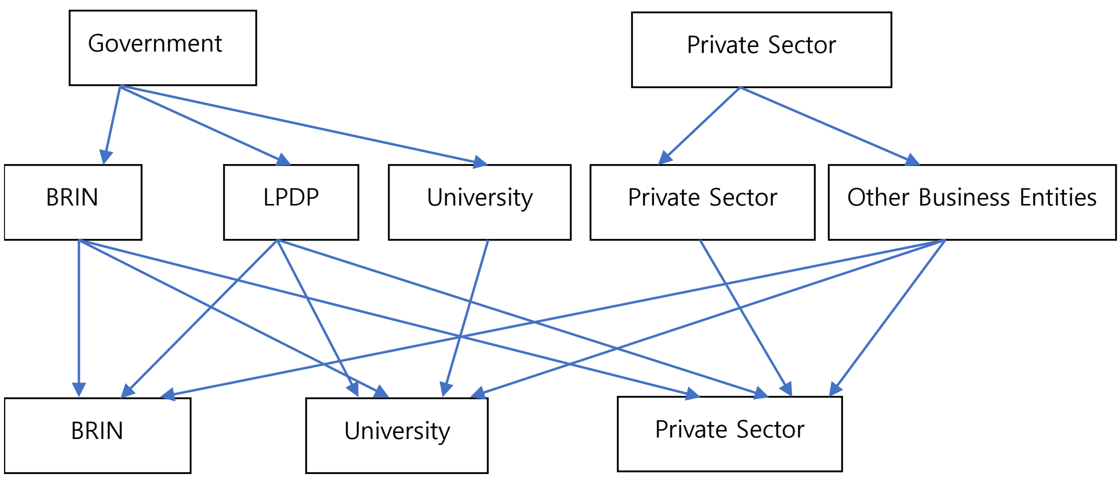
Let's summarize the roles of the agencies involved in government R&D budgeting.

Bappenas is responsible for program and activity planning and budgeting, but limited to indicative budgeting. The Ministry of Finance (MoF) plans and determines the final budget allocation and coordinates the program plans of various ministries and agencies. The Budget Directorate within the MoF is responsible for reviewing and approving budget plans submitted by ministries and agencies. BRIN is responsible for managing research infrastructure, human resources, and research funding. In addition, each ministry and agency is required to develop and submit a budget plan annually to the MOF.

Major funding agencies for R&D include the Indonesia Science Fund (ISF) (In Indonesia is called DIPI) and the Palm Oil Plantation Fund Management Agency (POPFMA) (In Indonesia is called BPDPKS). Indonesia Science Fund (ISF) is an independent research funding agency under the Indonesian Academy of Sciences (IAoS) (In Indonesia is called AIPI) that aims to improve the national research ecosystem through competitive, flexible, and sustainable research funding. The Palm Oil Plantation Fund Management Agency (BPDPKS) is an operational agency under the Ministry of Finance that manages the Palm Oil Plantation Fund and manages the fund in accordance with the policies set by the Steering Committee while considering government programs.

Indonesia's R&D budget comes from the National Revenue and Expenditure Budget (NREB) or it called as APBN and the private sector. The government R&D budget funds researchers through BRIN, LPDP (Indonesian Endowment Fund for Education Agency), and universities, while the private sector funds various research projects required by industry.

[Figure 5-1] Indonesia's R&D Budget Flow



Source: BRIN (2024)

The National Research Plan (RIRN), The National Medium-Term Development Plan 2020-2024, and The National Long-Term Development Plan are some of the policies that should be considered in the government's R&D budgeting process.

2.3 Improvements to BRIN’s R&I Funding System

2.3.1 Current Status

Let's take a look at the current status and composition of Indonesia’s R&D funding. In 2021, the national R&D budget was IDR 41.43T, accounting for 0.28% of Indonesia's GDP. Of the national R&D budget, the public sector contributed IDR 36.85T (88.95%) and the private sector contributed IDR 4.57T (11.03%). In 2021, the government R&D budget was IDR 34.69T, accounting for 0.23% of the total government expenditure. According to KSI (2019), the government R&D budget was composed of 81.58% by the central government and 2.16% by local governments. According to Katadata (2018), the government R&D budget was composed of 80.97% by the central government, 2.91% by local governments, 2.65% by universities, 9.15% by manufacturing industries, and 4.33% by private companies.

Let's take a look at the current status and detailed composition of Indonesia’s R&D funding. In 2022, national R&D expenditure was IDR 17.74T, accounting for 0.1% of Indonesia's GDP. National R&D expenditure was composed of IDR 11.74T (66%) from government (BRIN/LocalGov't/LPDP), IDR 3.03T (17.1%) from universities, and IDR 2.96T (16.7%) from industry/enterprises. Government R&D expenditure amounted to IDR 11.74T, accounting for 0.06% of GDP, composing of central government (BRIN) 10.51T IDR (89.5%), local government 0.98T IDR (8.4%), and LPDP 0.25T IDR (2.1%).

(※ Source : Book of Indicators of Science, Technology, Research, and Innovation of Indonesia, BRIN, 2023)

In 2023, national R&D expenditure (2023) was IDR 20.39T, accounting for 0.1% of Indonesia's GDP. National R&D expenditure was composed of IDR 6.61T (32.4%) from government (BRIN/LocalGov't/LPDP), IDR 9.39T (46.0%) from universities, and IDR 4.40T (21.6%) from industry/enterprises. Government R&D expenditure amounted to IDR 7.93T, accounting for 0.04% of GDP, composing of central government (BRIN) 6.36T IDR (80.2%), local government 1.16T IDR (14.7%), and LPDP 0.41T IDR (5.2%).

(※ Source : Book of Indicators of Science, Technology, Research, and Innovation of Indonesia, BRIN, 2024)

In 2023, the original R&D budget allocated to the BRIN was 6.35 T IDR, with a research budget of 2.43 T IDR and a management budget of 3.91 T IDR. However, the actual R&D budget managed by BRIN was IDR 5.17 T, of which the research budget was IDR 1.62 T and the management budget was IDR 3.54 T. The actual R&D budget managed by BRIN is composed of BRIN (DFRI) 32B IDR (3%), APBN 439B IDR (70%) managed by LPDP, and cooperation with foreign funding organizations 100B IDR (27%).

(※ Sources: Yearly Report of BRIN 2023)

Initially, the budget provided by LPDP to BRIN was IDR 1.06T, but the actual R&D budget managed by BRIN was IDR 633B, which was 59.4% of the budget provided by LPDP, and the actual budget spent by BRIN was IDR 117B, which was only 11% of the budget provided by LPDP.

2.3.2 Issues and Challenges

(Issue 1) The size of the national R&D budget as a percentage of GDP is very low, and the size of the government R&D budget is also very low as a percentage of total government expenditure.

※ The case of South Korea

In 2021, the national R&D budget was USD 89.282M, accounting for 4.96% of GDP. The national R&D budget is composed of USD 23.856M (26.7%) from the public sector and USD 65.426M (73.3%) from the private sector. The government R&D budget was USD 23,856M, accounting for 4.9% of total government expenditure.

(Issue 2) There is a large gap between the budget allocated to BRIN and the actual amount of budget managed and utilized by BRIN.

(Issue 3) There is no evidence of policies or regulations that specify the governance of research and innovation funding. And how well are R&D investment priorities aligned with the direction of national science and technology policy in Korea? For reference, since 2023, BRIN is drafting major BRIN regulations on research and innovation funding through the

Directorate of Research and Innovation Funding, which is expected to serve as the regulatory basis for the national research and innovation management system.

※ The case of South Korea

In South Korea, there are two laws that provide the basis for budgeting for government R&D programs: the National Finance Act and the Basic Act on Science and Technology.

The National Finance Act includes the establishment of the national financial management plan and the budget for the next fiscal year. The Ministry of Planning and Finance (MoF) is required to establish a "National Financial Management Plan" for a period of next five years in order to efficiently and soundly manage the country's finances and submit it to the National Assembly 120 days before the start of the fiscal year, and to present a financial allocation plan and investment direction for each of the 12 major sectors. The Ministry of Planning and Finance must also notify the ministers of each ministry by March 31 of each year of the budget formulation guidelines and spending limits for each ministry in order to prepare the next year's budget. The ministries must submit their budget requests for the next fiscal year to the minister by May 31 of each year, and the MOF must submit the budget to the National Assembly no later than 120 days before the start of the next fiscal year.

The Basic Act on Science and Technology includes the establishment of a mid- to long-term investment strategy for national R&D and the allocation and coordination of budgets for major national R&D programs. The Ministry of Science, ICT and Future Planning (MSIP) is required to establish a five-year national R&D mid- to long-term investment strategy for strategic investment of the national R&D budget. The MSIP includes the mid- to long-term investment goals and directions of national R&D, the allocation of investment resources by field and stage of national R&D, and the division of roles between national R&D programs and private R&D programs. In addition, the MSIP must announce the direction and criteria for national R&D investment by March 15 of each year in order to allocate and adjust the budget for major programs in the national R&D program. All ministries must submit the next year's budget requests to the MoF by May 31 of each year, and the Minister of Science and ICT must notify the MoF by June 30 of each year of the results of the next year's budget allocation and adjustment for major programs among national R&D programs.

(Issue 4) BRIN's working relationship with Bappenas is unclear. All deliverables from the RIIM program managed by BRIN are supposed to contribute to the development planned by Bappenas. However, there is no legal basis for Bappenas' involvement in the funding program (RIIM), and the pattern of coordination of the RIIM program by Bappenas is not clear. The lack of direct involvement of Bappenas in the utilization of research outputs generated from the RIIM Scheme raises questions about the RIIM research outputs and their provenance. The lack of a clear relationship between RIIM research outputs and Bappenas' planned development initiatives calls into question the effectiveness and validity of RIIM funding in relation to the purpose of the research outputs.

2.3.3 Recommendations for improvement

(Issue 1) The size of the national R&D budget as a percentage of GDP is very low, and the size of the government R&D budget is also very low as a percentage of total government expenditure.

As a way to improve the first issue, it is necessary to adopt science and technology-based national development as a national priority policy, continuously increase the size of the government R&D budget every year, and strengthen support for global engagement and collaborative R&D.

The S&T Law Number 11/2019 aims to increase the utilization of science and technology for sustainable national development, quality of life, and public welfare, and suggests that science and technology should be developed through the national science and technology system as the foundation and comprehensive part of the national development planning system. Therefore, based on this content, S&T-based sustainable national development should be adopted as the top national policy and reflected in major policies. In addition, the sustainable national development based on S&T adopted as the top national policy should be reflected in major policies such as the National Research Plan (RIRN), The National Medium-Term Development Plan 2020-2024, and The National Long-Term Development Plan.

※ The case of South Korea

In 1962, the government established the First Five-Year Plan for Technology Promotion to reflect the recognition that science and technology are essential for national economic development.

In 1973, the government declared the industrialization of heavy chemicals as a top policy priority and promoted a policy of introducing foreign technology while strengthening domestic research and development. The government also enacted the Technology Development Promotion Act (1972) to promote research activities of private companies, encouraged private companies to build research facilities, commercialized new technologies, and promoted the protection of domestic new technology products. With these active government supports, industry's technology development activities increased in the late 1970s.

In the 1980s, the government promoted the development of core strategic technologies through the Technology Drive Policy based on economic growth. The President himself presided over the Technology Promotion Expansion Meeting, and more than 200 people, including members of the State Council, politicians, and representatives of industry, academia, and research institutes, participated.

Based on these policies, mid- to long-term targets for national R&D (as a percentage of GDP) and government R&D budget (as a share of total government expenditure) should be set and continuously reflect them every year. For example, a target could be set to increase the size of the national R&D budget as a percentage of GDP from 0.28% in 2021 to 0.10% in 2023, and to increase the share of the government R&D budget in total government expenditure from 0.23% in 2021 to 0.03% in 2023.

※ The Case of South Korea

The government has continuously increased the government R&D budget every year to build a science and technology-centered society and strengthen national competitiveness. The share of government R&D budget in total government expenditure increased from 3.6% in 2004 to 5.0% in 2014, and the year-on-year growth rate of government R&D budget was more than 10% until 2010.

In addition, R&I funding schemes should be developed in areas that are of interest to foreign countries and where Indonesia has strengths, such as adding value to local natural resources and (biological, resource, and cultural) diversity, and support should be strengthened to link with global collaborative R&D and accelerate the transfer of relevant knowledge and technologies through such collaborative research. For example, R&I funding schemes could be developed that link the development and export of key strategic natural resources with support for overseas technical cooperation. In addition, R&I funding schemes can be developed that link overseas technical cooperation support with CCS, hydrogen supply, and other areas related to carbon neutrality.

(Issue 2) There is a large gap between the budget allocated to BRINs and the actual budget managed and used by BRINs.

As an improvement direction for the second issue, the gap between the two budgets should be reduced as much as possible by analyzing the causes of the gap and devising solutions, and reflecting the results in the next year's budget.

For this purpose, it is necessary to analyze the results of the executed budget compared to the budgeted budget every year, and analyze the causes of the gap between the two budgets and devise a solution. In addition, the results of this analysis and solutions should be fully reviewed, and the results should be reflected in the next year's budget to reduce the gap between the two budgets as much as possible.

(Issue 3) There is no evidence of policies or regulations that specify the governance of research and innovation funding. And how well are R&D investment priorities aligned with the direction of national science and technology policy in Korea?

As a way to improve the third issue, it is necessary to reflect the main functions of national R&I funding in relevant laws or regulations and develop R&I funding support and management regulations including the overall role and functions of BRIN's R&I funding.

First, it is necessary to reflect the main functions of R&I funding at the national level in relevant laws and regulations. The next year's R&D funding is efficiently promoted in conjunction with the mid- to long-term investment strategy direction, and the next year's national R&D investment direction should be presented at the beginning of each fiscal year to match the strategic investment direction at the national level with the major research directions of relevant research institutes. National R&D funding priorities should be well aligned with the direction of national science and technology policy. To accomplish this goal, the implementation entities and timing for the establishment of the national R&D mid- to long-term investment strategy and the establishment of the next year's R&D investment direction are necessary to reflect in relevant laws and regulations.

In addition, BRIN's overall role and functions related to R&I funding should be reflected in the R&I funding support and management regulations being developed by the Directorate of R&I Funding within BRIN since 2023. It is necessary to draft detailed regulations on the steps and procedures related to BRIN's R&I funding function reflected in laws and regulations, the implementation system and roles, and details of major functions. The developed draft regulations should be finalized after sufficient coordination with relevant ministries such as Bappenas and MoF. This is expected to be the regulatory basis for the national R&D management system.

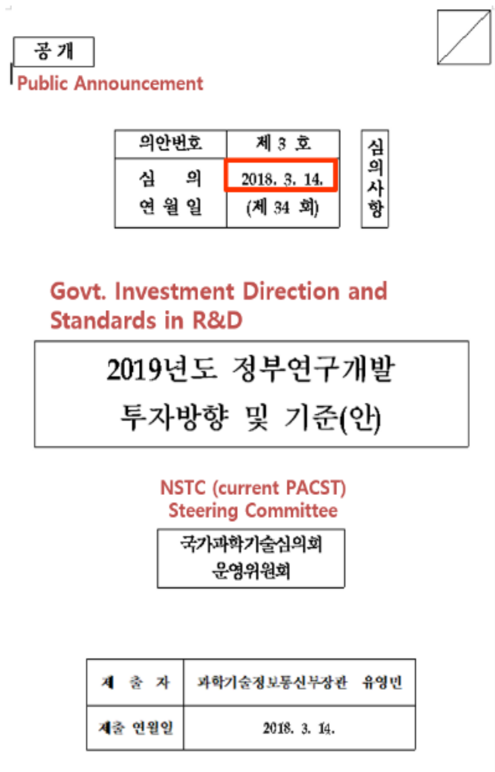
In addition, the national S&T policy should be well connected to the R&D investment direction and priorities. When setting R&D investment directions and priorities, it should be well coordinated with the National Research Plan (RIRN), the National Medium-term Development Plan 2020-2024, and the National Long-term Development Plan.

※ The case of South Korea

In South Korea, key features of R&D funding are reflected in the law.

The National Finance Act reflects the need for a financial resource allocation plan and investment direction for each sector, including R&D, for the next five years.

The Basic Act on Science and Technology requires the establishment of a five-year national R&D mid- and long-term investment strategy for strategic investment of the national R&D budget, and the direction and standards for national R&D investment must be announced by March 15 of each year. The following are sample cover pages for the next year's government R&D investment direction and standards agenda.



공 개
Public Announcement

의안번호	제 3 호
심 의 연 월 일	2018. 3. 14. (제 34 회)

심의사항

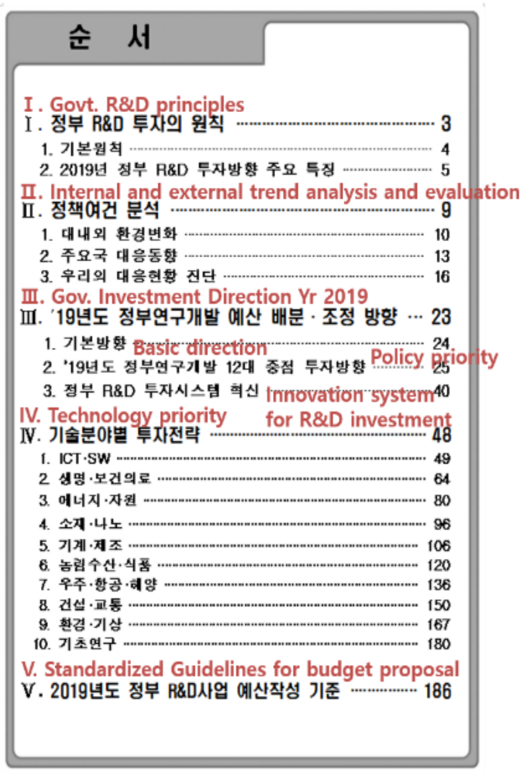
Govt. Investment Direction and Standards in R&D

2019년도 정부연구개발 투자방향 및 기준(안)

NSTC (current PACST) Steering Committee

국가과학기술심의회 운영위원회

제 출 자	과학기술정보통신부장관 유영민
제출 연월일	2018. 3. 14.



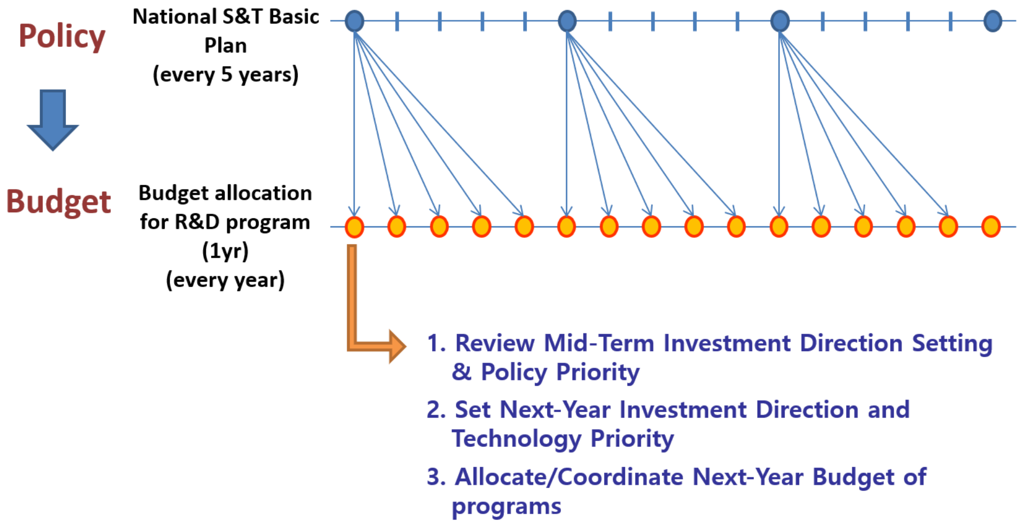
순 서

※ This report can be downloaded at http://www.pacst.go.kr. Korean version is only available.

※ The case of South Korea

The picture below shows the relationship between S&T policy and R&D budget.

Each government establishes the National Science and Technology Basic Plan, the top-level policy for science and technology, every five years, and each year, the government R&D budget is formulated while reflecting the direction of the National Science and Technology Basic Plan. Budgeting for R&D programs includes setting the mid-term investment direction and reviewing policy priorities, setting the next year's investment direction and technology priorities, and budgeting and adjusting the next year's programs.



(Issue 4) BRIN's working relationship with Bappenas is unclear.

As an improvement to the fourth issue, the link between the development planned by Bappenas and the support and management of the related funding program (RIIM) should be clarified. For this purpose, the relationship between the development planned by Bappenas and the support and management of the RIIM and the scope of Bappenas' involvement should be more specifically reflected in the relevant laws and regulations. In addition, the scope of involvement, including the roles and responsibilities of Bappenas and BRIN in supporting and managing the RIIM program, should be reflected in BRIN's regulations for supporting and managing research and innovation funding, which are currently under development.

2.4 Improvements to BRIN’s R&D Project Management System

2.4.1 Current Status

BRIN's R&D project management procedures are as follows.

First, a call for proposals is issued. The Directorate of Research and Innovation Funding issues a call for proposals for funding schemes that are managed according to a predetermined schedule. This call for proposals is intended to collect research proposals from researchers across Indonesia on predefined themes. The proposals received will then undergo an administrative and substantive selection process.

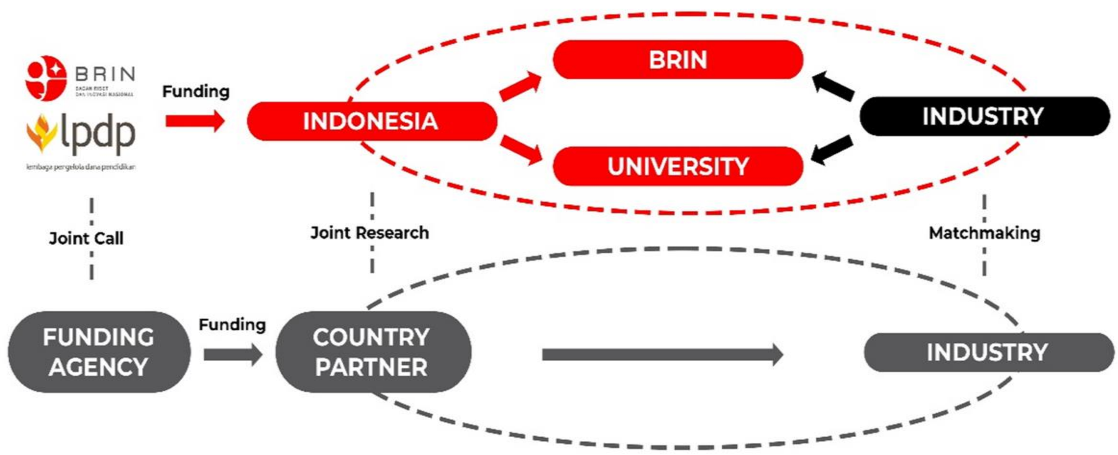
The administrative selection involves document verification, including checking the proposal's formatting and ensuring the completeness of the submitted files. The substantive evaluation is conducted by reviewers and/or an assessment committee appointed by the Research and Innovation for Advanced Indonesia (RIIM) program organizer. The substantive evaluation aims to assess the novelty of the research theme, its scientific merit, and the methods proposed to solve the problem. The substantive evaluation covers the following aspects:

A. Research planning, including the quality of the research team's track record, qualifications, reputation, and consistency of research experience in the relevant field, such as their education, work, and research activities; the background of the research and the problem being addressed; and the research roadmap;
B. The research framework, including data collection methods, testing, measurement, and analysis of research results;
C. The quantity and quality of outputs each year, which should be clearly stated and measurable both quantitatively and qualitatively, as well as the potential for research sustainability.

Next, the budget proposal evaluation is conducted by an evaluation team from the Directorate of Research and Innovation Funding at the National Research and Innovation Agency. This budget assessment aims to recommend the amount of funding to be allocated to each proposal.

The final recommendations are reviewed in a plenary leadership meeting to determine the recipients of the RIIM funding. The RIIM funding recipients are then formally announced through the issuance of a Decision Letter.

[Figure 5-2] BRIN's R&D Project Management Procedure



Source : BRIN (2024)

Monitoring and evaluation (Monev) can be carried out in two ways: direct and indirect. Direct monitoring and evaluation are carried out for researchers by the funding manager (BRIN). Meanwhile, indirect monitoring and evaluation are carried out for researchers by the funding recipient agency; then, the agency is monitored by the funding manager (BRIN).

Monitoring and evaluation of research and innovation funding includes substantive and financial aspects. In the substantive aspect, the performance of research activities is measured, including:

A. Ensure the output is produced in terms of type, quantity and status;
B. Calculate the percentage of output produced compared to the target in the contract/Cooperation Agreement;
C. Provide recommendations based on the implementation of research activities and the percentage of target achievement.

Meanwhile, financial report evaluation is carried out by the Research and Innovation Funding Directorate team to measure financial performance, which includes:

Confirm the type of expenditure that has been made;

A. Ensure the conformity of the documents presented with the provisions for each type of expenditure;

B. Calculate the amount of funds disbursed that do not comply with the provisions (potential for refund) And

C. Ensure whether there are funds that have been used at the end of the contract, and ensure the amount of funds that have not been used are returned.

2.4.2 Issues and Challenges

(Issue 1) System interoperability issues between BRIN and LPDP can cause significant delays in the planned disbursement of funds.

The use of the two systems for disbursement management requires smooth and secure interoperability, and the challenge is that the synchronization and data interoperation process between the two systems requires a significant amount of time.

(Issue 2) There have been multiple policy changes including the areas covered by program/scheme objectives, which delays the competition process and research activities, so countermeasures are needed. Also, how does Korea respond to changes in the S&T policy environment?

For example, the RIIM Competition Program was initially able to support all fields, but the selection process was delayed due to the exclusion of some fields from the selection process. This can also make it difficult for researchers to design long-term research.

(Issue 3) This issue is related to the quantitative and qualitative limitations of human resources. Excellent managers are needed to manage R&D projects, and the number and expertise of review experts participating in the selection evaluation and monitoring and evaluation stages of research projects are limited.

BRIN conducts R&D project management through the Directorate of Research and Innovation Funding. In 2023, 43 human resources will be assigned to the Directorate of R&D Funding to manage eight R&D funding schemes, which is not enough to manage R&D funding schemes with large budgets. In addition, the management of R&I funding schemes is conducted through an online system (website-based), so the staff in charge of managing the Directorate of R&I Funding must be skilled in this system.

The number and expertise of reviewers required at this stage is still limited, as reviewers are involved in the selection and evaluation of research proposals and the monitoring and evaluation of research activities. This makes the proposal selection process quite time-consuming, and as the number of proposals increases, the time required for evaluation will also increase. The limited number of reviewers is an obstacle to the evaluation process, which should take place immediately at the end of the research activity.

2.4.3 Recommendations for improvement

(Issue 1) System interoperability issues between BRIN and LPDP may cause significant delays in the planned execution of funds.

To address the first issue, BRIN needs to maintain a close working relationship with LPDP in order to smoothly manage the RIIM program funded by the research endowment fund managed by LPDP, and the interoperability of the two agencies' management systems needs to be improved.

In the short term, the two organizations should work together to develop an operational manual to ensure maximum compatibility between their management systems. In the long term, it is necessary to develop an integrated system that can be jointly operated by both organizations.

(Issue 2) There have been multiple policy changes including the areas covered by program/scheme objectives, which delays the competition process and research activities, so countermeasures are needed. Also, how does Korea respond to changes in the S&T policy environment?

As a way to improve the second issue, decisions and subsequent changes to program/scheme objectives should be confirmed in writing and made public through formal procedures.

Program objectives should be confirmed in a formal document through a letter between relevant agencies after the decision is made and made public as necessary, and task management should be conducted through formal management channels based on this letter. If policy changes are required, the relevant organizations should request them in writing, and the results of the decision should be approved internally by BRIN, responded to the requesting organization in writing, and disclosed externally as necessary.

※ The case of South Korea

In Korea, the R&D investment direction and investment strategy for the next year are provided in March every year through analysis of domestic and international environmental changes and policy situations. Considering these results, R&D investments in related fields are strengthened for the next year, and each program also conducts detailed research planning in related fields in consideration of this direction and announces the results, including the fields to be supported and the amount of support.

(Issue 3) This is an issue related to the quantitative and qualitative limitations of human resources.

As a way to improve the third issue, investment and management of human resources should be prioritized.

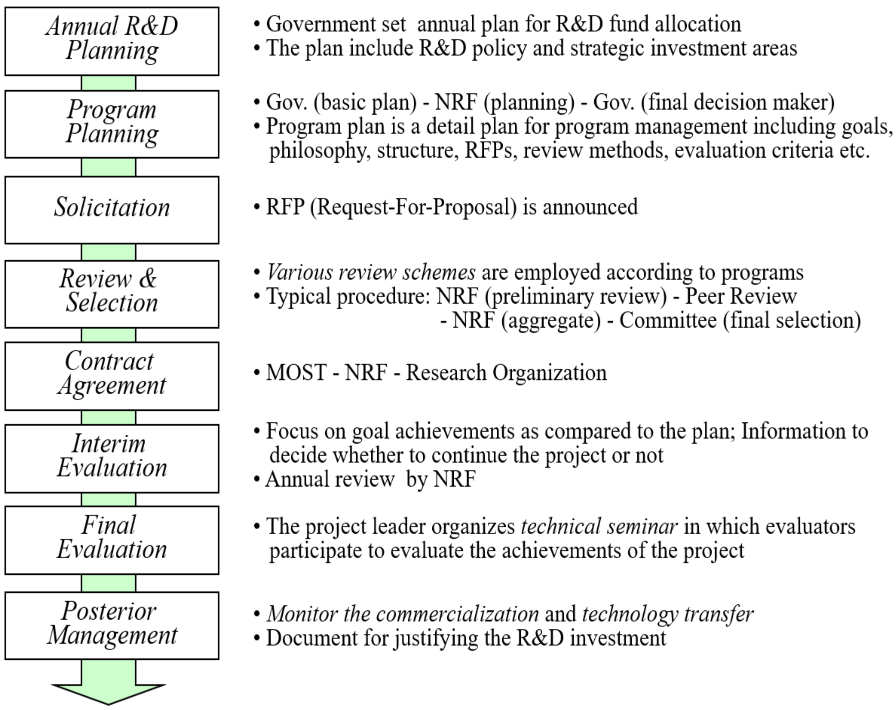
BRIN has 43 staffs in the Directorate of Research and Innovation Funding to manage eight funding schemes, which is not a small number, but additional support is needed for effective management and evaluation. It could consider utilizing review experts in the evaluation process, and hiring temporary or short-term support staff as needed.

The administrative staff of the Directorate of Research and Innovation Funding within BRIN should be proficient in online systems and have expertise in project management. For this purpose, manuals should be created and distributed for each task, and periodic internal training should be organized.

In order for BRIN to secure a certain number of expert reviewers, it is necessary to operate an expert reviewer’s database. The reviewer’s database should be updated and operated annually to include reviewers who have participated in the past, current heads of government R&D projects, and industry, academia, and research experts with research experience over a certain period of time.

※ The Case of South Korea

This is a diagram of the project management process for the NRDP program of the Ministry of Science and Technology of South Korea.



First, the annual plan for R&D funding is finalized by the Ministry of Science and Technology.

The NRF, as the funding management agency, prepares a detailed program plan, including objectives, structure, RFPs, and evaluation criteria, which is finally approved by the Ministry of Science and Technology.

RFPs are released and projects are selected through various reviews and evaluations. In general, the final selection of projects is made by the project-specific promotion committee after preliminary review by the NRF, expert peer review, review by specialized organizations, and prior consultation with the coordinator.

2.5 Measures to Activate Industry's Participation in R&D

2.5.1 Current Status

In 2021, the national R&D budget was IDR 41.43T, accounting for 0.28% of Indonesia's GDP. Of the national R&D budget, the public sector contributed IDR 36.85T (88.95%), or 0.25% of GDP, and the private sector contributed IDR 4.57T (11.03%), or 0.03% of GDP.

R&D in Indonesia is government-driven, with very little participation from industry or the private sector in contributing to the research budget. Therefore, BRIN aims to increase the private sector's participation, starting by promoting schemes that provide funding and facilitation.

The Directorate of Research and Innovation Funding in BRIN is continuing to work to form funding schemes that can include the industrial sector, including co-funding schemes with industrial research collaboration centers (PKR). PKR Type II (Industrial Research Collaboration Center) is a proposed research collaboration center in a company that includes BRIN researchers and may also include universities. The Industrial Research Collaboration Center emphasizes the development of products and services that can be utilized by companies that originate from research and innovation.

2.5.2 Issues and Challenges

In 2021, the R&D budget provided by the industry is only 11.03% of the national R&D budget and 0.03% of the national GDP, so it is urgent to devise measures to encourage the industry's participation in R&D.

※ **The case of South Korea**
In 2021, the national R&D budget was USD 89.282M, accounting for 4.96% of GDP. The national R&D budget was split between the public sector at USD 23,856M (26.7%), or 1.32% of GDP, and the private sector at USD 65,426M (73.3%), or 3.64% of GDP.

Through the RIIM Collaboration scheme, the Directorate of Research and Innovation Funding seeks to collaborate with national and international funders, especially with partners who are well established in research activities managed by BRIN. However, the RIIM Competition scheme, which manages more than 1,000 research proposals, has yet to demonstrate the expected collaboration with corporate funders, with less than 1% of the 1,000 proposals involving corporate partners in research activities.

2.5.3 Recommendations for improvement

A variety of policy measures are needed to increase industry participation in national R&D.

First, consider enacting a technology development promotion law that includes R&D investment, tax support, technology protection, and privileged access to government-owned patents and R&D infrastructure.

Second, establish specialized research institutes for strategic industry sectors at the national level to train human resources needed by industries and promote technology transfer and collaborative research.

Third, a public procurement system should be promoted to facilitate the purchase of SMEs' products and support the sales of SMEs' products created and invented through national R&D support, improving SMEs' competitiveness.

Fourth, the R&I funding scheme for BRINs should be strengthened to support corporate R&D. Industrial firms should be given preference in project announcements, such as designating them as lead research organizations or requiring them to participate in project components. In 2024, BRIN developed the PKR Industry scheme (Industrial Research Collaboration Center) to support company-focused research.

Fifth, BRIN needs to develop and support collaborative R&D programs to resolve technology challenges for companies.

Sixth, it is necessary to strengthen collaborative research and joint utilization of infrastructure in research institute such as BRIN by companies.

※ The case of South Korea

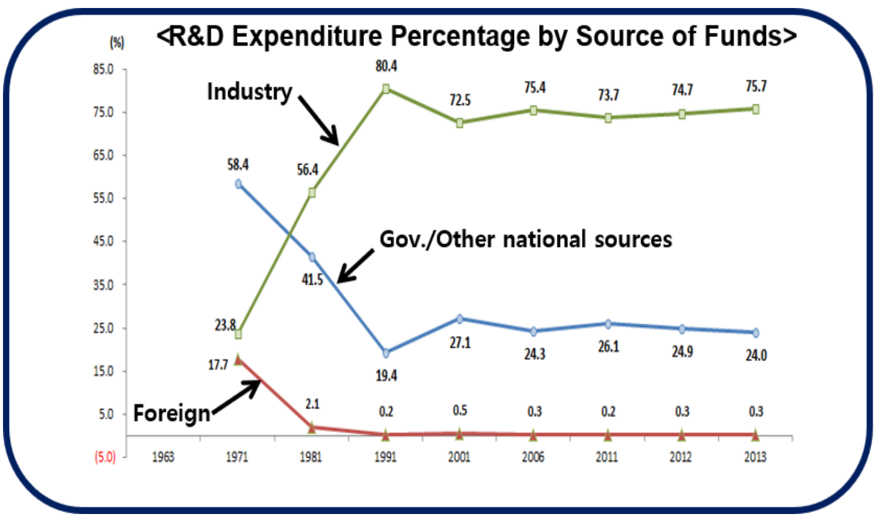
The South Korean government has mobilized a variety of policy instruments to increase industry participation in national R&D, including R&D investment, tax support, and exemption from military service for specialized research personnel.

In 1972, it enacted the Technology Development Promotion Act, which included tax incentives, technology protection, R&D support, and special use of government-owned patents.

To drive private R&D, the government established a number of specialized research institutes in major industrial sectors, each of which actively attracted overseas researchers and secured the latest scientific brains. The Korea Institute of Science and Technology (KIST) was established in 1966, the first government-funded research institute, to support industrial technology development.

The government rapidly increased the enrollment of engineering high schools and universities of science and technology to expand the supply of human resources for industrial sites. Universities focused on training human resources for industrial companies.

The government's investment in R&D and tax incentives have led to a rapid increase in the number of corporate R&D centers, from 100 in 1982, 1,000 in 1991, and over 30,000 in 2014. As a result, private R&D investment in 2014 was 180 times higher than in 1982.



The Korean government is promoting the public procurement system to expand the purchase of SMEs' products by public institutions as stipulated in Article 2 of the Act on the Promotion of SMEs' Products and Support for their Marketing to improve the competitiveness of SMEs. The Public Procurement Service designates R&D results conducted by central administrative agencies, pre-commercial prototypes, and technology-recognized products that are recognized as innovative by the Procurement Policy Review Council as innovative products, and the designated innovative products can be contracted for three years after designation. In addition, through the Innovative Product Pilot Purchase Project, the Public Procurement Service purchases innovative products with the Public Procurement Service's budget and provides them to demanding organizations such as public institutions, and evaluates feedback on the results after conducting pilot use.

The Korean government is also actively engaging industrial companies through R&D funding programs. It is promoting joint technological cooperation by supporting R&D for SMEs using infrastructure such as universities and research institutes. In addition, it is utilizing methods such as designating companies as lead research institutions when announcing projects, and requiring companies to participate in project components. If a company is a nationally recognized technological innovation company, if it is affiliated with an excellent corporate research institute, or if it has an excellent track record of paying technology fees, it is given extra points when selecting projects.

3. Policy Advisory Outputs and Utilization

3.1 Outputs and Expected Effects

This study developed draft guideline for the Indonesian R&I funding scheme.

First, to develop the draft guideline, we conducted a survey on the main areas of interest of the Directorate of R&I Funding in BRIN and proposed three areas: improvements to BRIN's R&D funding scheme, improvements to BRIN's R&I project management system, and measures to activate Indonesian industry's participation in R&I.

The main contents of the draft guideline developed based on the three areas proposed by the Directorate of R&I Funding are organized as follows.

- Overview of the Guidelines: Purpose, Need, and Scope of Application
- Laws and policies related to R&I funding and BRIN's mission based on them
- Current status and operation system of R&I funding
- Improvement of BRIN's R&I Funding System: Identification of current status, issues, and challenges, and suggestions for improvement based on Korea's advanced R&D funding experience
- Improvement of BRIN's R&I Project Management System: Identification of current status, issues, and challenges, and suggestions for improvement based on Korea's advanced R&D funding experience
- Activation of industrial participation in R&I: Identification of current status, issues, and challenges, and suggestions for activating industrial participation in R&D based on Korea's advanced R&D funding experience

The expected effects of this study are as follows.

First, it is expected to serve as a comprehensive guideline for BRIN to fulfill its mission related to R&I funding. It can also serve as a reference for the R&I funding support and management regulations recently developed by the Directorate of R&I Funding in BRIN.

By identifying the current status, issues, and challenges in the three areas of interest to the Directorate of R&I Funding in BRIN and providing specific recommendations for improvement based on Korea's advanced R&D funding experience, it is expected to provide direction for BRIN's advanced mission on R&I funding and strengthen BRIN's R&I funding capabilities.

It is expected to improve the consistency of Indonesia's R&I funding with its science and technology policy and to enhance the efficiency and effectiveness of R&I funding, ultimately contributing to the government's R&I productivity and national economic development.

CHAPTER 6

Lesson for Indonesia from the South Korea's Experience

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Chapter 6. Lesson for Indonesia from the South Korea's Experience

1. Current Condition of Information System of Research and Innovation Infrastructure in BRIN

1.1 The Perception of ELSA Users (Based on Survey Result)

BRIN established in 2021 has re-improved the management in delivering service for research and innovation (R&I) infrastructure through information system called ELSA-BRIN. This system is aimed to manage the BRIN's R&I infrastructures which are located around Indonesia. BRIN is the public R&D institution in Indonesia which has various R&I infrastructure such as laboratories, science parks, botanical gardens, scientific collections, and research vessels. Before BRIN established, those R&I infrastructures were previously under management of several public R&D institutions like the Indonesian Institute of Sciences (LIPI), the National Institute of Aeronautics and Space (LAPAN), the National Nuclear Energy Agency (BATAN), the Agency for Assessment and Application of Technology (BPPT), and other public R&D units under the Indonesian ministry management.

The ELSA system managed by BRIN is aimed to integrate all R&I services for internal and external users. Not limited to BRIN users, the external users of ELSA are from private R&D institutes, universities, enterprises, and other institutes. Previously, the users who utilise R&I infrastructures in Indonesia register in many dispersed R&D units in Indonesia. Other ELSA goals are to simplify and to accelerate the services for users and to make an efficient service management.

Referring to the survey described on the chapter 2 previously, most of respondents show the positive results concerning the BRIN's R&I infrastructures service. The respondents are from universities, enterprises, and BRIN researchers who use R&I infrastructure in the areas of engineering, health/medicine, and natural sciences with experiences more than 1 years. Respondents revealed that this system made "an easy access" to register and use with the normal waiting time between less than 1 month up to less than 6 months. Though, they said that the long turnaround time was the most challenging thing during the waiting process of ELSA system.

At most, the respondents were satisfied with the ELSA system in delivering R&I infrastructure services. Though, they encountered small problem with the BRIN's R&I infrastructures while conducting research. This survey result indicated that the BRIN's ELSA system is proper in delivering R&I infrastructure services. Nevertheless, this system requires improvement in the function and management to respond the future.

According to survey results, the R&I infrastructure service of the ELSA system is categorized as medium-high value with achievement of efficient-efficient enough. The ELSA system was responded as the easiness for the BRIN's R&I infrastructure users. The survey result provided several inputs and suggestions to improve the ELSA's system for answering the future challenges. Even though, this system needs to be improved in the part of features to get detailed information such as the use of equipment, comprehensive specifications or performance information, a maintenance schedule, and tool availability schedule.

Other suggestions to improve of the R&I infrastructure are streamlining the queue process, re-location of R&D laboratories and its equipment to be close with researchers/users, rental equipment scheme, transparency of schedule maintenance, preventive maintenance, the speed of repair and materials procurement process, flexible and planned resource allocation, and communication style from the ELSA operators/managers to users. Communication is important to respond the emerging issues when users is employing R&I infrastructures.

They highly expect that the accessibility of BRIN's R&I infrastructure through the ELSA system technically such as improving ELSA's user interface, improving timeliness of support services, simplifying access procedures, and spreading out infrastructure across different regions. On the specific figures of the ELSA system, improvement of this system can be added such as monitoring of tools usage, resource allocation statistics, equipment feedback on equipment from operators/managers to users, additional equipment needed to support R&D activities. The points of these improvements are to provide wider R&I infrastructure service to users and to shorten process in using those infrastructures.

In sum, the ELSA system has provided a breakthrough to deliver R&I infrastructure service for users from internal and external BRIN. Improvement of management function and system is highly required to answer the users need as well as to answer upcoming challenges. Learning from South Korea practice is one of strategy for BRIN to improve the ELSA system function and its management especially focusing on delivering R&I infrastructures in Indonesia.

1.2 The Improving R&I Infrastructure Service through Information System Management

Republic of Korea (RoK), in this context also called South Korea, is one of developed economies which is relied on the research-based economic growth. Research and Innovation (R&I) infrastructure in South Korea is systemically managed through Zone of Research Equipment Utilisation Service (ZEUS). Korea's ZEUS System, a research infrastructure information system, is being built and operated to promote efficient investment and systematic utilization of research infrastructure. As mentioned previously in chapter 4, ZEUS provides R&D infrastructure service to all users (Korean and non-Korean) in accessing and using all R&D infrastructures and facilities in South Korea.

In Indonesia, there is the ELSA system to deliver R&I infrastructure. ZEUS and ELSA have similar goals namely to provide excellent services for wider users, to provide research-infrastructure-based performance information inquiry and management services, and to improve function with efficient budget and utilization. Though, features of ZEUS and ELSA are different since both are initiated and developed in the distinctive region, culture, and environment. Again, the existing politics-bureaucracy system between RoK and Indonesia becomes an influencing factor to create an information system in delivering public services, including R&I infrastructure services.

The ELSA system can imitate partly or fully from the ZEUS operation system and then adjust it into ELSA system in a new manner. In the Chapter 4, the main issues of R&I infrastructure service are equipment procurement process which is not linked to the existing infrastructure system, no monitoring process and information system in place, no supply and demand data between research facilities, and no have a procedure to measure the utilization of its research infrastructure. Consequently, there are duplication of equipment use, limited utilization due to limited information access, lack of integrated information monitoring and evaluation, poor and delayed maintenance of R&D infrastructures. Hence, the improvement of business process of ELSA is required to answer those issues.

Referring to chapter 4, getting lesson from South Korea through ZEUS, several actions and policies can be promoted to strengthen management R&I infrastructure services in Indonesia through ELSA system. At macro level, there are two main strategies to improve the ELSA system; 1) establishment of R&I infrastructure information collection system; and 2) establishment of an efficient joint utilization system. Therefore, the BRIN’s policy should support both strategies like 1) policy for data-based procurement management, 2) policy for ensuring the systemic maintenance of existing R&I infrastructure, 3) policy for getting information easily, 4) policy for providing reservation and utilization of open R&I infrastructures with high usage, 5) policy for promoting procurement of preventive maintenance through integrated monitoring, 6) policy for identifying new demand and supply for R&I infrastructure.

At the meso level, the improvement of the R&I infrastructure information system can be embarked ranging from: 1) improving information system function, 2) improving information collection and management system, 3) Strengthening the monitoring and integrated management function, 4) expanding services and improving accessibility for users, 5) supporting the establishment, development, and performance evaluation of R&I infrastructures, and 6) streamlining procurement and budgeting. At micro level, the ELSA system can be improved in some stages covering; 1) efficient research infrastructure procurement and budget execution, 2) maximizing the utilization of research infrastructure, 3) establishing a maintenance system, 4) data-based performance management and evaluation, and 5) user accessibility and service expansion.

The contexts of Indonesia and South Korea are key clues that the ELSA practice is distinctively different compared to the ZEUS practice in South Korea. Hence, the Government of Indonesia can consider factors such as politics, geography, economics-market, socio, and local culture which have directly impact to the formation of an information system. Again, the working style of bureaucracy in Indonesia and South Korea are other factors to make a governmental information system workable. Those factors are essential for improving the ELSA information system. At least, we can start from at micro level by which BRIN has authority to regulate, re-shape, repair, and execute an improved change on the ELSA system.

2. Current Condition of Innovation Funding Scheme and Project Management in Indonesia

2.1 Research and Innovation (R&I) Funding Scheme

Research and Innovation (R&I) funding schemes in Indonesia are managed by public and private institutes. There are the Indonesian public institutes have direct authority to support R&I funding scheme namely the National Research and Innovation Agency (BRIN) through competitive research scheme, the Indonesian Ministry of Education, Culture, Research and Technology through universities, and the Indonesian Ministry of Finance through the education endowment fund (LPDP). As well, there are two categories of private institutes in supporting R&I funding namely private R&D institutes and R&D unit in private enterprises.

As described in Chapter 3 and 5 previously, the Indonesian Government contributes to the national R&I funding at most, while the support of R&D funding initiated from private institutes is limited. The issues of R&I funding in Indonesia is not merely a portion allocation between government funding and private funding, but also the government commitment and weak coordination among public-private institutes. Surely, it will influence the direction of S&T support to an economic development.

At general, the issues of R&I funding scheme in Indonesia (referring to Chapter 3) can be classified into 5 (five) categories namely: 1) the small R&D budget compared to the Indonesian Gross Domestic Product (GDP); 2) the less contribution of R&D funding from private sector; 3) no clear measurement of R&I funding and outputs at national scale; 4) weak policy which connects between R&I funding and business investment program; 5) bureaucratic process in managing public R&D funding scheme.

2.2 Research and Innovation Project Management

Research and innovation (R&I) project management is slighlty similar to R&I funding management. The most distinguished is on the technical and operational program in supporting the national R&I funding. Briefly, R&I funding scheme talks about a whole R&I funding at national scale, while R&I project management has more incapacious scope, it is a specific program with particular targets managed by an institute. In this discusión, we focuses on the BRIN’s R&I project management. The Directorate for R&I funding – BRIN is a key actor in managing R&I projects. This unit is not limited to manage BRIN’s R&I funding, but also to manage external R&D funding through mandatory schemes and collaboration schemes.

In Chapter 3 mentioned that the Directorate for R&I funding meets some technical and operational challenges in managing R&I project management. They are a complex administrative process, limited quantity and quality human resources in managing R&I funding, no standardization of financial procedures, weak coordination among public institutes, and incoherence of researchers need and managerial tasks. Those challenges need to be solved rapidly through the BRIN’s management initiatives in collaboration with BRIN’s researchers and external partners.

2.3 Improving Research and Innovation Funding Scheme and Project Management in Indonesia

Republic of Korea (RoK) has a plenty of experiences in formulating and implementing R&I funding programs connected to the Korean national development. This country is identical with a knowledge-based economy (KBE) growth by putting S&T as the main pillar at each program designed by the RoK Government. Case of R&I funding scheme in Indonesia, there are key points from the South Korea's practices that can be adjusted into the Indonesian context. First, learning from South Korea, the Indonesian Government should formulate the long term science-technology-innovation (STI) policy directed to national economics development. Second, there are indicators and measurements of R&I funding with detailed outputs at each governmental program. Third, the strength of national S&T act/law becomes the basic policies which is supported with operationable and measureable instruments. Fourth, improving a collaboration among public and private institutes to increase the national R&D budget allocation. Fifth, providing incentives to those who do R&D activities (public and private institutes).

To improving BRIN's R&I project managament, there are several measures that can be adopted from South Korea practices into Indonesian practices. Summarizing from Chapter 5, there are 3 (three) strategic measures namely: 1) increasing collaboration between BRIN and other related public institutes in managing R&I projects; 2) making a formal procedures of R&I project management and publishing it to public; 3) investing in human resources to get talents more qualified with an increased number of people (including staffs, managers, reviewers of research proposal).

Recommendation to improve R&I funding scheme and project management is provided by an expert from South Korea. Recommendations are aimed to increase industry participation in funding R&D activities in Indonesia. In practice, the Government of Indonesia has formulated policies in supporting R&I funding scheme such as tax deduction to industries or private R&D industries, long term policy coined into the national long & middle term development plan (RPJMN), partnership with business sectors, involving private sector to collaboration in R&D activities, technology transfer through turn-key projects, and optimizing the government R&I infrastuctures for public users. Indeed, BRIN has strategic policies to enlarge R&I competitive schemes and promote private sectors in involving into R&D activities using the BRIN R&I infrastructures. In addition, BRIN has specific progams to develop human resources in improving their skill in R&I fields.

At the national implementation stage, the Government of Indonesia need to have operational and detailed measurement of formulated R&I funding policies. Often, the derivative and technical laws are not present to translate a policy document into practice so that parts of policies are not workable to support R&I activities in Indonesia including bureaucracy culture and business culture. For example, BRIN has strategic policies for supporting collaboration in sharing R&D funding between public R&D institutes and private sectors. That policy is less implementable in Indonesia since the national government regulations and business climate is not coherent.

CHAPTER 7

Conclusion

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Chapter 7. Conclusion

1. Summary

1.1 The Improvement of R&I Infrastructure through Information System in Indonesia

The improvement of research and innovation (R&I) infrastructure and R&I funding scheme in Indonesia is aimed to provide public services optimally to users. In this report, the improvement of R&I infrastructure focused on the BRIN's ELSA system that provides detailed information system related to application, utilization, and operation of R&D laboratories and facilities to all users. They are derived from BRIN's researchers, academicians from universities, researchers in private R&D institutes, engineers at enterprises and other users. Each of them has purpose in employing BRIN's R&I infrastructures through the ELSA system. While, the R&I funding system focuses on the BRIN's R&I funding system interlinked into the national funding system in Indonesia.

The experiences and best practices from South Korea or Republic of Korea (RoK) in operating R&D infrastructure and funding system become references for the Government of Indonesia in improving R&I infrastructure service and funding scheme in Indonesia. In this case, BRIN as a public R&D institute as well as STI stakeholders in Indonesia is the focal point to créate R&I innovation ecosystem through the improvement of R&I infrastructure and funding system at national level.

Specific to R&I infrastructure, referring to the South Korean ZEUS system, the BRIN ELSA system has similar purposes; to identify, to manage, and to evaluate the R&I infrastructure operation. In addition, both systems are functioned to detect the erroneous or malfunction equipment and to minimize human mistake when they are doing activities in R&D laboratories. Learning from the ZEUS, the ELSA is a different system. The R&I infrastructure service of information system can be improved into the ZUES system through many technical and operational requirements, as well as the support of BRIN management.

BRIN has developed the ELSA system since at the end of 2021 when BRIN just was established, while the ZEUS system has been under construction and operation by the Korean Government since 2013 with starting on 2.000 equipments reservations. Nowadays, that system has provided more tan 70.000 equipments reservations around South Korea. Compared to the ELSA system, the time differentiation scale of the ZUES system is almost 10 years, and the ELSA system covers limited equipment reservations in the BRIN organization only, not included the equipment reservations in universities, private R&D units, and enterprises in Indonesia. Though, the ELSA system covers all equipment reservations all public R&D laboratories and facilities result from integration of public R&D institutes in Indonesia such as LIPI, LAPAN, BATAN, and BPPT.

The ELSA system is a forward way to advance and to improve the R&I infrastructure information system in Indonesia. The more advanced steps of the ZUES system can be learned and adjusted for the ELSA system to close the advanced levels between ELSA and ZEUS. It means that the ELSA system can develop to be a similar system of ZEUS covering all equipment reservations in a country and supported by a systemically-managed information operation. In sum, the ELSA system is the early initiation to be a robust information system providing all information and management about R&I infrastructure in Indonesia, like the ZEUS system in South Korea.

1.2 The Improvement of R&I Funding System in Indonesia

Learning from the South Korean experiences, several suggestions are formulation and implementation a prority R&I funding policy with long-term scheme. To attract the private investment in the R&I activities, government needs to créate a private-public funding collaboration scheme through several R&D projects. Creating a strategic R&D-based

industry hub involving private sector, universities, and government agencies is a real measure to increase public-private funding in Indonesia. Other measures to increase the national R&I funding allocation are tax allowance or incentives for R&D-based enterprises, technology protection, exclusive access of public-owned patents, and the use of public R&D infrastructures. The connection of goods purchasing system to local enterprises is one of strategies to manage R&I funding system.

So does the South Korean Government, the Government of Indonesia has had programs to attract private R&D investment and to increase public R&D allocation at national level such as providing tax allowance or incentive to private companies, arranging national R&I policies, creating a collaborative project involving R&D institutes, universities, enterprises, and government agencies (for example the PKR program initiated by BRIN), public procurement for R&D goods, the intelectual property rights protection, and the support for R&D-based industries.

Learning from the Government of South Korea, the Korean STI policies including the national R&D funding programs are designed into the long-term policy with the measurable indicators and operational instruments at each program. Government of South Korea strongly relies on S&T-based economics to boost the Korean national development. This policy is supported by qualified human resources, public incentives, robust regulations/laws, an organized bureaucracy system, and good public infrastructures. All supports are required to design S&T growth-based policy in South Korea. The South Korean government commitments at each stage of STI policy is a key to corroborate the national R&D funding heading to an innovation ecosystem.

The Government of Indonesia through STI stakeholders can make a shared commitment to design a long term STI policy by proritizing on public-private R&D funding scheme. Formulating a grand design of national STI policy requires measurable indicators and operational instruments at each of program implementation stage. Surely, the political direction of the Indonesian Government to prioritize STI policy as an engine for national economics development is a must.

2. Policy Recommendations

2.1 Measures to improve R&I Infrastructure service through Information system

In the practice, BRIN can create new figures in the ELSA system to respond user needs and streamline the business process to make service efficient and effective. But it is not enough to improve function of the R&I infrastructure system in Indonesia. Identifying problem, collecting data and information, coordinating with internal operators, staffs, and managers, shortening registration process, maintaining equipment, linking to procurement process, monitoring in place at the time, evaluating process regularly, turning around mistakes and re-structuring division of organization are inevitable efforts to make the ELSA system responsive to the user demands.

Human resources such as operators, staffs, and managers in managing the ELSA information system need to paid more attention by the BRIN management. BRIN policy can target them as the first agent of change in delivering R&I infrastructure system through short training and education, socialization, internship practice, and other activities to improve their skill and knowledge about R&I infrastructure service through an information system like ZEUS in South Korea. Collaborating with the ZEUS management is one of feasible options to improve the ELSA system in delivering R&I infrastructures services and managing R&I infrastructures efficiently.

Gathering information from users are valuable inputs to improve the ELSA functions. In one hand, ELSA is aimed to support optimum R&D activities to users, in another hand, BRIN management need to make efficient business process of R&I infrastructure service. Therefore, BRIN should find an appropriate scheme to bridge two goals namely, improvement of R&I infrastructure service quality and efficient business principle. Consideringly, regulation and bureaucracy are not separated from the BRIN as a public agency in Indonesia. Improving the ELSA functions has to adjust to the higher-level regulations in the scope of information law. Even, other regulations about public service, goods procurement, information access, task force of organization, revenue system, and taxation are necessarily considered to develop a ELSA system.

2.2 Improving the Performance of R&I Funding Mechanism

The stakeholders of R&I funding in Indonesia are many public institutes such as the National Research and Innovation Agency (BRIN), the Ministry of Education, Culture, and Research and Technology, and the Ministry of Finance through LPDP institute and universities. Each of them has collaborated through a joint R&I project to connect between R&D activities and business activities. Harmonizing the funding regulations aimed to a big STI project in Indonesia is a breakthrough to involve public private partnership. Those projects are related to researchers, enterprises, and other public institutes dealing with R&I funding. Thereof, the Indonesian long-term grand design of STI policy is required to créate a national innovation ecosystem.

Specifically, there are three suggestions adopted from South Korea to the Indonesia practice related to R&I funding scheme. First, the Indonesian Government (in this case – BRIN), can collaborate with other public institutes, universities and business sector to develop some specific research centers directly connected into the industries. This centers can provide specific trainings, education programs, technology transfer and other activities that provide direct impact to the economics. Second, BRIN can corroborate quality and quantity of human resources in the R&I fields through upgrading education system regularly. Third, BRIN in joint with other public and private institutes can identify and purchase local products through procurement process which is linked to the R&I funding directorate. This scheme can impact to the improvement performance of R&I infrastructure system through ELSA system.

A great decisión need to be made to minimize cumbersome practices of R&I project management. BRIN has sufficient resources and authority to solve issues in the R&I project management. The high-medium level managers of BRIN have authority to direct new ideas and breakthroughs through BRIN regulations. Legally and hierarchically, this strategy is appropriate to address these issues, but it is not sufficient to answer R&I market dynamics at national and global level. Hence, the involvement of users and external partners is a necessary to make an improved system in managing R&I Project management in BRIN.

The BRIN's initiatives have direct impactful contribution to the national development provided that there is a policy function to make R&I funding indicators which are correlated to national investment programs. BRIN can identify existing issues and address them by employing internal resources maximally. Nevertheless, BRIN cannot provide impactful result of R&D activities if other government institutes do not support the linkage between R&D results to industries. Hence, collaboration BRIN with various public institutes is a must to improve the performance of the BRIN's R&I project management.

2.3 Collaboration of STI Stakeholders

Developing science-technology-innovation (STI) policy in Indonesia requires basic principles that involves multiple actors from public and private institutions. Improving R&I funding and infrastructure in Indonesia is not merely attached on the BRIN' authority but also from other public and private institutions in Indonesia. Since at the end 2024, the Government of Indonesia has launched the Indonesian Ministry of Higher Education, Science and Technology (MoHEST) that has a little bit similar function to BRIN in formulating STI policy. It means that creating a STI ecosystem is part of a whole system of the government agencies involving BRIN and MoHEST.

As parts of STI stakeholders, other ministries such as Ministry of Industry, Ministry of Finance through the funding endowment for research (LPDP), and the Ministry of national development and plan (BAPPENAS) are crucial to improve the R&I infrastructure service and funding mechanism in Indonesia. In addition, the involvement of business enterprises, universities, private research and development (R&D) institutes is inevitable part to improve R&I infrastructure and funding governance in Indonesia. Some suggested to reinforce the existing policies in Indonesia as follows:

1. STI stakeholders should formulate a joint policy to improve and to widen functions of R&I infrastructure and funding at national level.
2. STI stakeholders should issue a regulation to attract private sectors in investing in R&I infrastructure system and R&I funding. For example: incentives, tax holiday, exclusive program, and simpilify of business permit. This policy can be executed by involving other related ministries.
3. STI stakeholders should formulate and implement consistently a long-middle term STI plan (at least 5 years) with clear indicators and measurements of achievement at each stage.
4. STI stakeholders should widen and augment the scheme of the research collaboration center/PKR initiated by BRIN.

STI stakeholders should formulate a joint policy to créate a standard regulation for managing R&I funding in Indonesia. It can be conducted by BRIN, MoHEST, and Ministry of Finance.

REFERENCES

Chapter 1.

[1] Asmara, A.Y. and Kusumastuti, R. (2021). Innovation Policy Implementation in Indonesia: Perspective of Triple Helix. Journal of STI Policy and Management, 6 (1) (2021), pp. 1–19. DOI: 10.14203/STIPM.2021.290.

[2] Huda, Nur; Pawennei, Irsan; Ratri, Andhina; Taylor¸ Veronica L. (2020). Making Indonesia's Research and Development Better Stakeholder Ideas and International Best Practices. Retrieved from https://www.ksi-indonesia.org/assets/uploads/original/2021/02/ksi-1613637314.pdf.

[3] Jin, Seunghoo & Lee, Kangwon. (2020). The Government R&D Funding and Management Performance: The Mediating Effect of Technology Innovation. J. Open Innov. Technol. Mark. Complex. 2020, 6, 94; doi:10.3390/joitmc6040094.

[4] Kim, Yongjoo. (2023). South Korea's R&D Infrastructure Policy. Policy Consultation on the Design and Implementation of Research & Innovation Infrastructure and the Funding Policy for Improvement of the Science, Technology and Innovation System in Indonesia. Published by STEPI.

[5] Mani, S. (2002). Government, Innovation, and Technology Policy. Edward Elgar, Cheltenham-UK.

[6] Patanakul, P., & Pinto, J. K. (2017). Examining the Roles of Government Policy on Innovation. Journal of High Technology Management Research 25: 97–107.
https://doi.org/10.1016/j.hitech.2014.07.003

[7] WIPO. (2023). Republic of Korea ranking in the Global Innovation Index 2023.
https://www.wipo.int/edocs/pubdocs/en/wipo-pub-2000-2023/kr.pdf.

[8] World Bank. (2010). Innovation Policy: A guide for Developing Countries. World Bank Publishing, Washington DC.

[9] Yoon, Donghun. (2018). The policy conflict research of interested parties for the efficient management of research equipment: With focus on the government and the scientist, Cogent Business & Management 5, 147587.

Chapter 2.

[10] BRIN. (2023). National Research Infrastructure Report 2023. Jakarta: BRIN.

[11] Brousseau, E., & Penard, T. (2007). The Economics of Digital Business Models: A Framework for Analyzing the Economics of Platforms. Communications & Strategies.

[12] Cressey, D. (2011). Research infrastructure: facilities shared. Nature.

[13] Georghiou, L., Halfpenny, P., Procter, R., & Hewitt, B. (2010). Management of Research Infrastructure in the 21st Century. Research Policy.

[14] Katz, J. S., & Martin, B. R. (2007). What is research collaboration? Research Policy.

[15] OECD (2017). Enhancing the Performance of Research Infrastructures: A Guide to Best Practices. OECD Publishing.

[16] Pillay, R., Hoskisson, R. E., & Tihanyi, L. (2015). Research Infrastructures in Emerging Economies. Academy of Management Review.

[17] Ponds, R., Oort, F., & Frenken, K. (2007). The geographical and institutional proximity of research collaboration. Research Policy.

[18] Schmidt, P., & Teräs, M. (2013). Long-term strategic planning for research infrastructures. Journal of Science Policy & Governance.

Chapter 3.

[19] BRIN. (2023). Performance Report of Government Agencies, BRIN 2022.

[20] BRIN. (2024). Performance Report of Government Agencies, BRIN 2023.

[21] Book of Indicators of Science, Technology, Research, and Innovation of Indonesia, BRIN, 2023 & 2024.

[22] Directorate of Research Funding. (2024, September 24–25). Presentation material delivered at the Workshop of The 2023 K-Innovation Partnership Program. National Research and Innovation Agency (BRIN).

[23] BRIN and LPDP (2023).

[24] Katadata (2018).

[25] Bappenas (2024).

Chapter 4.

[26] BRIN (2024).

[27] ZEUS (2024).

Chapter 5.

[28] BRIN (2023). Book of Indicators of Science, Technology, Research, and Innovation of Indonesia.

[29] BRIN (2024). Book of Indicators of Science, Technology, Research, and Innovation of Indonesia.

[30] BRIN (2024), BRIN and STEPI Workshop Material (23-24 Sep. 2024, Jakarta)

[31] Ministry of Science and ICT (2022). Framework Act on Science and Technology, Act No. 18863, June 10, 2022, Some revisions.

[32] Ministry of Strategy and Finance (2023). National Finance Act, Act No. 19430, June 9, 2023, Amendment to other laws.

[33] KSI (2019).

[34] Katadata (2018).